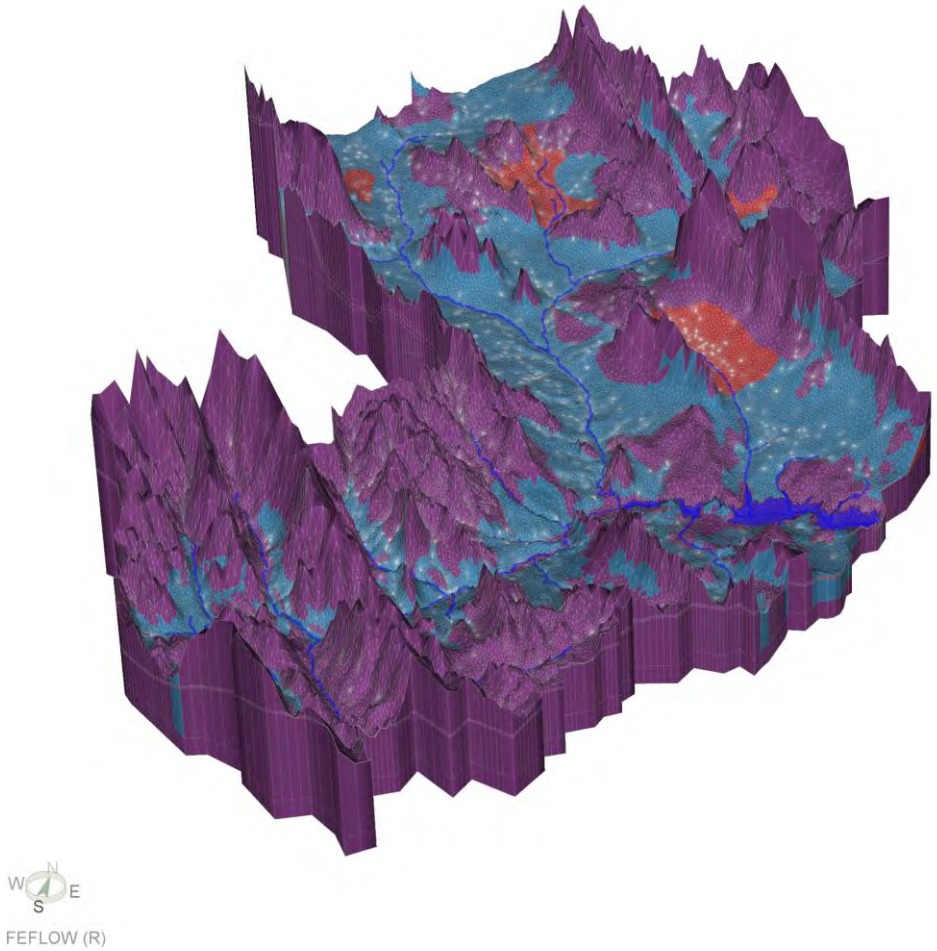
IWRM Zayandeh Rud Iran – Phase 2

Hydrogeological structural model for the upstream part of the Zayandeh Rud River basin



Ministry of Energy - Isfahan Regional Water Company

Report

May 2016

The expert in WATER ENVIRONMENTS

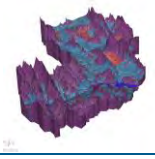
This report was prepared as part of the DHI business management system.



IWRM Zayandeh Rud Iran – Phase 2

Hydrogeological structural model for the upstream part of the Zayandeh Rud River basin

Created for	Ministry of Energy - Isfahan Regional Water Company
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1 Introduction

1.1 Aim of the study

A detailed understanding of the hydraulic relevant structures in the sub-surface is a major cornerstone for hydrogeological studies and a sustainable groundwater management. The understanding of the hydraulic relevant structures can be documented in a 3D-“Hydrogeological Structure Model” (HSM). A HSM itself can be used for planning and operating of wells and for groundwater management issues like creating risk maps for groundwater contamination from anthropogenic substances.

The aim of the present study was to develop a conceptual HSM for the relevant part of the Zayandeh Rud river basins upstream the Chardegan dam, based on the availability of data and the relevance for the WMT .A conceptual HSM in this case means, that the HSM should divide the major hydraulic relevant layers inside the alluvial aquifer system. In the next step, the HSM will be used to build up the 3D-Groundwater model for the same model area. The HSM was built up with the groundwater simulator FEFLOW and will be delivered as 3D-fem-file.

1.2 Model area

The HSM is located in the eastern part of the province of Esfahan and covers also some parts of the province Chahar Mahaal & Bakhtiari. The HSM has an area of 4115 km² and includes the sub catchments with the codes 4211 to 4216. The extension of the HSM is shown in Figure 1.

The model area was mainly defined by the availability of data, which were delivered by the Isfahan Regional Water Company (IRWC). Upstream the Chardegan reservoir, there are two sub catchments, which are located in the catchment area of the Zayandeh Rud river, but belong are managed by the provinces Chahar Mahaal & Bakhtiari. The model area should cover these sub catchments also, but the model quality depends on the availability of data.

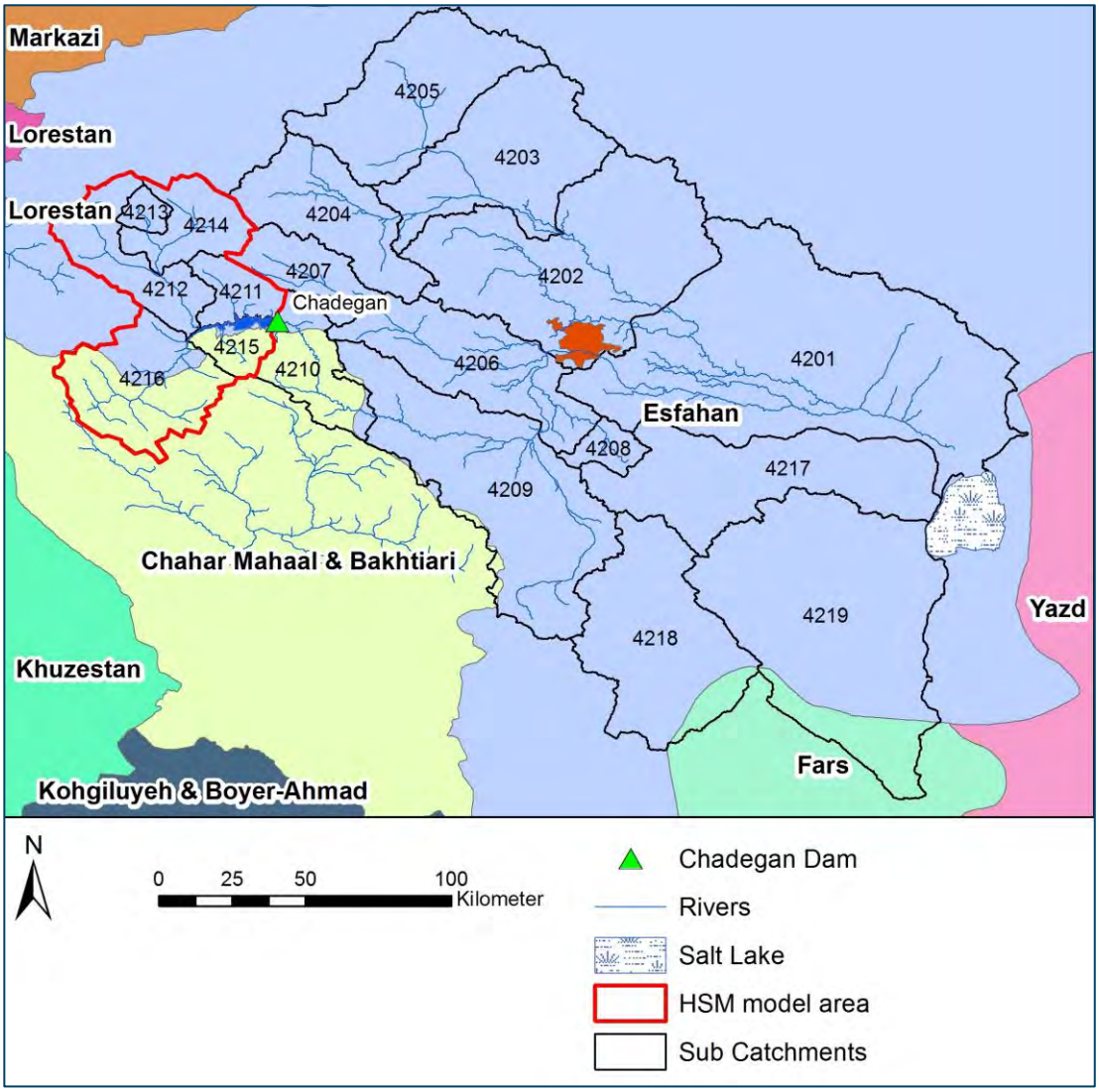


Figure 1: Location of the HSM

2 Data and data processing

2.1 Geological well logs – Delivered data

During the first phase of the project, several geological well log data were delivered by the IRWC (data delivery at 28.08.2012). This data could not be used for the development of the HSM. The reason is the data format. The data are mainly written in Farsi language and do not consist of any editable data format. An example for the delivered well log data of the first phase is presented in Figure 2. Because of the data format and the data language the information is not accessible for the German group. Therefore this kind of well log information was not considered by the actual study for the development of an HSM.

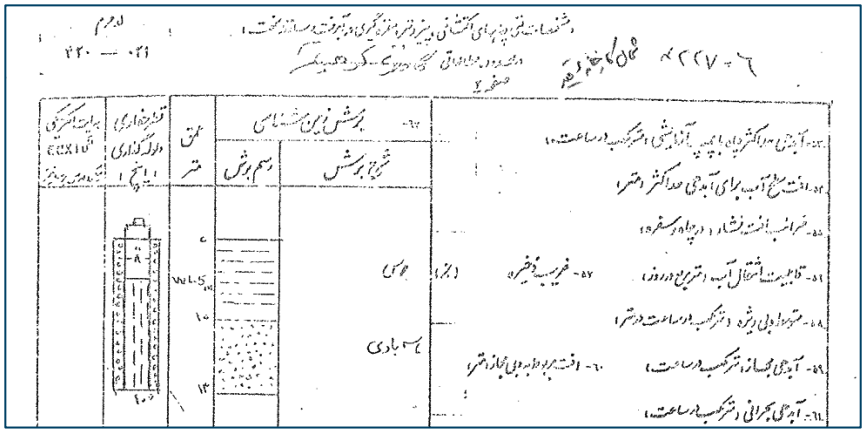


Figure 2: Geological well log data which were delivered during the first project phase

Additional data for the second project phase were delivered by IRWC at 25.01.2016 within the excel-file “log_fereidan.xlsx”. The excel-file consists of the coordinates and geological well log descriptions and follows the requested format for data delivery from the German partners.

For the sub catchments 4211 – 4214 a total sum of 38 well log data were available. For the sub catchments 4215 and 4216 no data were available. An overview of the available well log data is presented in Table 1.

Table 1: Overview of the available drilling logs per sub catchment

Sub catchment code	Name	Number of well logs
4211	Chardegan	5
4212	Daran	5
4213	Chehel khane	3
4214	Damaneh	25
4215		No data
4216		No data

The location of the available well logs is shown in Figure 3. It is shown, that for some sub catchments the data is scarce. On the one hand, the total number of drillings is limited. On the other hand, the distance between the data is rather variable.

The map also shows the extension of the quaternary alluvium (Q) and the aquiver extension. The aquifer extension was delivered by the IRWC and was created by analyzing the presence of wells and groundwater from different data sources.

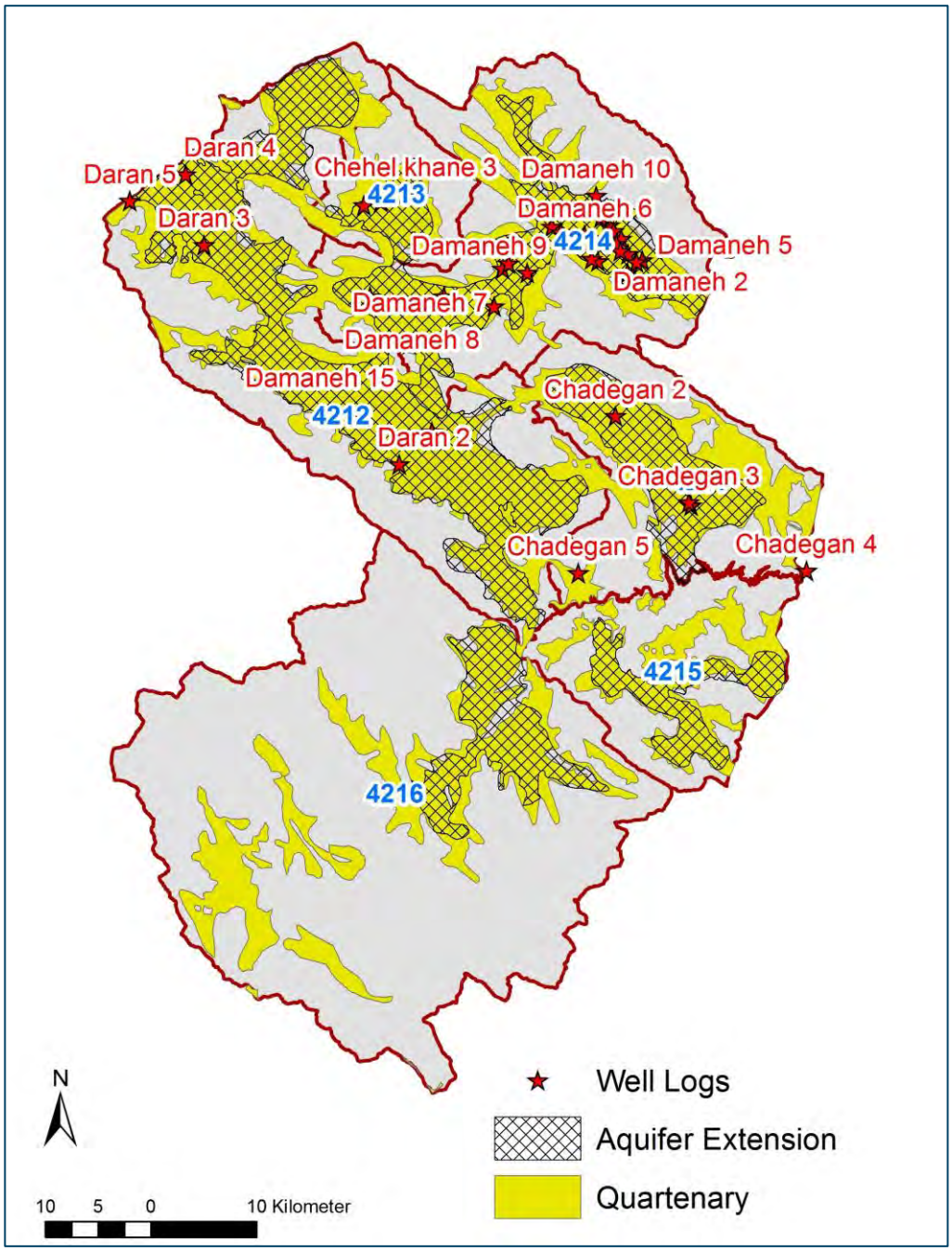


Figure 3: Location of the available well log data for the model area upstream the main dam

2.2 Geological well logs – Data processing

The well log data were delivered as excel-file with a grain size description of the sample. This data were processed by the borehole data management program GeODin ©. The program generates visual drilling profiles, which were used for manual geological correlation. The drilling profiles from GeODin are documented in the Appendix.

Based on the percentage distribution of clay, sand and gravel, the abbreviation in GeODin was chosen. The grain sizes with the dominant proportion become the major component. In Table 2 an example for the well log chehel khane 1 is shown. The first 12 meter consists of gravel, which is very clayey and slightly sandy. The descriptive attributes are defined by the percentages. The attribute “slightly” stands for 10 %, “moderately” stand for 20 %, “very” stands for 40 %. No descriptive attribute was uses if a grain size class represents around 30 % of the material.

Table 2: Example of well log data of well ID 4213001 - Chehel khane 1

Depth [m]	Formation	Clay [%]	Sand [%]	Gravel [%]
12	Alluvium	35	10	55
21	Alluvium	20	10	70
30	Alluvium	90	5	5
41	Alluvium	65	5	30

The delivered well log data can be visualized by using standard symbols and the drilling depth information. An example for the well log of Chehel khane 1 is shown in Figure 4.

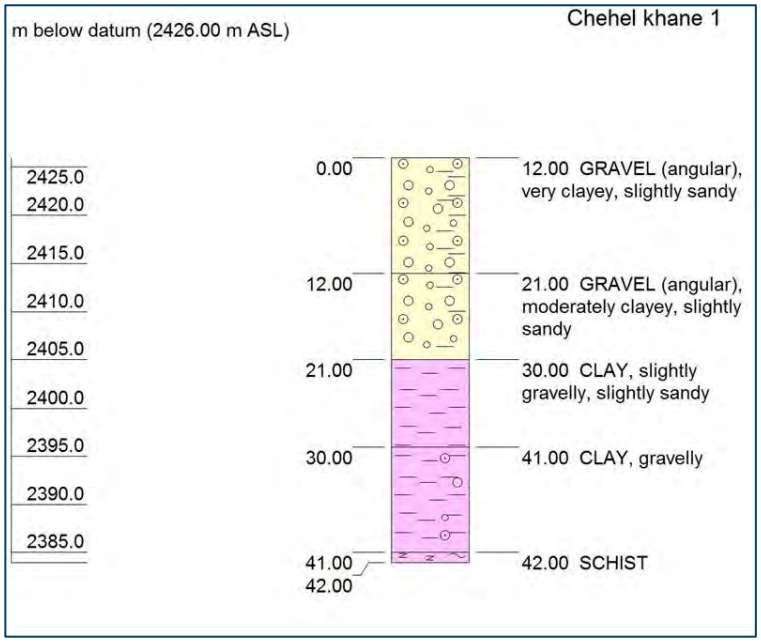


Figure 4: Well log of Chehel khane 1

2.3 Geological maps

Several geological maps were delivered in TIFF- and SHP-format and different scales by the IRWC. The data was delivered on 28.08.2012. An overview of the relevant geological maps in the study area upstream of the dam is shown in Figure 5. The provided geological maps were used to get a general understanding of the study area. Next to the occurrence of geological formations, the maps provide structural information like the occurrence of faults.

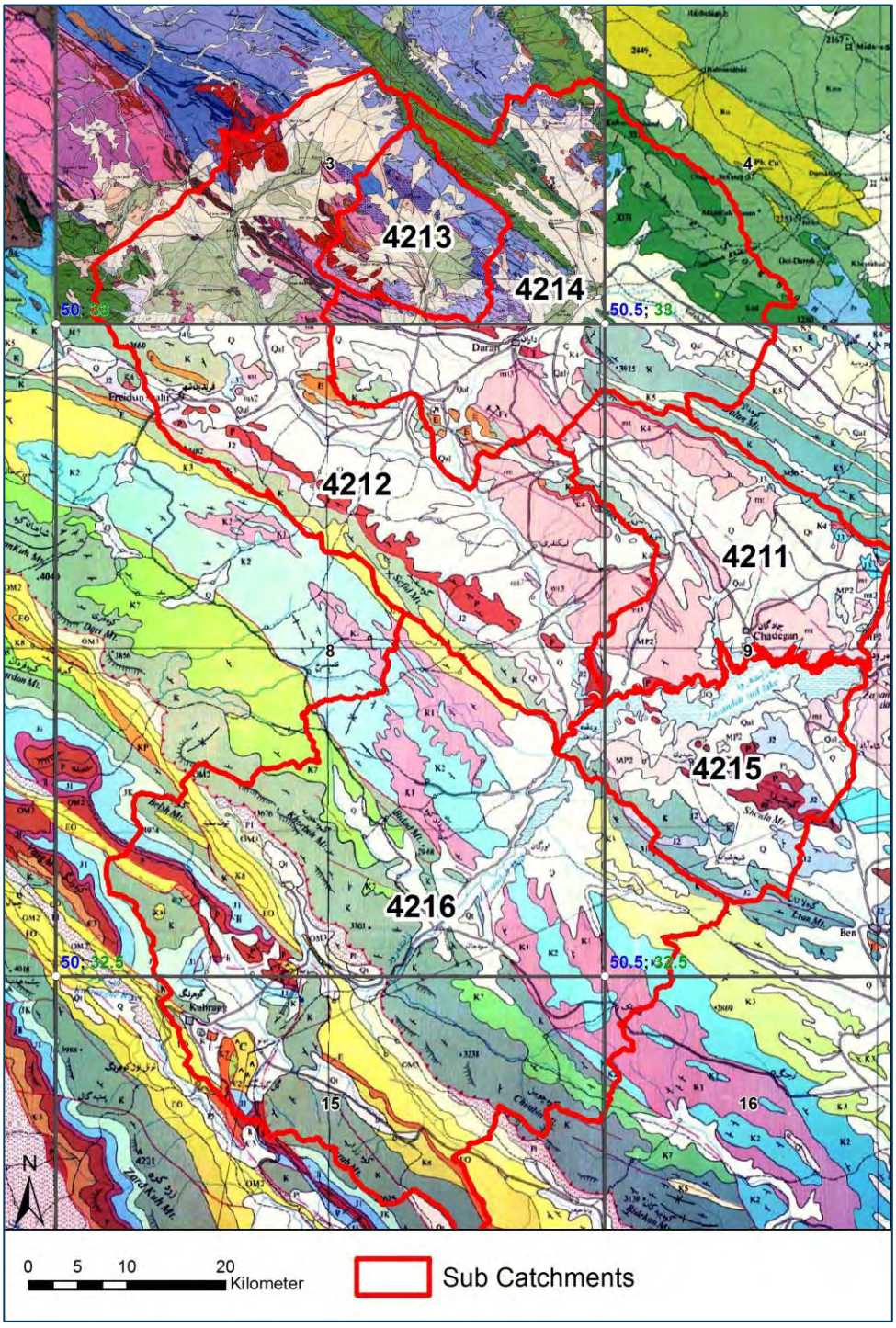


Figure 5: Available geological maps

The delivered geological map in SHP-file format is shown in Figure 6. The map covers the entire model area and therefore provides a good overview of the near surface geology. Additionally to the well logs mentioned in Chapter 2.2 this map was used for the development of the HSM to divide hard-rock formations from the occurrence of the quaternary alluvial sediments (Q). The extension of the quaternary alluvial sediments is not similar with the extension of the aquifer, which is shown in Figure 3. The difference is the water saturation. Inside the borders of the aquifer extension groundwater is present, while inside the borders of the quaternary alluvial sediments also just dry sediments could be present.

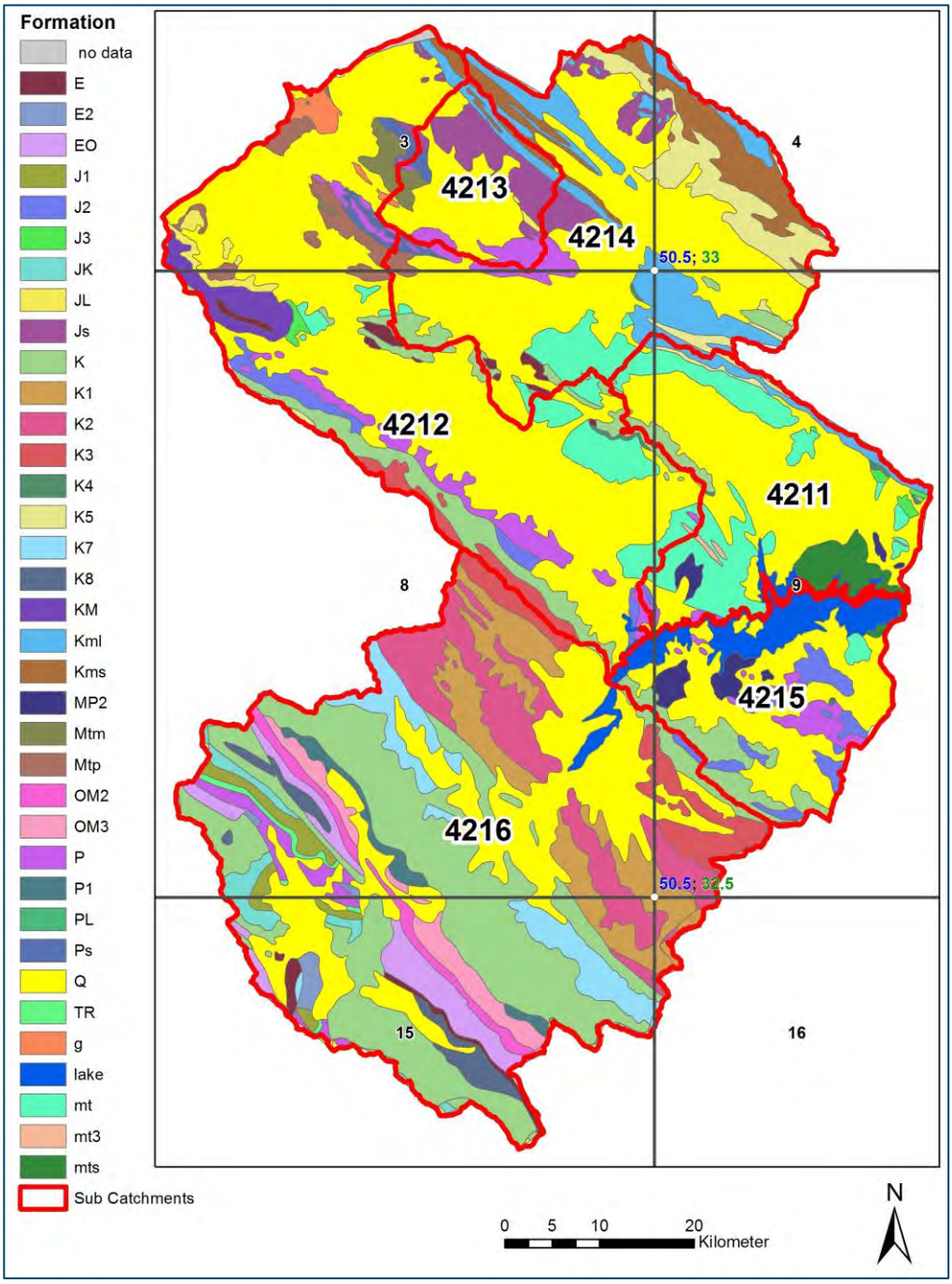


Figure 6: Geological map for splitting of the quaternary alluvium (Q) and hard rock formations

2.4 Geophysical data

During the first phase of the project, the IRWC has delivered two datasets of geo-electrical surveys. (Data delivery on 28.08.2012)

The first dataset consists of geo-electrical measurements for the provinces Meymeh and Moorchekhort (Sub catchment code 4205 and 4203). The data were delivered in excel-tables and surfer-files. Both analyzed provinces are not part of the actual HSM study. Therefore the data could not be used.

The second dataset consists also of geo-electrical data. A description of the data or a report written or translated in English was not delivered. The data consists of manually developed and scanned maps and cross sections. All data were delivered as TIF-files. The data covers several sub catchments. An overview of the available geo-electrical data is shown in Table 3. The location of the analyses was identified by using the coordinates at the map or by manually matching with the geological map. For one dataset the identification of the location was not possible.

None of the geo-electrical data could be used for the actual study for the development of an HSM, because the data location did not fit to the study area.

Table 3: Overview of geo-electric data

Data Directory	sub catchment code	Located by
1-motaleat-e-geoelektrik-e-mantagheye fereidan	4214	geologic map
6-motaleat-e-geoelektrik-e-dasht jarghuye esfahan	4217	geologic map
7-motaleat-e-geoelektrik-e-shomal wa shargh esfahan(tekrari14)	unknown	
27-motaleat-e-geofisik dasht mahyar	4208	geologic map
28-motaleat-e-geofisik mantagheye khpaye-segzi	4201	geologic map
30-barrasihaye geoelektrik mantagheye lenjanat	4206	coordinates
32-motaleat-e-geoelektrik-e-mantagheye jarghuye(tekrari44)	4217	coordinates
33-motaleat-e-geoelektrik-e-dasht mahyarjonubi(tekrari43-71)	4217	coordinates
48-motaleat-e-geoelektrik-e-mantagheye ardestan	4801	description “ardestan”
49-gozaresh geoelektrik dasht lenjanat esfahan	4206	coordinates
70-motaleat-e-geoelektrik-e-dasht ardestan	4132	coordinates

2.5 Topography

For the definition of the topography the IRWC has delivered a DEM with a raster size of 90 m during the first phase of the project. Additionally, a DEM with a raster size of 10 m was delivered at 19.08.2015. The DEM 10 for the area of the HSM is shown in Figure 7. The data was delivered in GRID-file format.

The date covers the entire area of the HSM. Because of the high resolution and the big model area, it was necessary to resample the data to a raster size of 20 m. The ESRI resampling function with cubic interpolation was used. In the next step the 20 m raster was converted into a point shape file.

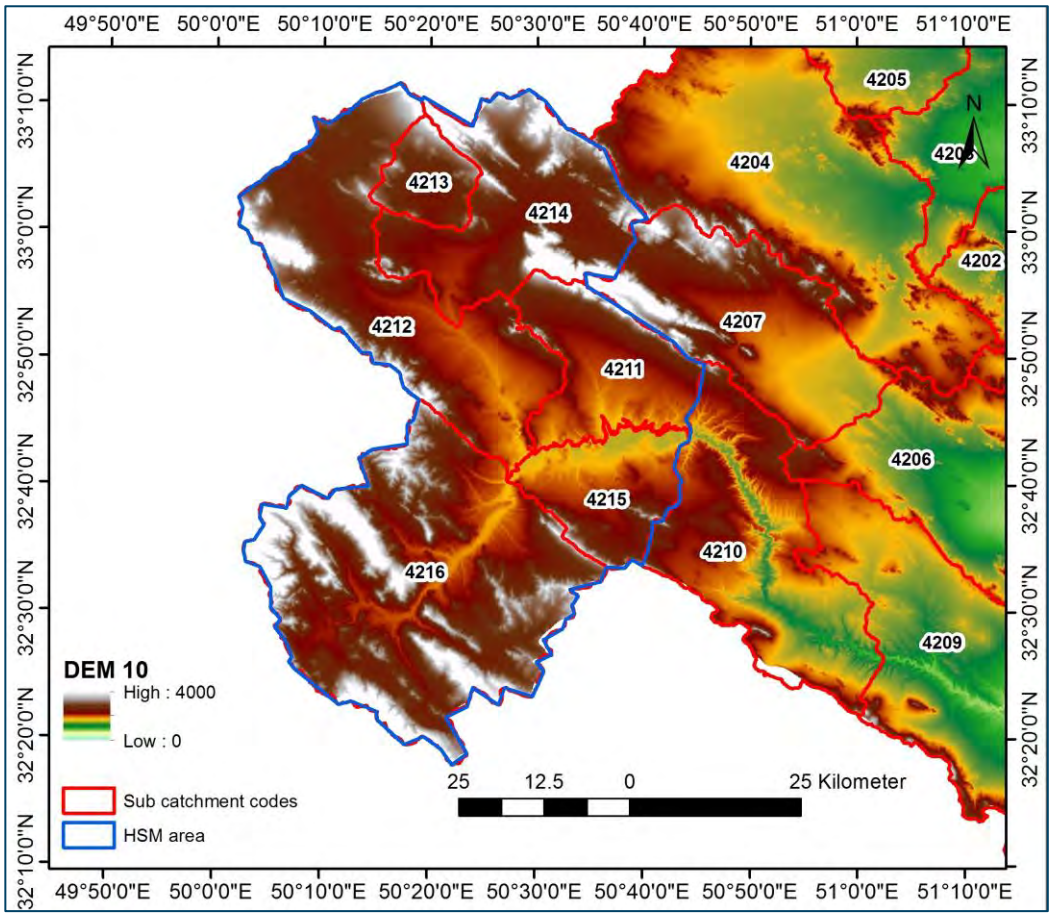


Figure 7: DEM 10 of the area of the HSM

2.6 Base of the quaternary alluvium

During the first phase of the project, data for the definition of the base of the alluvial aquifer were generated by the group of Dr. Safavi and delivered to the German group at 02.12.2012. The data define the contact between the alluvial aquifer and the underlying base rock. The raster data have a cell size of 110 m and were converted to a point shape file. The data do not cover the sub catchment codes 4215 and 4216.

To avoid interpolation effects between the aquifer base information and the borders of the HSM additional helping points were added along the border of the HSM area and inside the sub catchments with the codes 4215 and 4216.

In arrangement with the IRWC the sub catchments 4215 and 4216 are not analyzed in detail. The used simplifications are described in chapter 3.1.

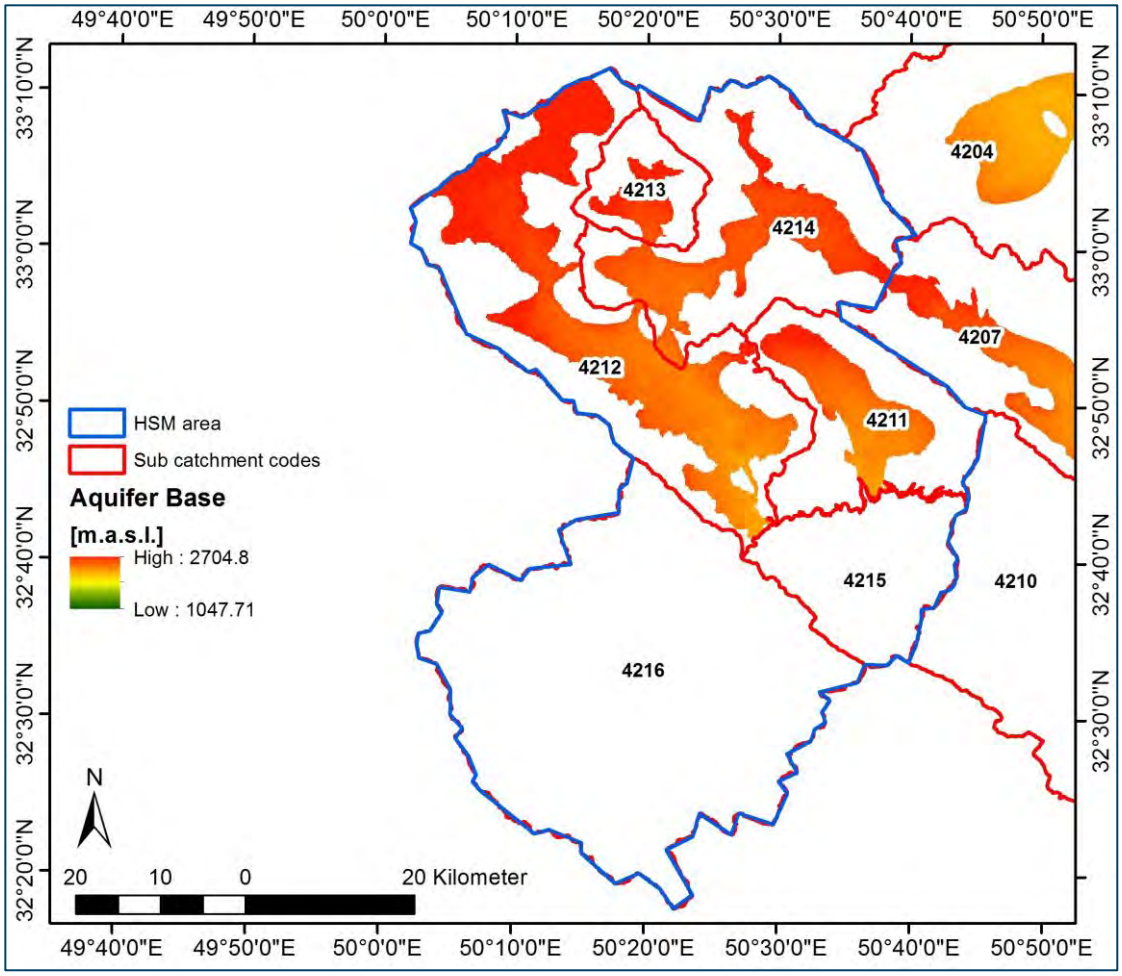


Figure 8: Base of alluvial aquifers

3 Results

In the following chapter the results of the HSM model will be presented.

3.1 Conceptual Overview

All 38 well log data were classified into two hydraulic conductive classes. Material that consists of 50 % or more clay corresponds to a poor hydraulic conductive layer and was summarized to “Clay”. The good hydraulic conductive class was named “Sand”. This class contains less than 50 % of clay, all kinds of sands and conglomerates. The good conductive class was colored in yellow; the poor hydraulic conductive class was colored in purple. The classification of all 38 well logs is shown in Table 4. The classification is related to the model layers, which are numbered from the most shallow layer (layer 1), to the deepest layer (Layer 5) of the alluvial aquifer. Layer 6 is related to the hardrock basement of the quaternary alluvium.

Because 28 of 38 well logs start with clay as first class, the first sand layer was set to layer 2. Except for of two well logs, all well logs show a sand layer at layer 2. At least 21 of 38 wells show a clay layer below the first sand layer. This second clay layer was related to layer 3. Seven well logs show a second sand layer below the second clay layer. This second sand layer was related to the structural layer 4. Because the base of layer 4 did not always fit exactly to the aquifer base, a fifth layer was set between layer 4 (sand) and the top of the basement. The hardrock basement was defined as layer 6.

Table 4: Classification of all 38 well log data by hydraulic conductivity

Code	Name	Layer 1	Layer 2	Layer 3	Layer 4	Layer 5	Layer 6
4211	Chadegan 1	Clay	Sand				Base
4211	Chadegan 2	Sand	Sand	Sand	Sand	Sand	Base
4211	Chadegan 3	Clay	Sand				Base
4211	Chadegan 4	Clay	Sand				Base
4211	Chadegan 5	Clay	Sand	Clay			Base
4212	Daran 1	Clay	Sand	Clay			Base
4212	Daran 2	Clay	Sand	Clay	Sand		Base
4212	Daran 3		Sand	Clay	Sand		Base
4212	Daran 4	Clay	Sand	Clay			Base
4212	Daran 5	Clay	Sand	Clay	Sand		Base
4213	Chehel khane 1		Sand	Clay			Base
4213	Chehel khane 2		Sand	Clay			Base
4213	Chehel khane 3	Clay	Sand	Clay	Sand	Clay	Base
4214	Damaneh 1	Clay	Sand				Base
4214	Damaneh 2	Clay	Sand	Clay	Sand		Base
4214	Damaneh 3	Clay	Clay	Clay	Clay	Clay	Base
4214	Damaneh 4	Clay	Clay	Clay	Clay	Clay	Base
4214	Damaneh 5	Clay	Sand	Clay			Base
4214	Damaneh 6	Clay	Sand	Clay			Base
4214	Damaneh 7		Sand	Clay	Sand		Base
4214	Damaneh 8	Clay	Sand	Clay	Sand		Base
4214	Damaneh 9	Sand	Sand	Sand	Sand	Sand	Base
4214	Damaneh 10		Sand	Clay			Base
4214	Damaneh 11	Clay	Sand				Base
4214	Damaneh 12	Clay	Sand				Base
4214	Damaneh 13	Clay	Sand				Base
4214	Damaneh 14	Clay	Sand				
4214	Damaneh 15	Clay	Sand				
4214	Damaneh 16	Clay	Sand	Clay			
4214	Damaneh 17	Clay	Sand				
4214	Damaneh 18		Sand	Clay			
4214	Damaneh 19	Sand	Sand	Sand	Sand	Sand	
4214	Damaneh 20	Sand	Sand	Sand	Sand	Sand	
4214	Damaneh 21	Clay	Sand				
4214	Damaneh 22	Clay	Sand	Clay	Sand		
4214	Damaneh 23	Clay	Clay	Clay	Clay	Clay	
4214	Damaneh 24	Clay	Sand				
4214	Damaneh 25	Clay	Sand				

The 38 individual well logs were summarized in a general classification per sub catchment. All four sub catchments show a lateral variation in the classification. The lateral variation is shown in Table 5.

Table 5: Horizontal variation of conceptual classes

Layer	4211		4212		4213		4214		
	NW	SE	SW	NE	SE	SW	SW	Center	NE
1	Aquifer	Aquitard	Aquitard	Aquitard	Aquifer	Aquitard	Aquitard	Aquifer	Aquitard
2	Aquifer	Aquifer	Aquifer	Aquifer	Aquifer	Aquifer	Aquifer	Aquifer	Aquifer
3	Aquifer	Aquifer	Aquitard	Aquitard	Aquitard	Aquitard	Aquifer	Aquitard	Aquifer
4	Aquifer	Aquifer	Aquifer	Aquitard	Aquitard	Aquifer	Aquifer	Aquitard	Aquifer
5	Aquifer	Aquifer	Aquitard	Aquitard	Aquitard	Aquitard	Aquifer	Aquitard	Aquitard
6	Bedrock	Bedrock	Bedrock	Bedrock	Bedrock	Bedrock	Bedrock	Bedrock	Bedrock

For each such catchment at least one representative cross section was interpreted. The locations of the cross sections are shown in Figure 9. The shown quaternary alluvium in blue consists in the layer 1 mainly of clay; only some parts consist of different sands (red).

For the sub catchments 4215 and 4216 neither geological well logs nor the definition of the aquifer base were available. Therefore, no vertical differentiation into aquifers and aquitards was possible. Inside the borders of the aquifer extension and the quaternary alluvial sediments the layers 1-5 were defined as aquifers with a total thickness of 50 m. Each layer has the same thickness of 10 m. In arrangement with the IRWC this part of the model will not be analyzed in detail.

In several sub catchments a clay layer covers the aquifer. An aquifer which is located below a sealing clay layer is a potentially confined aquifer. If the pressure level is higher than the top of the aquifer, the potential aquifer can clearly be called confined aquifer. As an effect of a strong groundwater level decrease, confined – unconfined aquifer conditions can change in several parts of an aquifer during time. Groundwater pressure analyses are not part of the actual study and are not relevant for the development of an HSM. As simplification all potential confined aquifers are called confines aquifer.

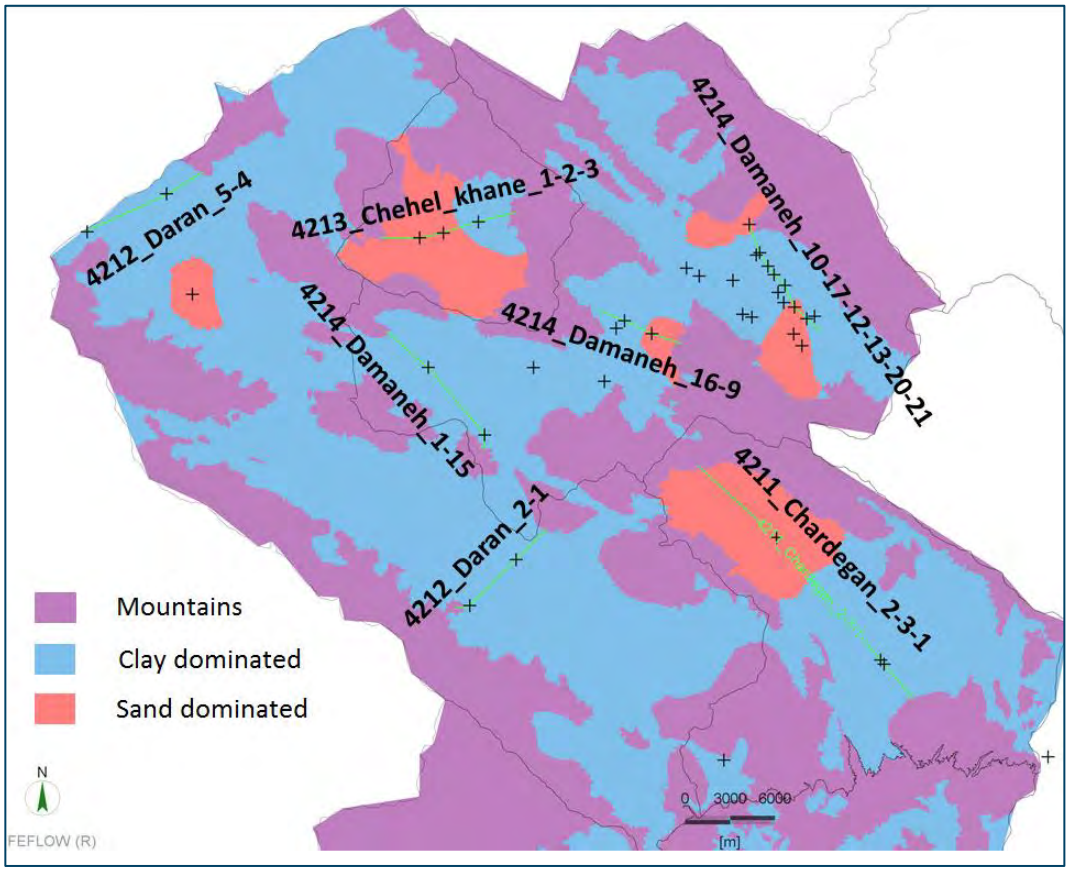


Figure 9: Location of interpreted cross sections

3.2 Sub catchment 4211 - Chadegan

For sub catchment Chadegan well log data for five wells were available. Three of these well logs were used for a geological correlation. The other two well logs are located in separated parts of the quaternary alluvium. The geological interpretation is shown in Figure 10. It is a schematic cross section from northwest to south east thought the sub catchment. 4211. In the northeastern part an unconfined aquifer is present, while in the southwest the presence of an overlaying clay layer causes confined conditions. In Figure 11 the correspondent cross section view is shown.

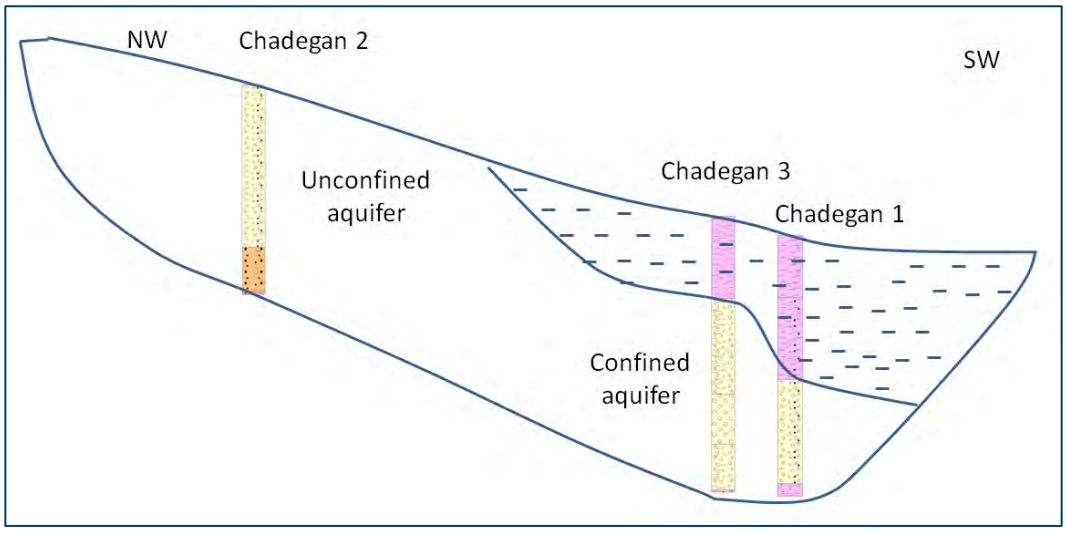


Figure 10: Schematic cross section though sub catchment 4211, Chadegan 2-3-1

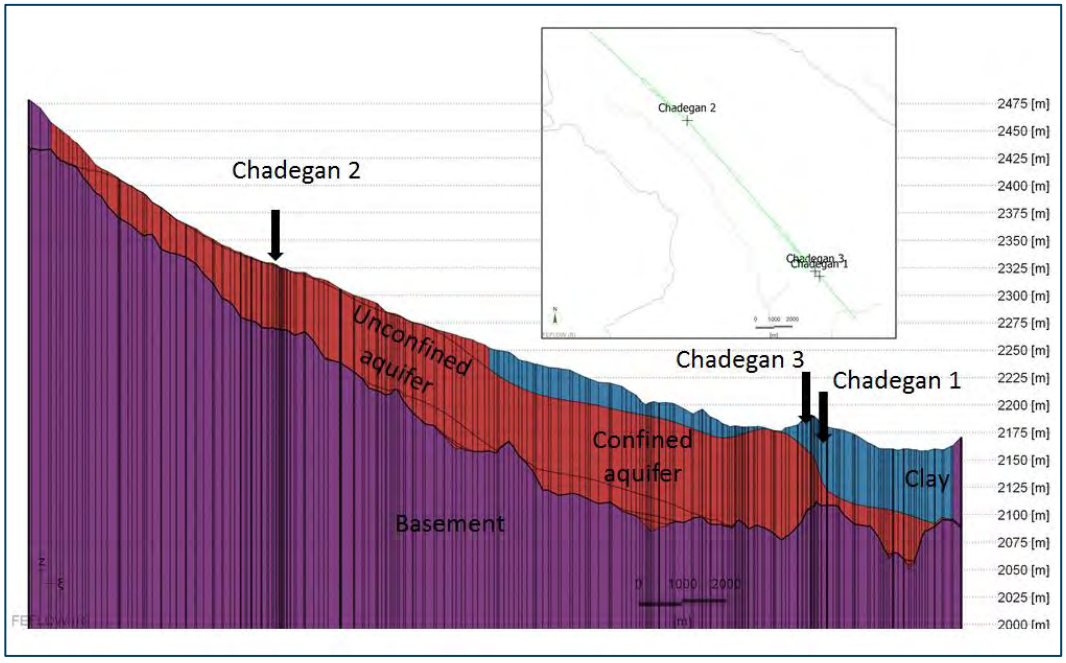


Figure 11: Cross section “4211_chardegan_2-3-1”, vertical exaggeration of 25

3.3 Sub catchment 4212 – Daran

For sub catchment Dran well log data of five wells were available. Most of the well logs show two confined aquifers. Between Daran 2 and Daran 1, the second aquifer ends. The same occurs between Dran 5 and Dran 4. The Shematic cross sections for the sub catchment are shown in Figure 12 and Figure 14, while the correspondent cross sections are shown in Figure 13 and Figure 15.

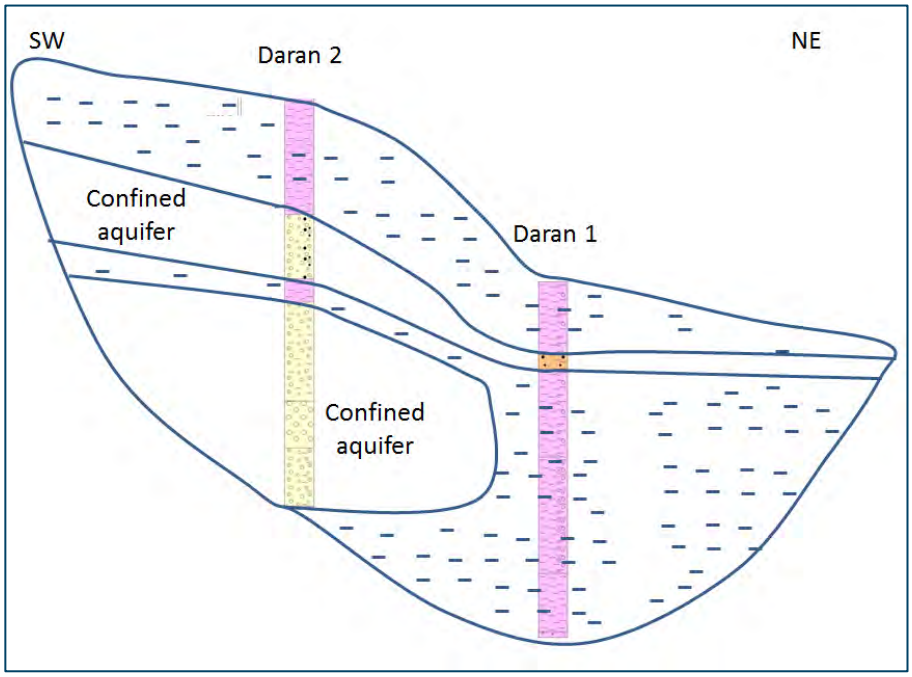


Figure 12: Schematic cross section though sub catchment 4212 southern part, Daran 2-1

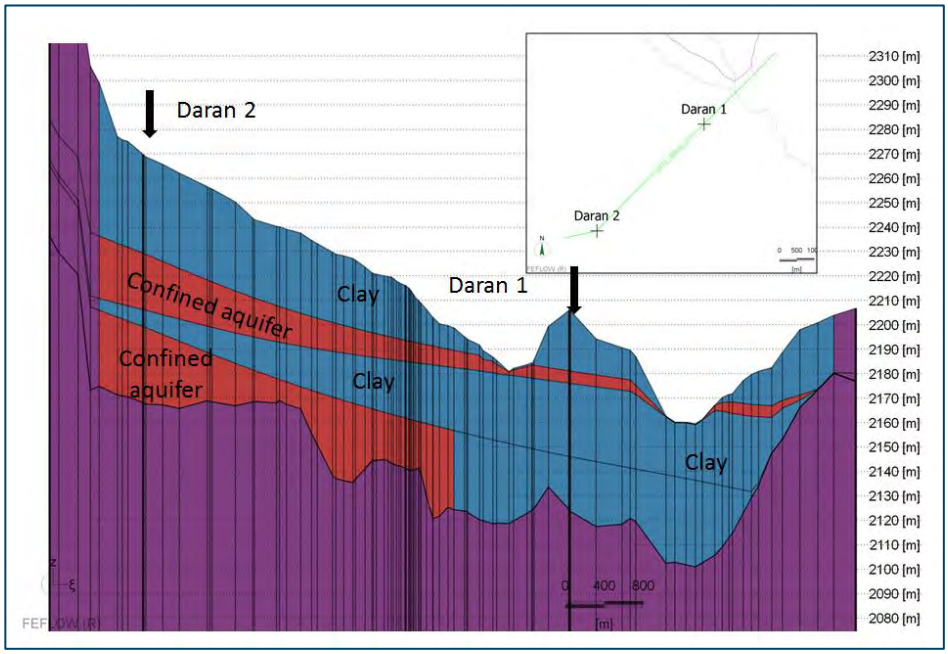


Figure 13: Cross section 4214_Daran_2-1, vertical exaggeration of 25

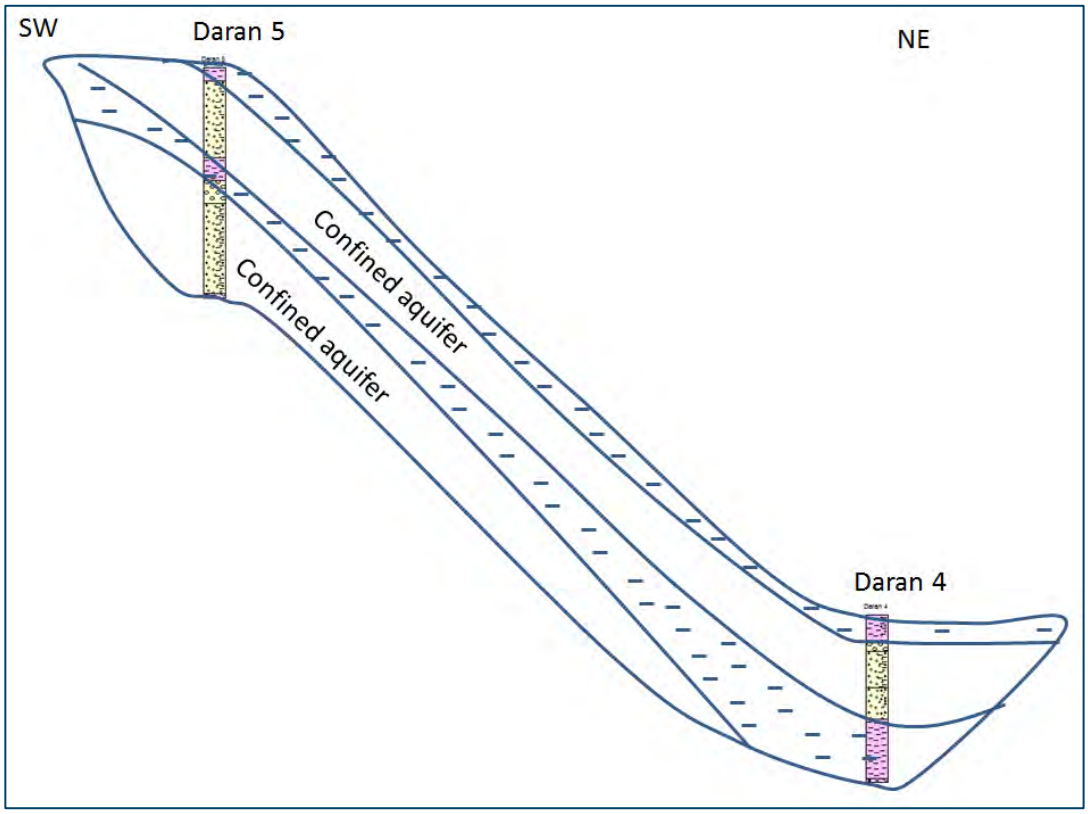


Figure 14: Schematic cross section thought sub catchment 4212 northern part, Daran 5-4

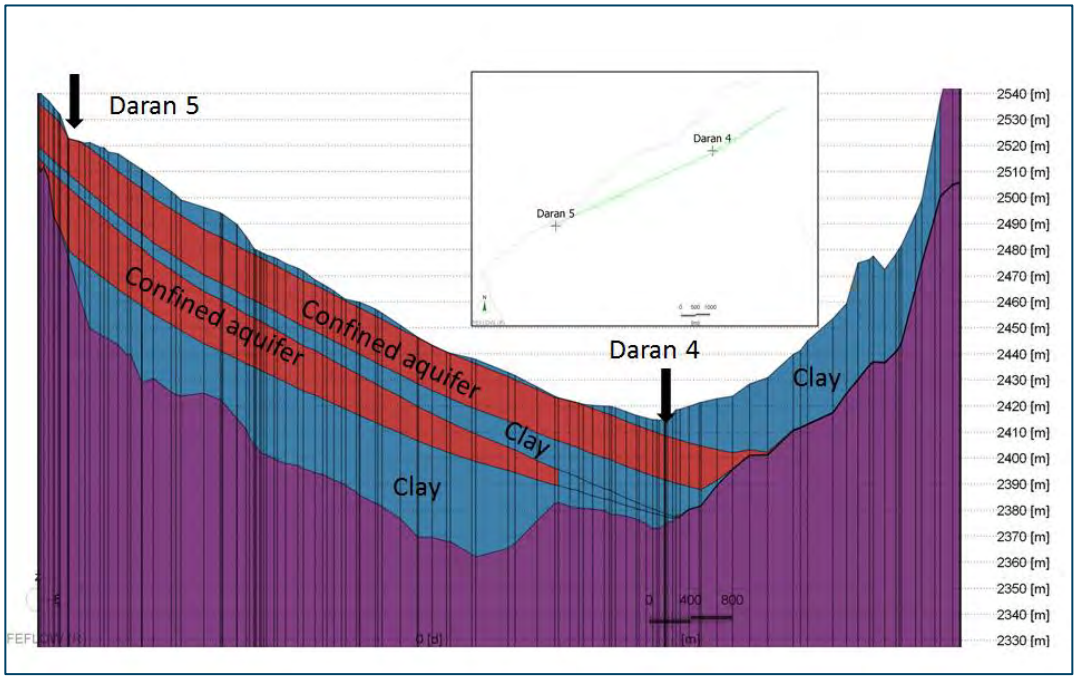


Figure 15: Cross section 4212_Daran_5-4, vertical exaggeration of 25

3.4 Sub catchment 4213 - Chehel khane

For the sub catchment Chehel khane three well log data were available. The schematic interpretation is shown in the cross section in Figure 16. In the western part of the sub catchment an unconfined aquifer is present, while in the eastern part a play layer covers the aquifer layer. The thin sand layer in the well log of Chehel khane 3 was interpreted as local sand lens and was not implemented in the conceptual HSM shown in Figure 17.

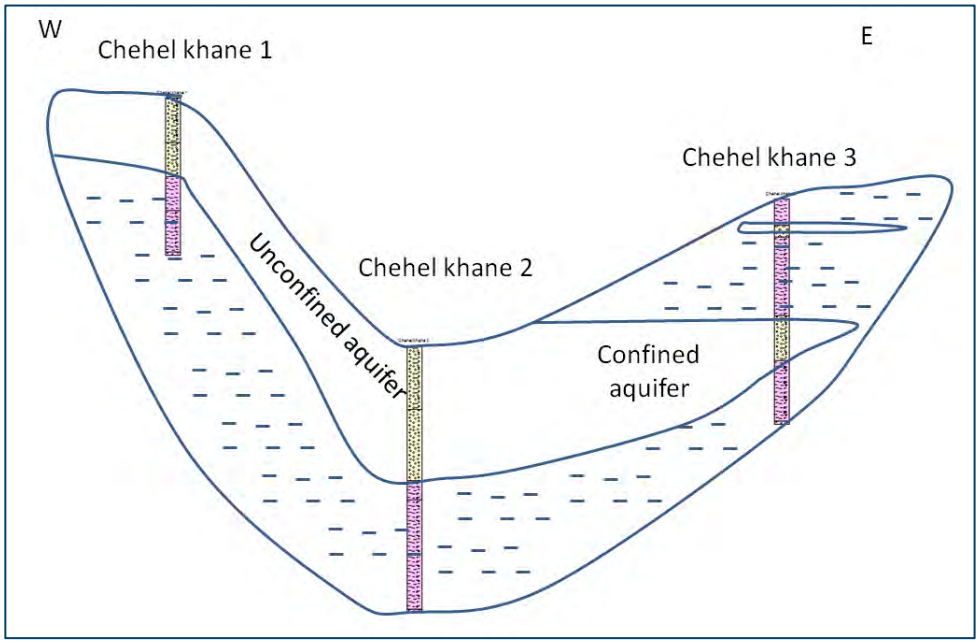


Figure 16: Schematic cross section though sub catchment 4213 Chehel khane 1-2-3

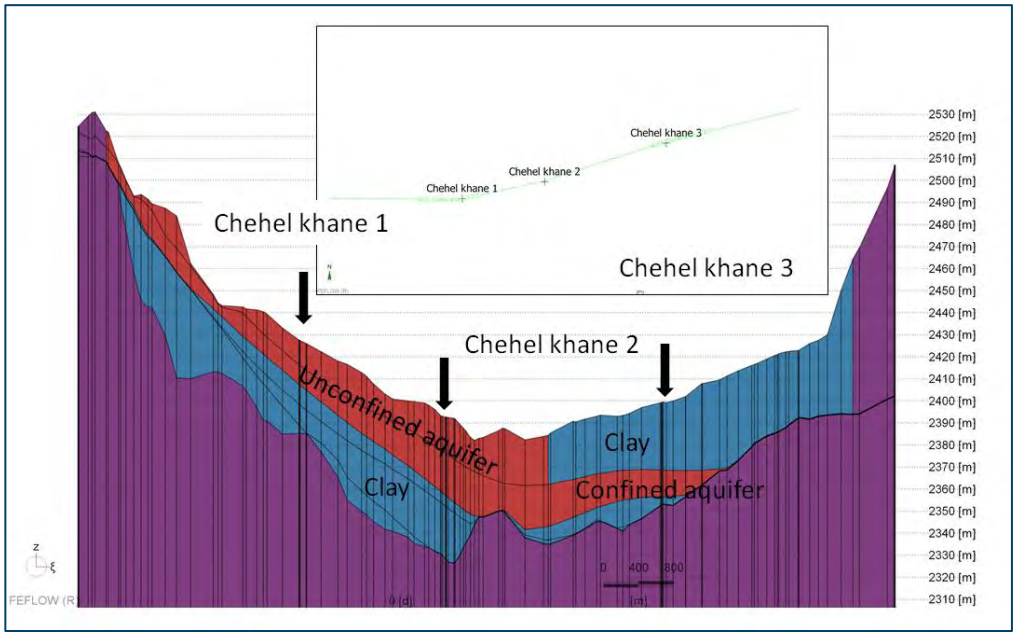


Figure 17: Cross section 4213_Chehel khane 1-2-3, vertical exaggeration of 25

3.5 Sub catchment 4214 – Damaneh

In the sub catchment Damaneh 25 well log data were available. By the interpretation of the geological well log data, the sub catchment becomes divided into three parts. In the south-western part, a thick clay layer is present above an aquifer system. The schematic cross section is shown in Figure 18, while the correspondent cross section of the HSM is shown in

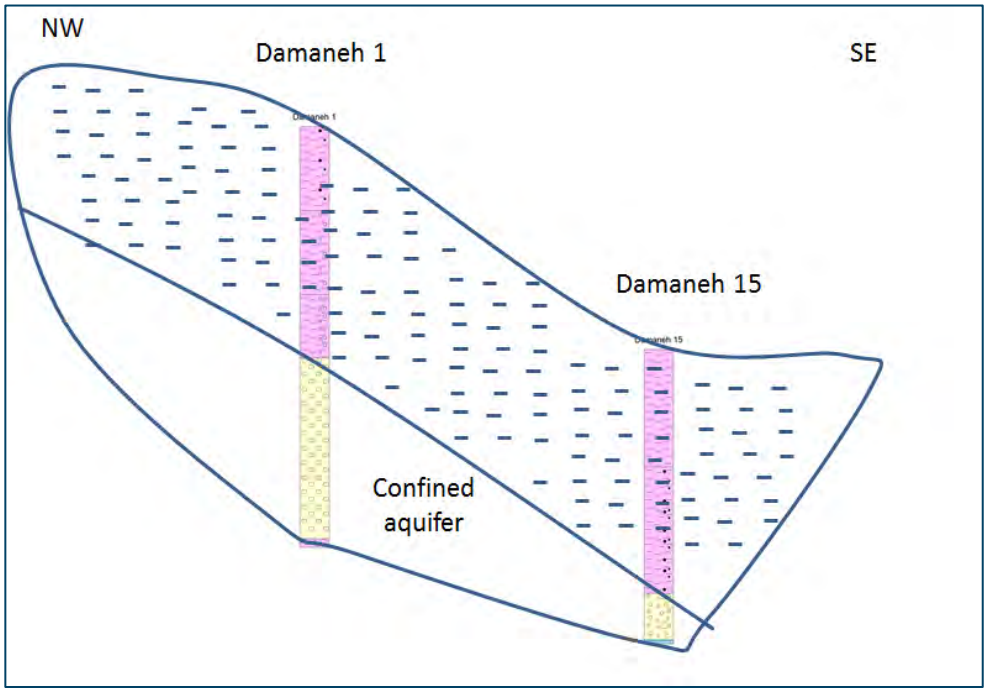


Figure 18: Schematic cross section though sub catchment 4214, Damaneh 1-15

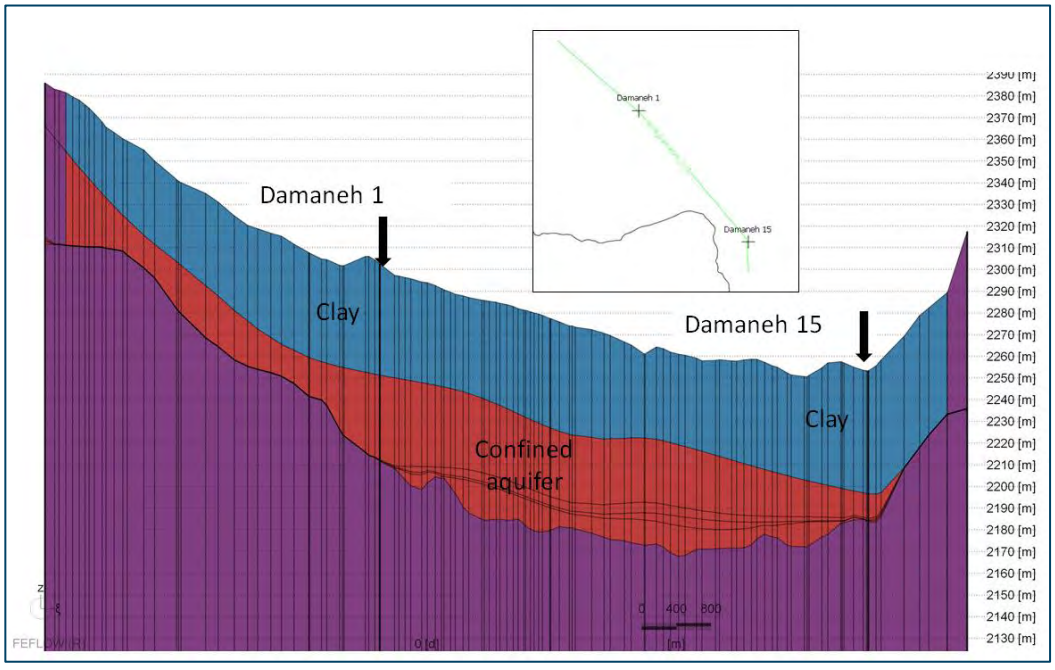


Figure 19: Cross section 4214_Damaneh_1-15, vertical exaggeration of 25

In the center part the first sand layer is not covered by a clay layer. This is documented in Figure 20. The correspondent cross section of the HSM is shown in Figure 21.

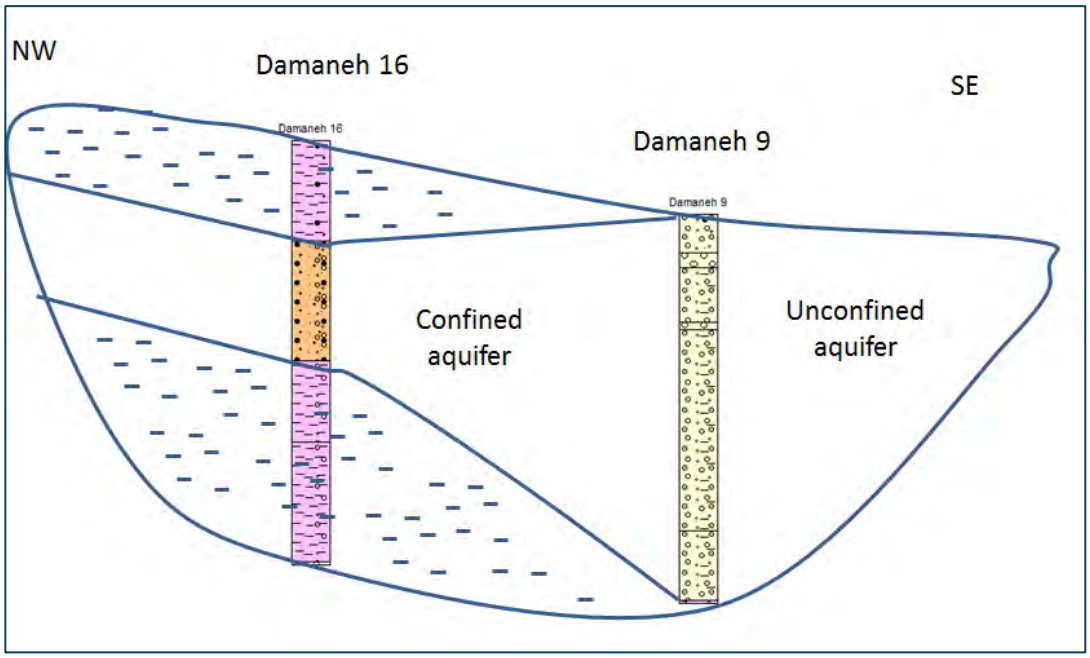


Figure 20: Schematic cross sections though sub catchment 4214, Damaneh 16-9

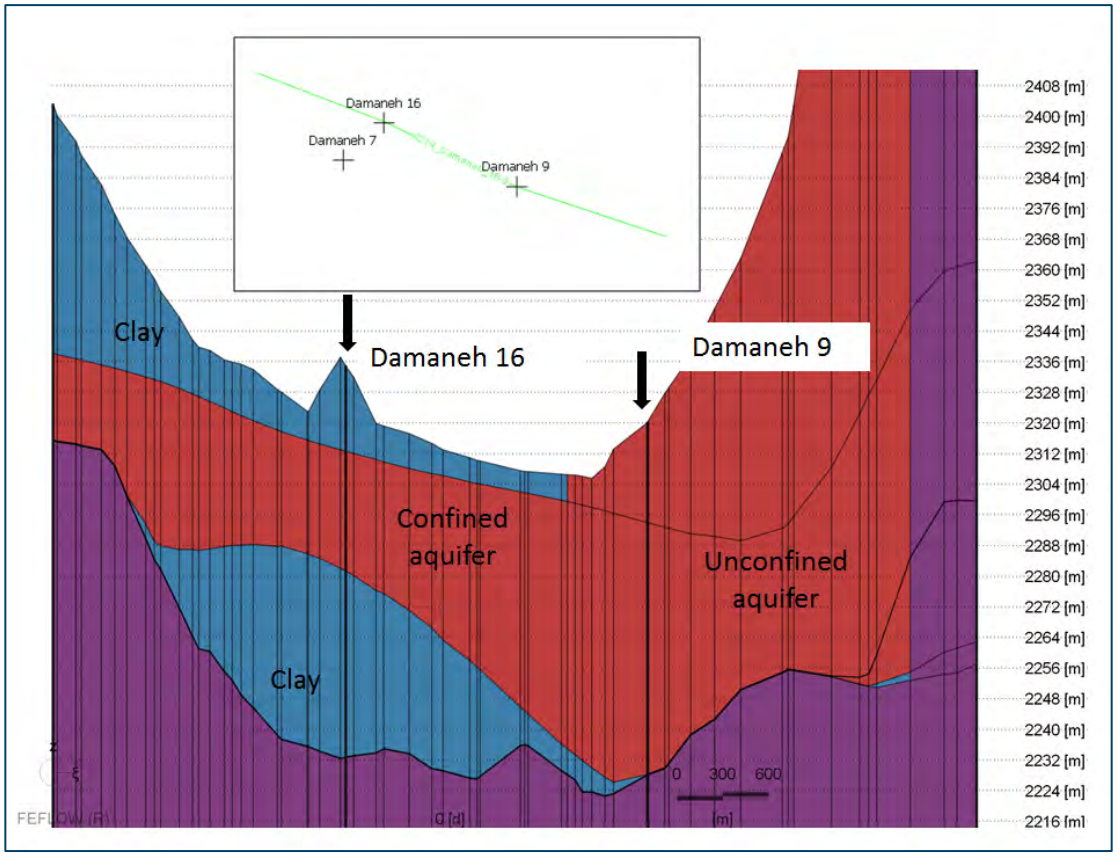


Figure 21: Cross section 4214_Damaneh_16-9, vertical exaggeration of 25

In the northern part of sub catchment Damaneh the presence of the first clay layer is variable. The neighbored well logs Damaneh 18-20 show on their entire length no clay layer. An explanation for this could be a single coarse grained alluvial fan, which was sampled by these three well logs. Especially at the flanks of the quaternary alluvial system coarse grained alluvial facies and the related sedimentary deposits are also plausible. The schematic cross section through the northern part of the sub catchment Damaneh is shown in Figure 22, while the correspondent cross section of the HSM is shown in Figure 23.

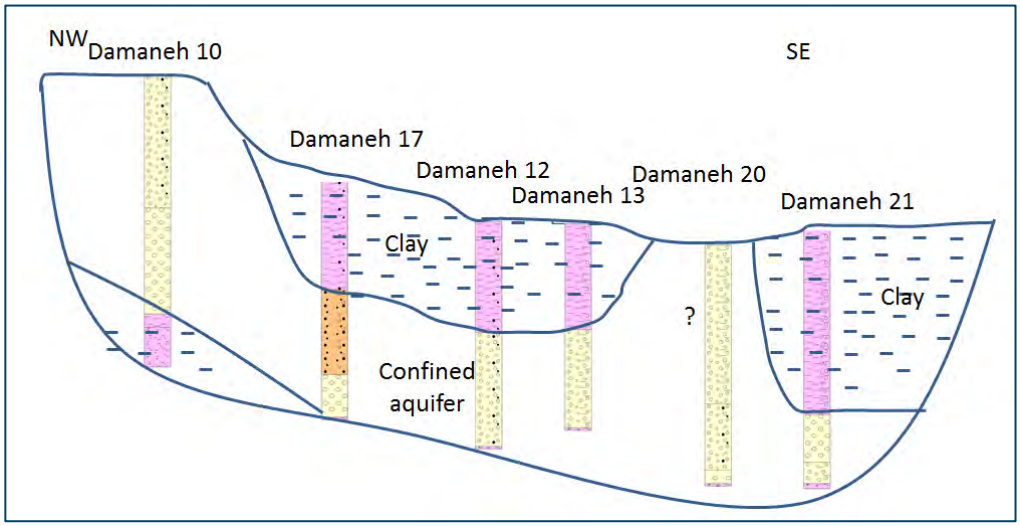


Figure 22: Schematic cross sections though sub catchment 4214, well logs 24-12-13-20-21

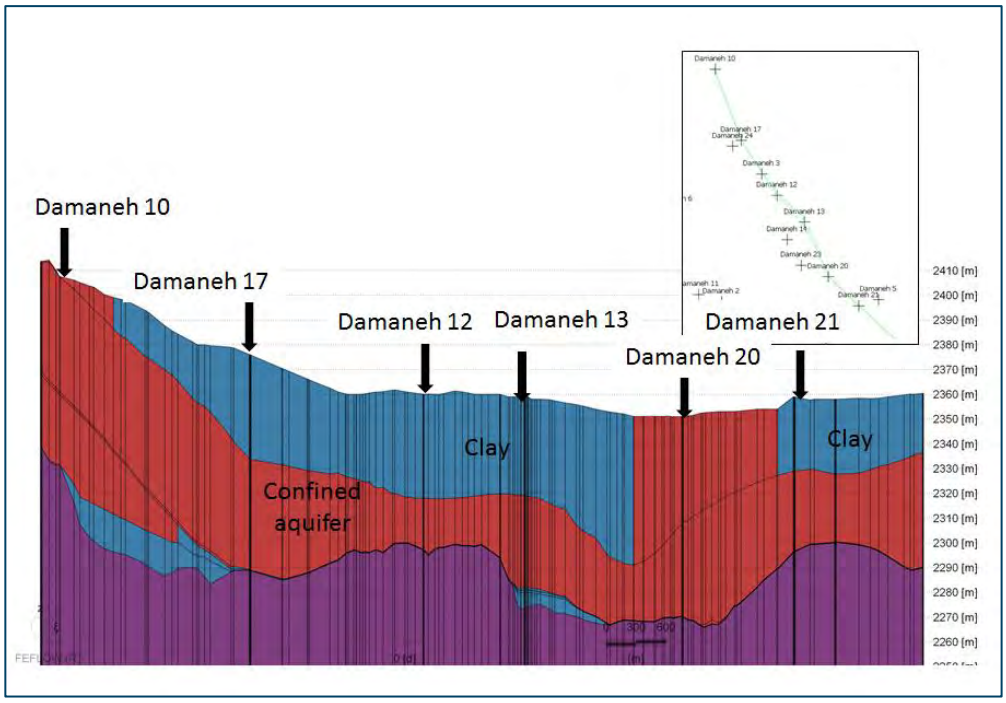


Figure 23: Cross section 4214_Damaneh_10-17-12-13-20-21, vertical exaggeration of 25

3.6 Hydraulic Layering

The HSM consists of 6 layers. The upper five are defined by well log data. Each layer consists of different hydraulic conductivities. The horizontal spreading of the three major materials was defined by the available geological map (Figure 6) which separates the hard rock mountains from the unconsolidated quaternary alluvium sediments (Q). The differentiation of the quaternary alluvium sediments was manually done by the interpretation of the geological well logs. The quaternary alluvium sediments were differentiated between clay dominated sediments and sand dominated sediments. s

The horizontal spreading of the three major material types is shown in Figure 24 - Figure 28.

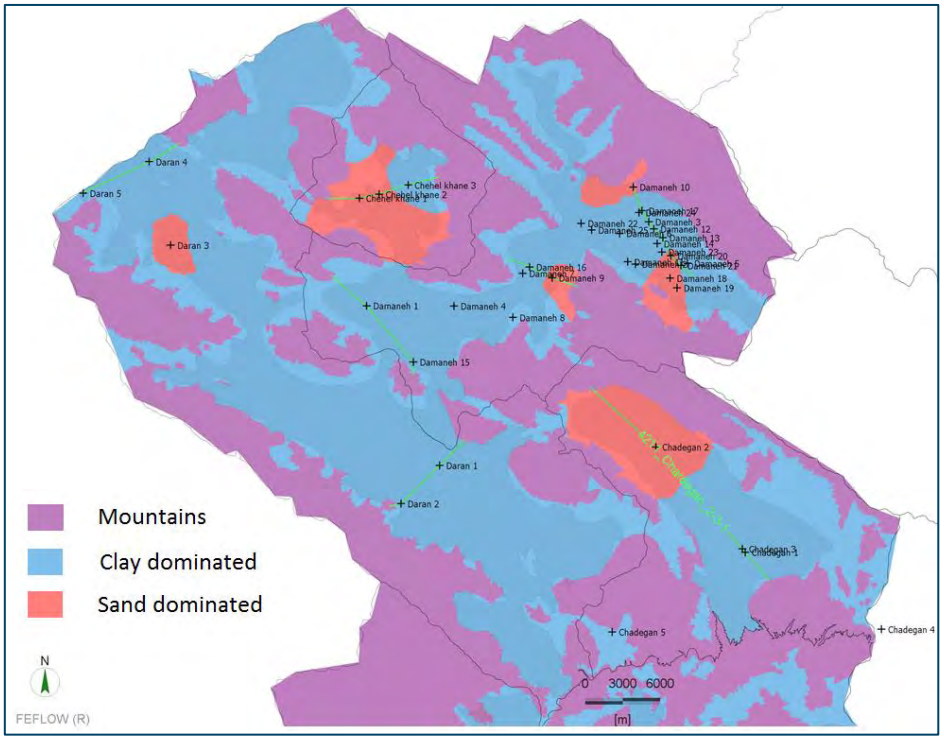


Figure 24: Horizontal spreading of the three major material types of layer 1

In layer 2 and layer 4 mainly sand dominated sediments are present. Both layers represent mainly aquifers. To avoid aquifer connections between non connected areas several parts of laser 2 and layer 4 were defined as clay. This causes a physical barrier between inside the HSM.

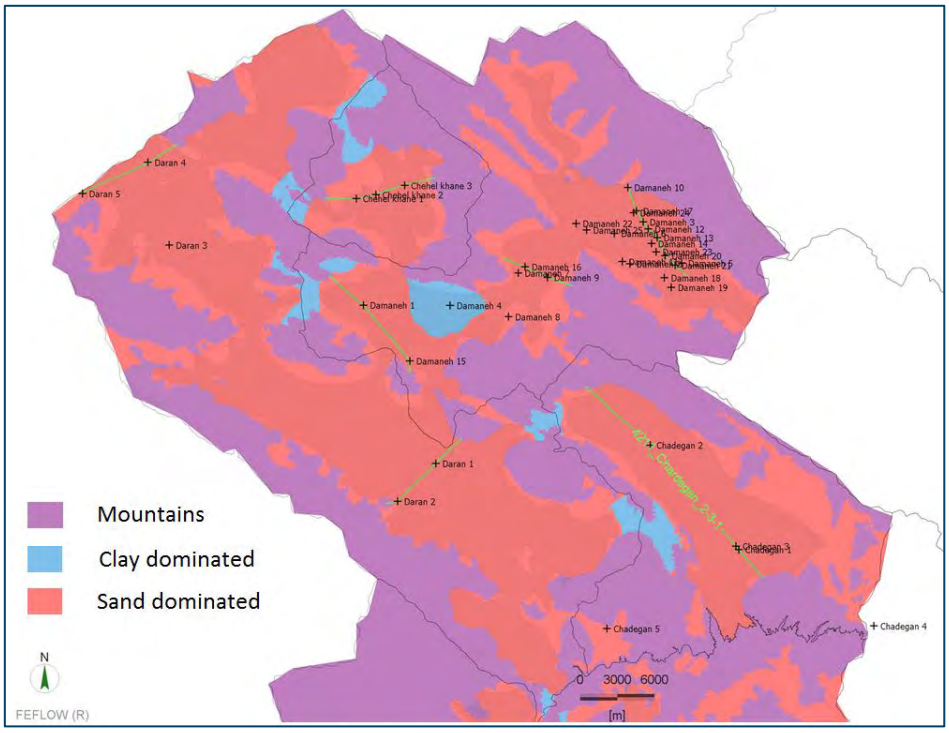


Figure 25: Horizontal spreading of the three major material types of layer 2

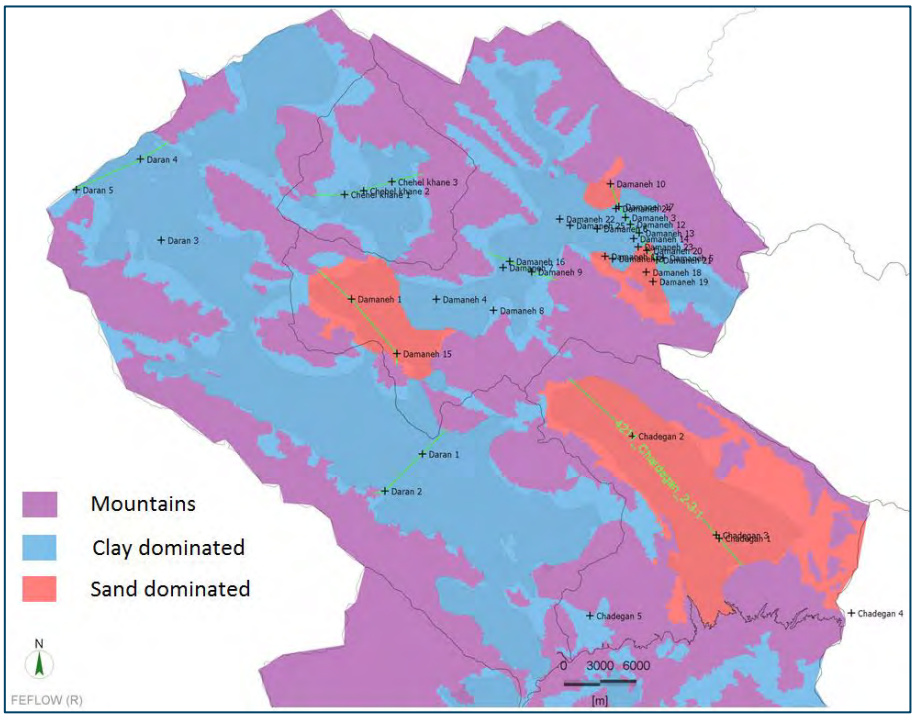


Figure 26: Horizontal spreading of the three major material types of layer 3

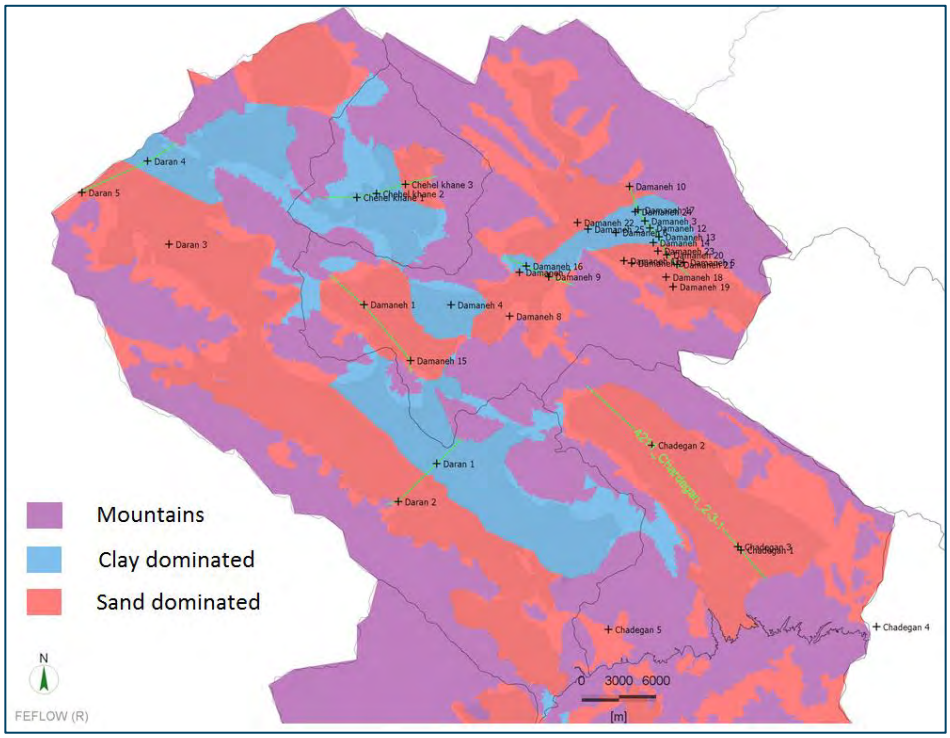


Figure 27: Horizontal spreading of the three major material types of layer 4

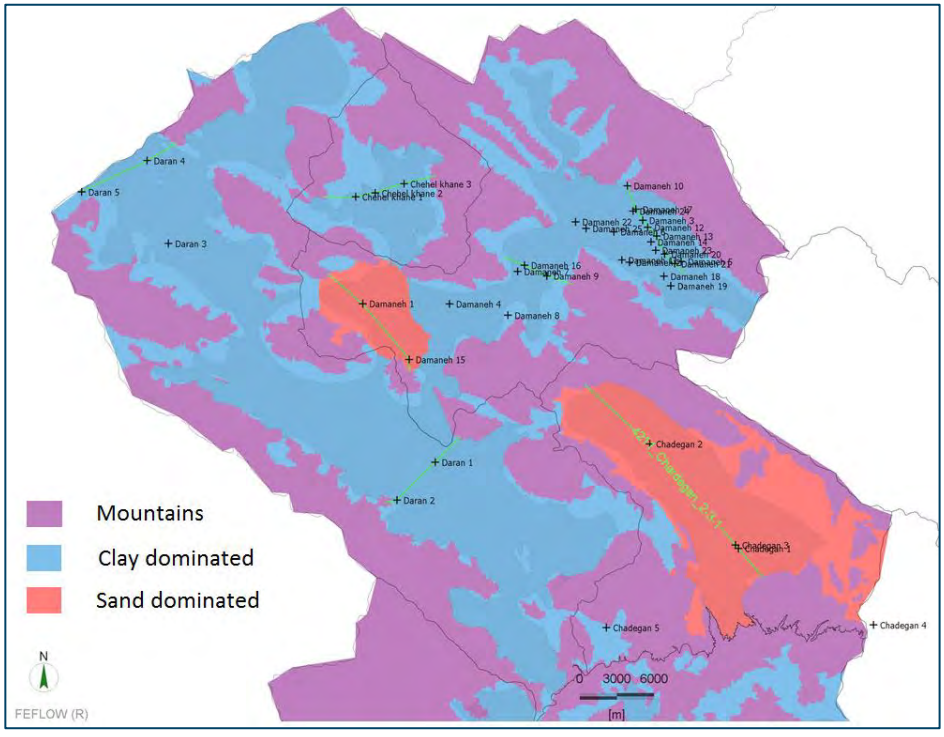


Figure 28: Horizontal spreading of the three major material types of layer 5

In Figure 29 a 3D-view the HSM is shown. In the sub catchment Daran, on the left side, two sand dominated alluvial aquifers are present. On the center left side, only one sand dominated aquifer is present, which was defined by the well log of Daran 1. In the right side a cross section of the sub catchment Damaneh is shown. Adjacent to the well Damaneh 7 there are sand dominated sediments above clay dominated sediments.

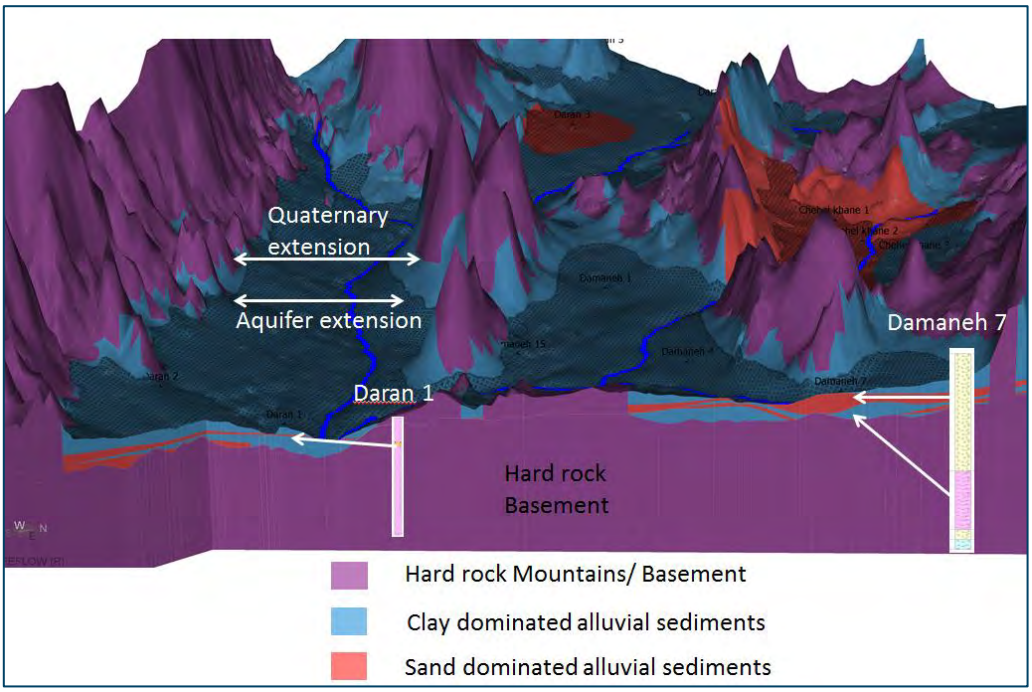
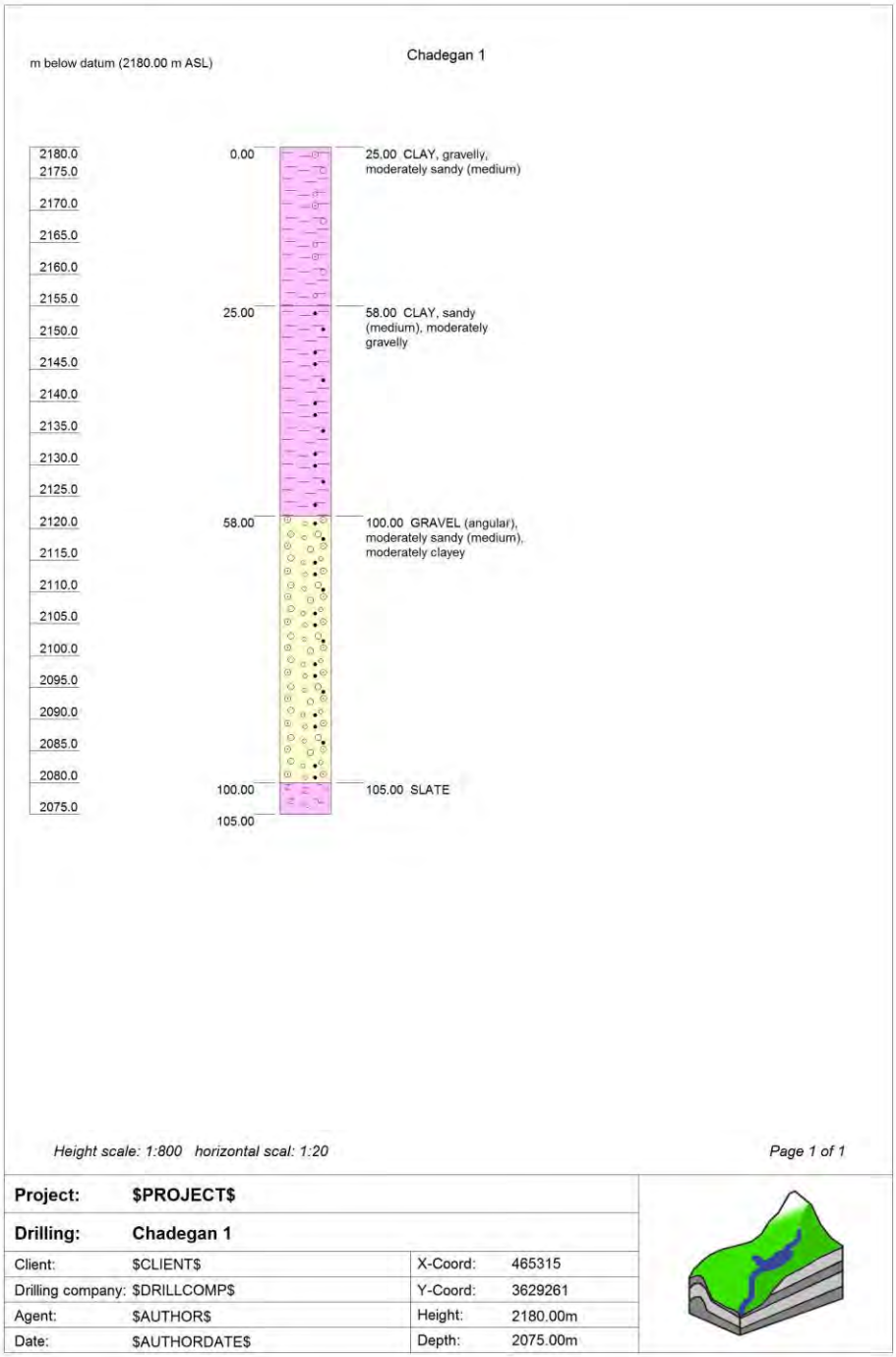
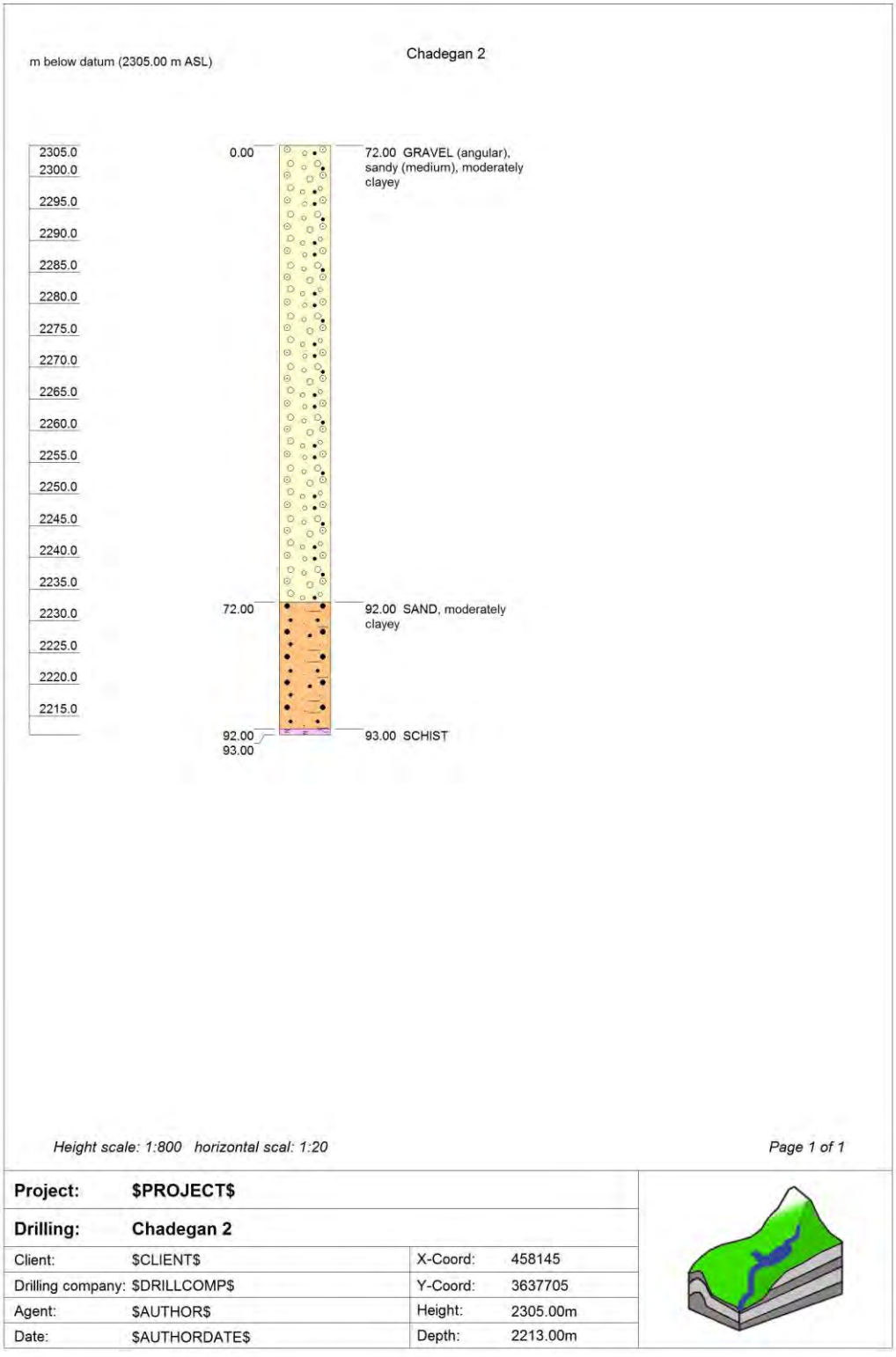


Figure 29: 3D-view of the HSM with cross section through the sub catchments Daran (left) and Damaneh (right); vertical exaggeration of 10

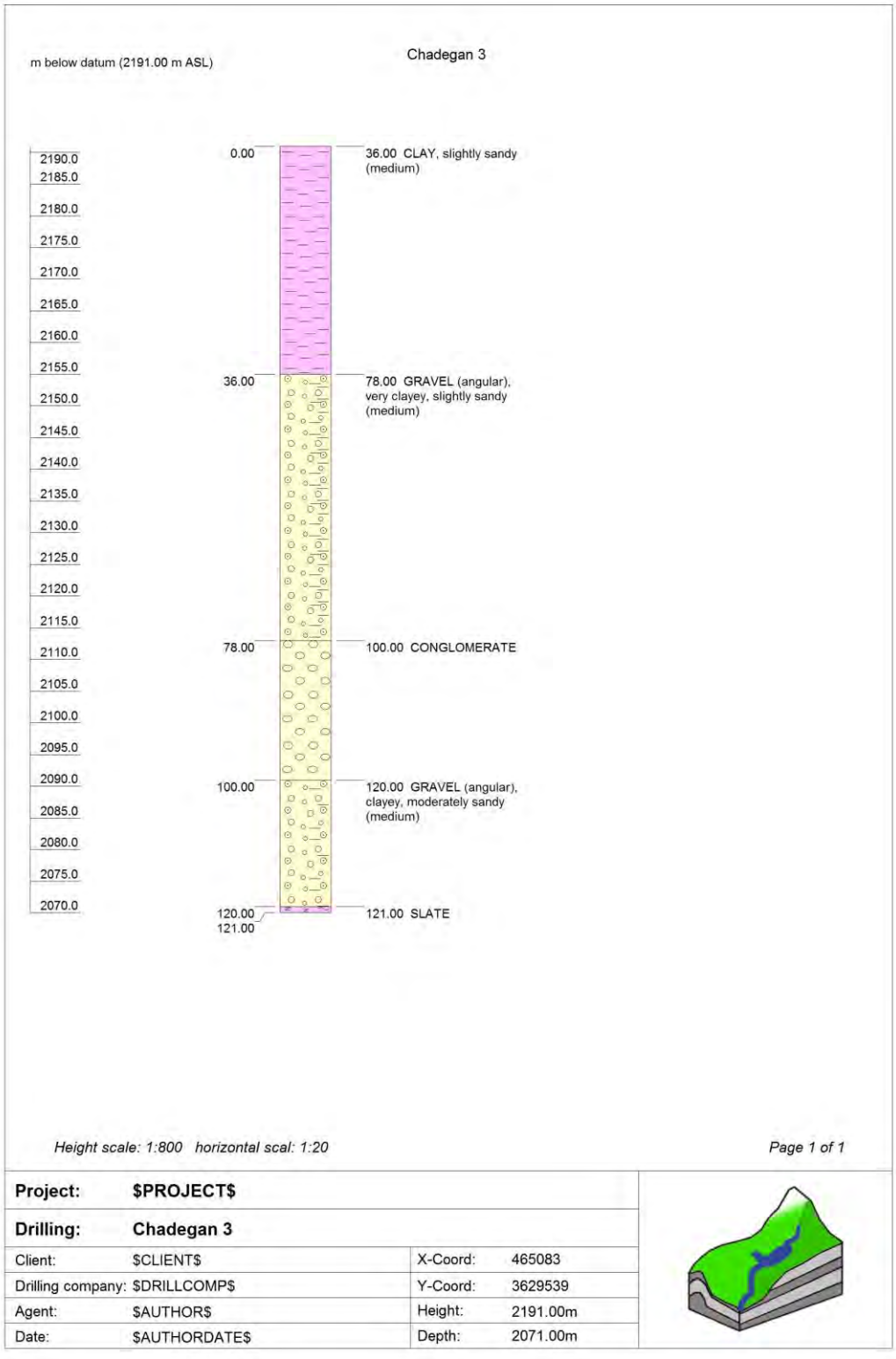
Appendix



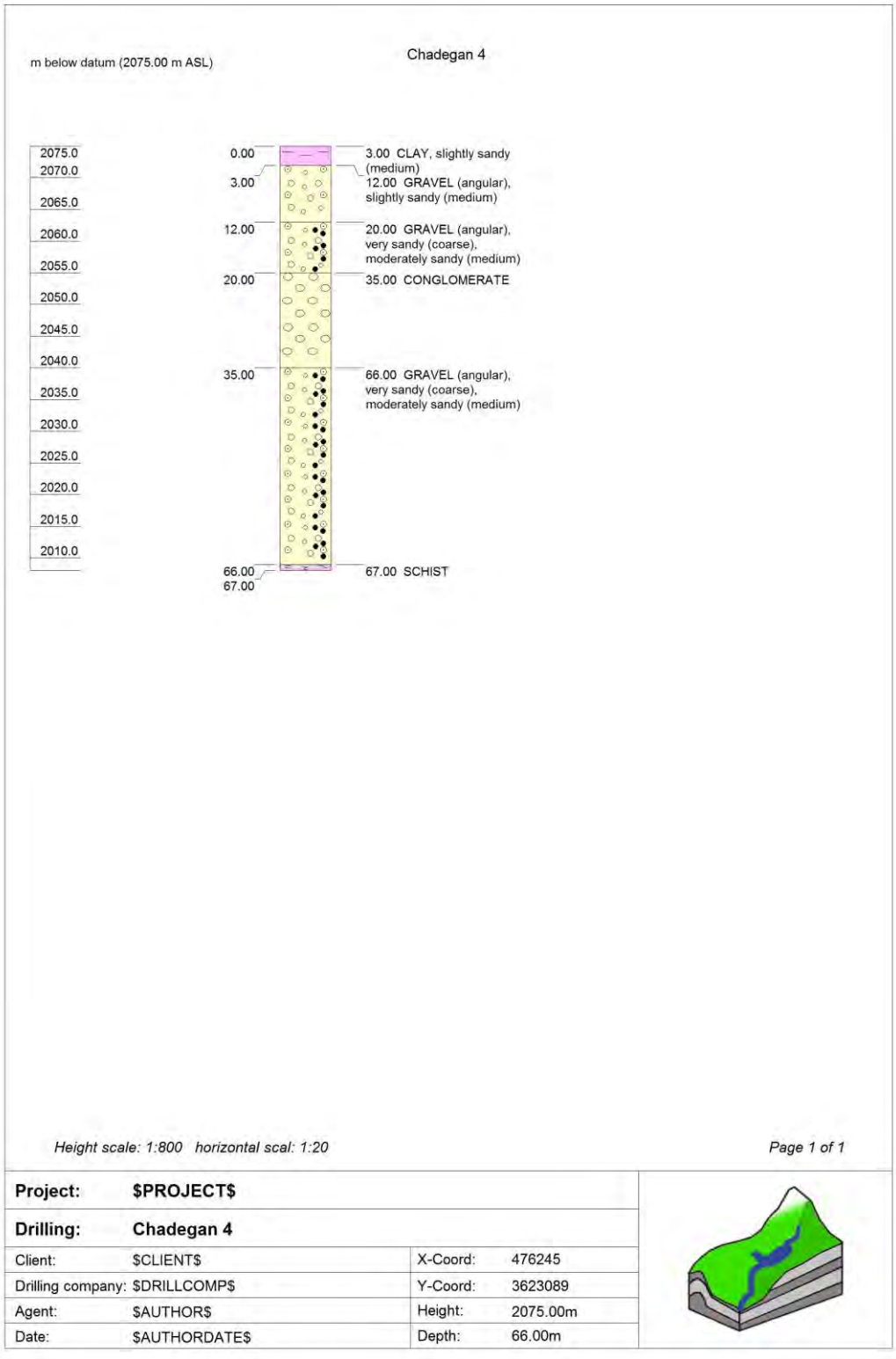
Appendix 1: Well log data of Chadegan 1



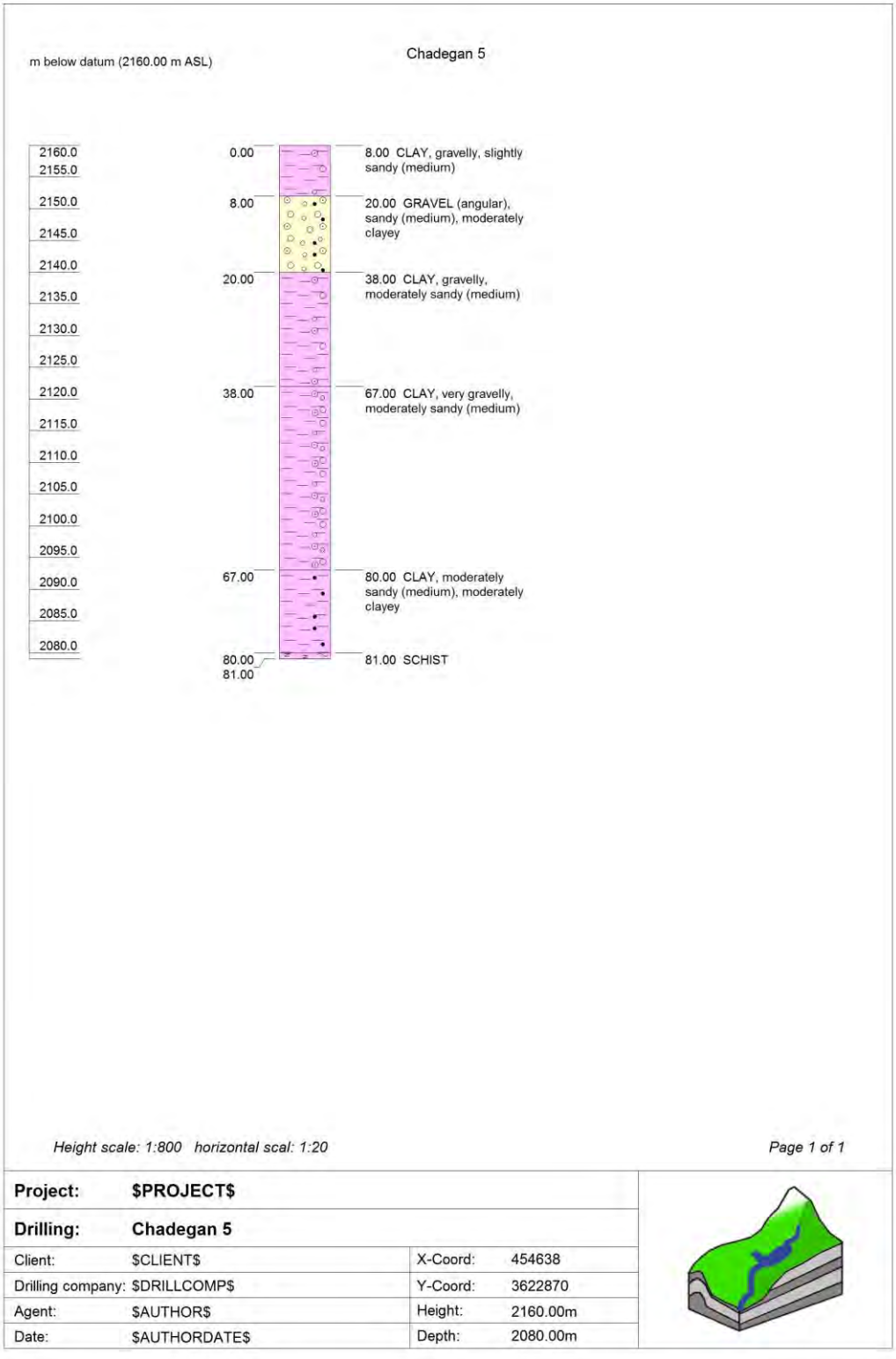
Appendix 2: Well log data of Chadegan 2



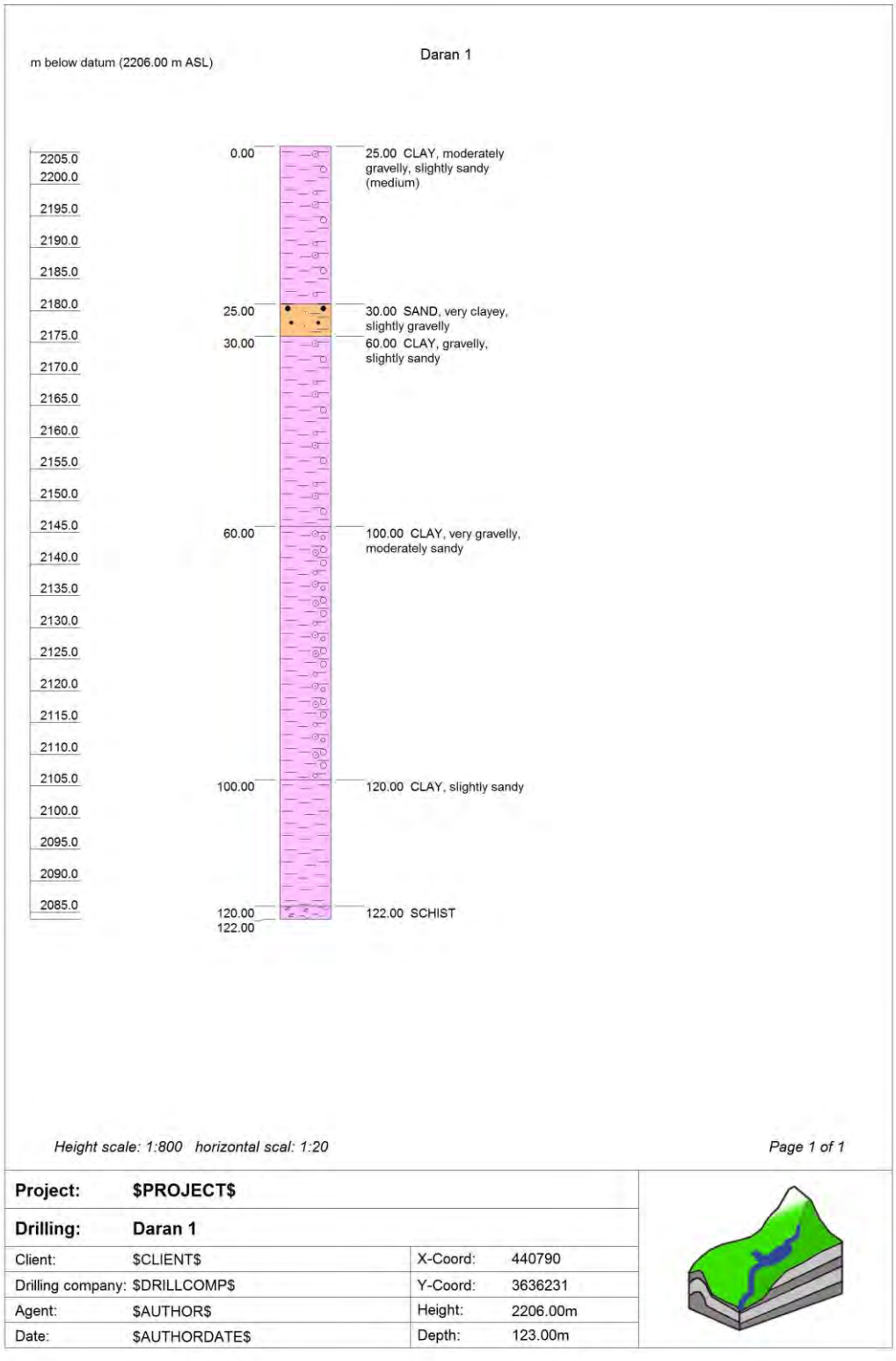
Appendix 3: Well log data of Chadegan 3



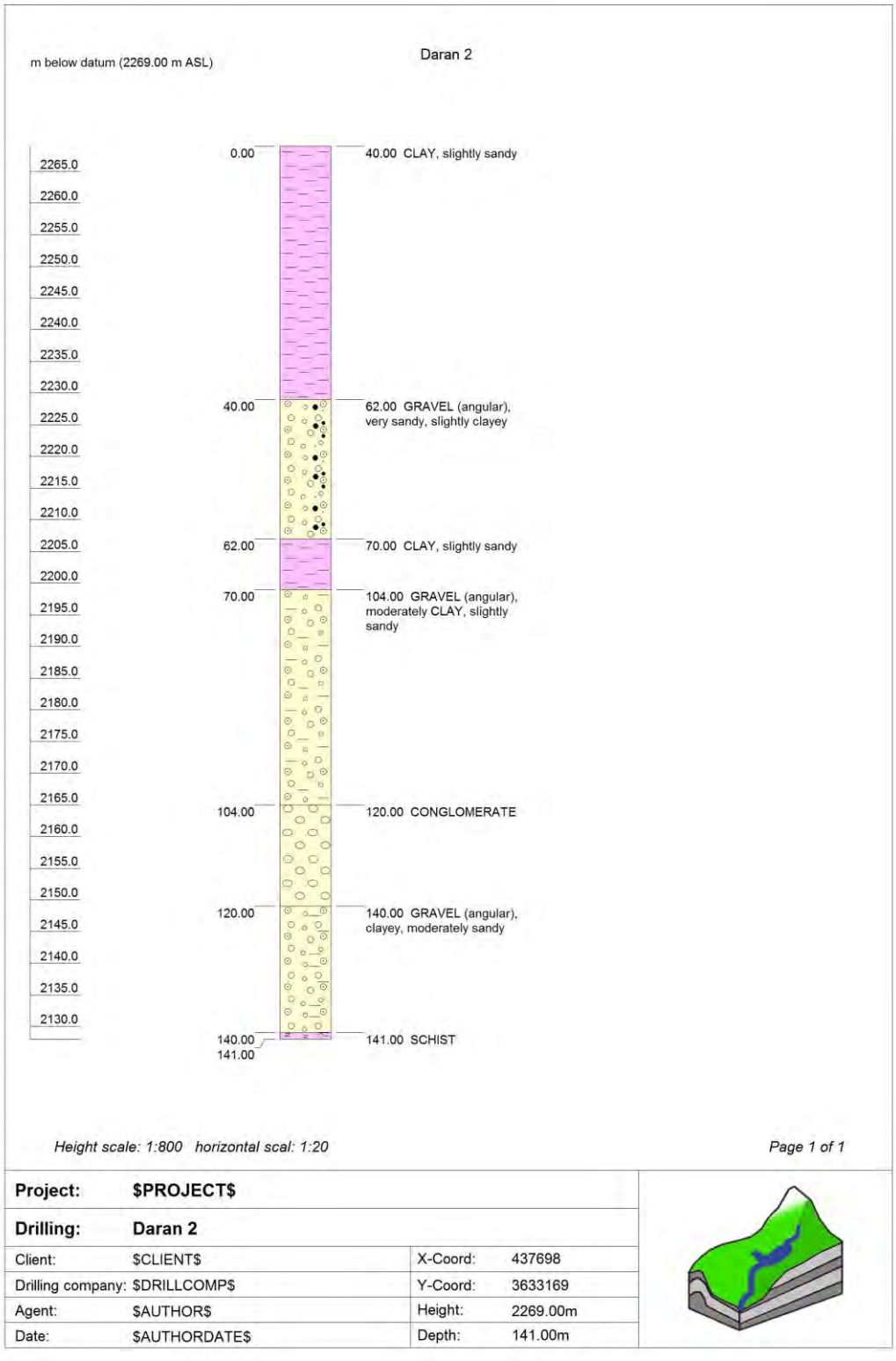
Appendix 4: Well log data of Chadegan 4



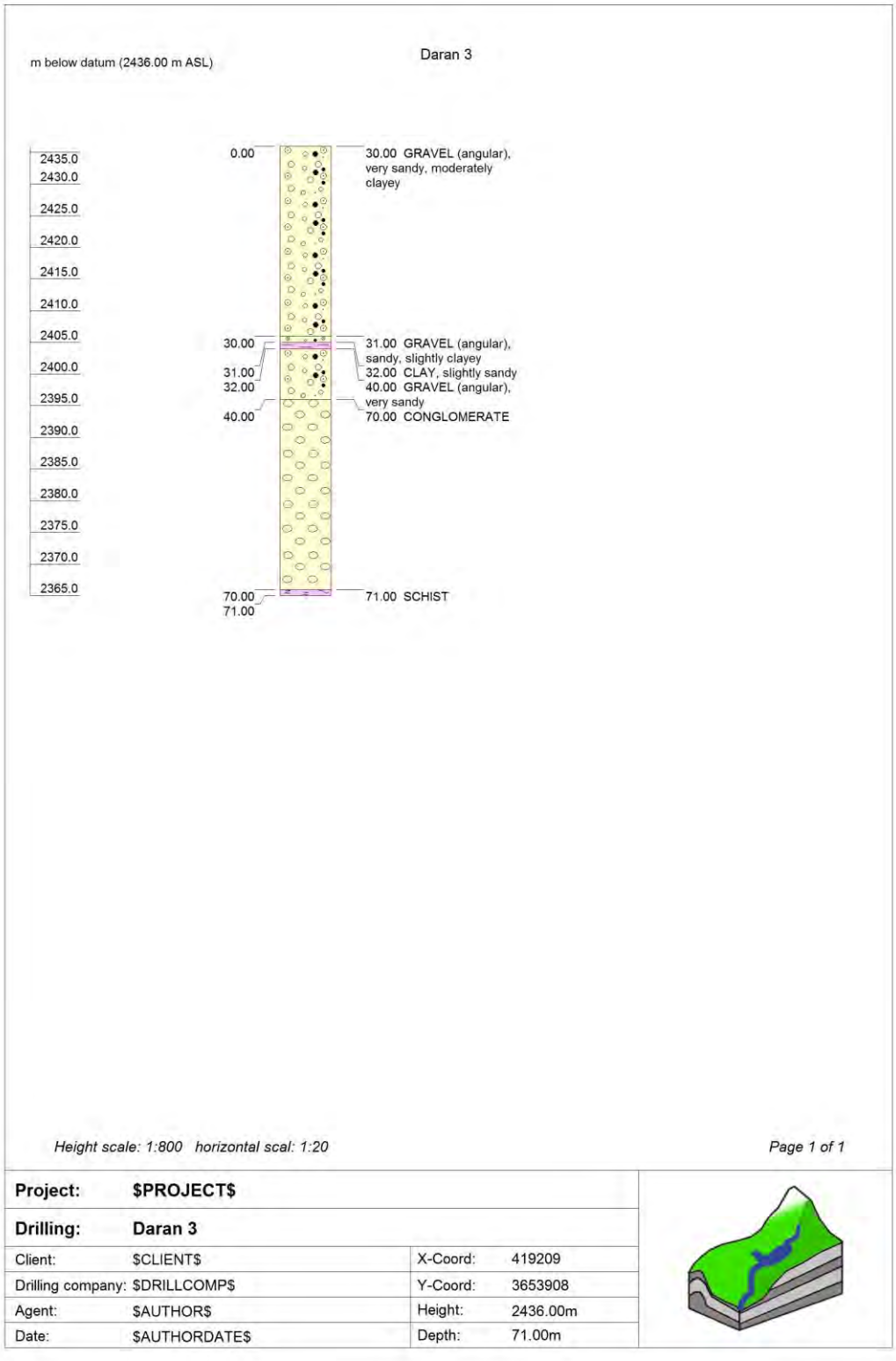
Appendix 5: Well log data of Chadegan 5



Appendix 6: Well log data of Daran 1



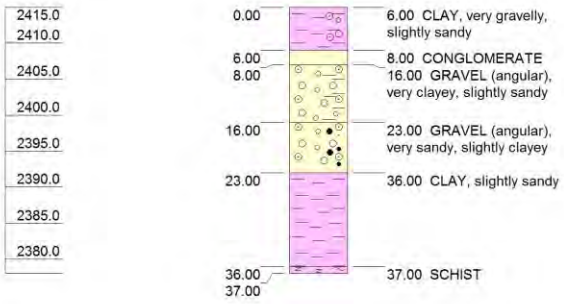
Appendix 7: Well log data of Daran 2



Appendix 8: Well log data of Daran 3

m below datum (2415.00 m ASL)

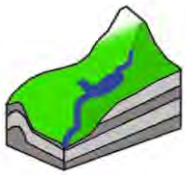
Daran 4



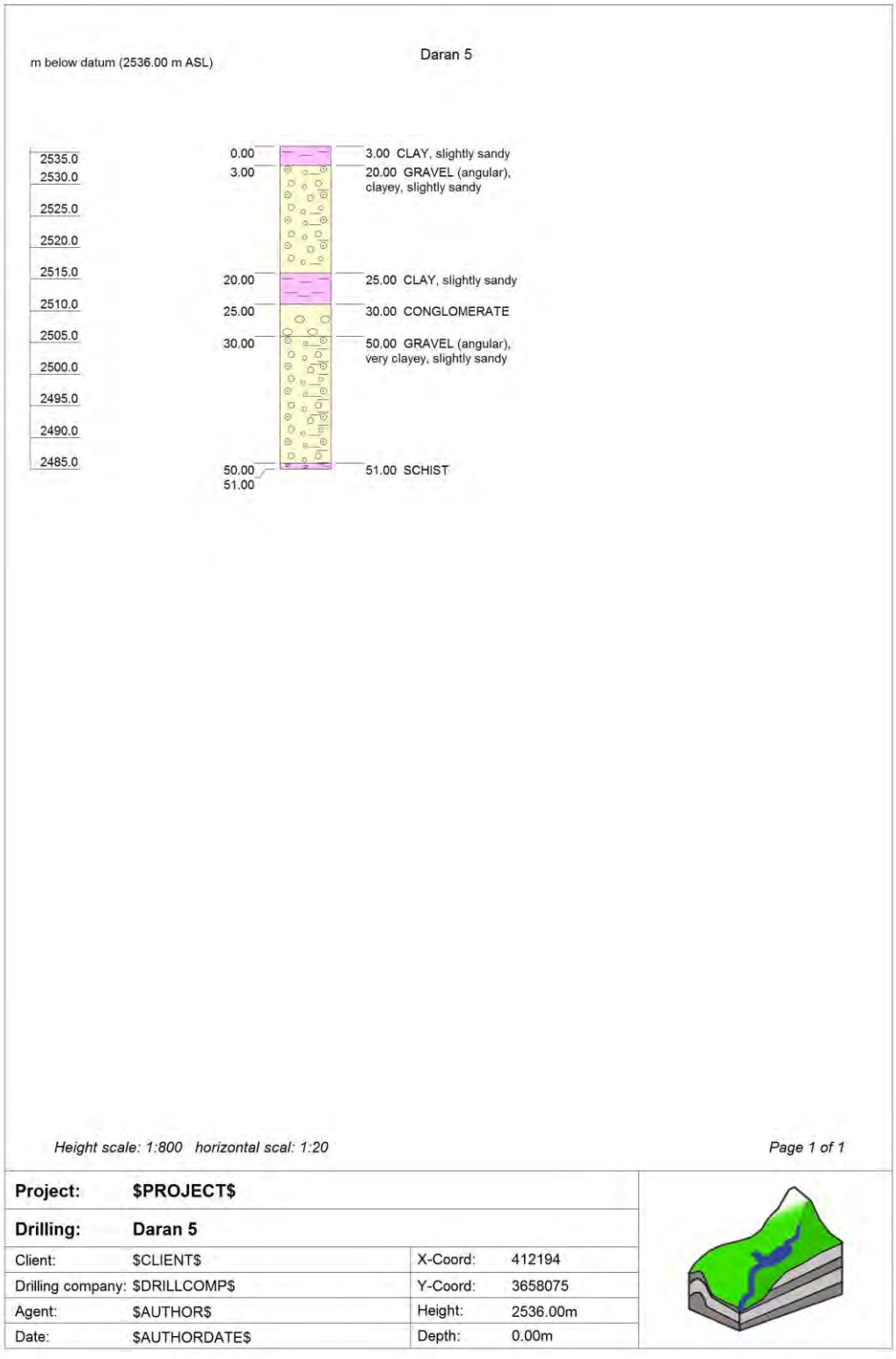
Height scale: 1:800 horizontal scal: 1:20

Page 1 of 1

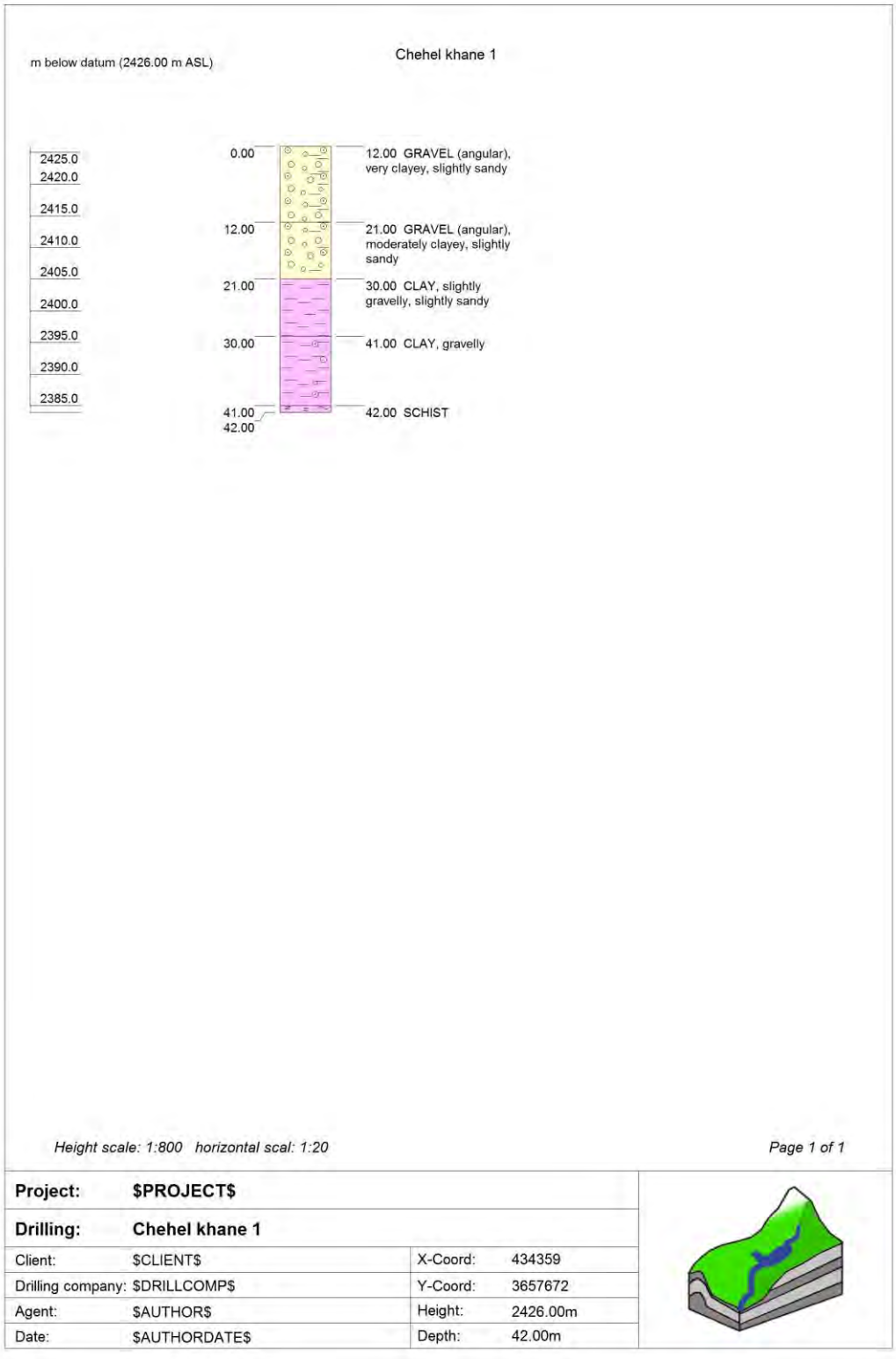
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Drilling:	**Daran 4**		
Client:	$CLIENT$	X-Coord:	417486
Drilling company:	$DRILLCOMP$	Y-Coord:	3660608
Agent:	$AUTHOR$	Height:	2415.00m
Date:	$AUTHORDATE$	Depth:	37.00m



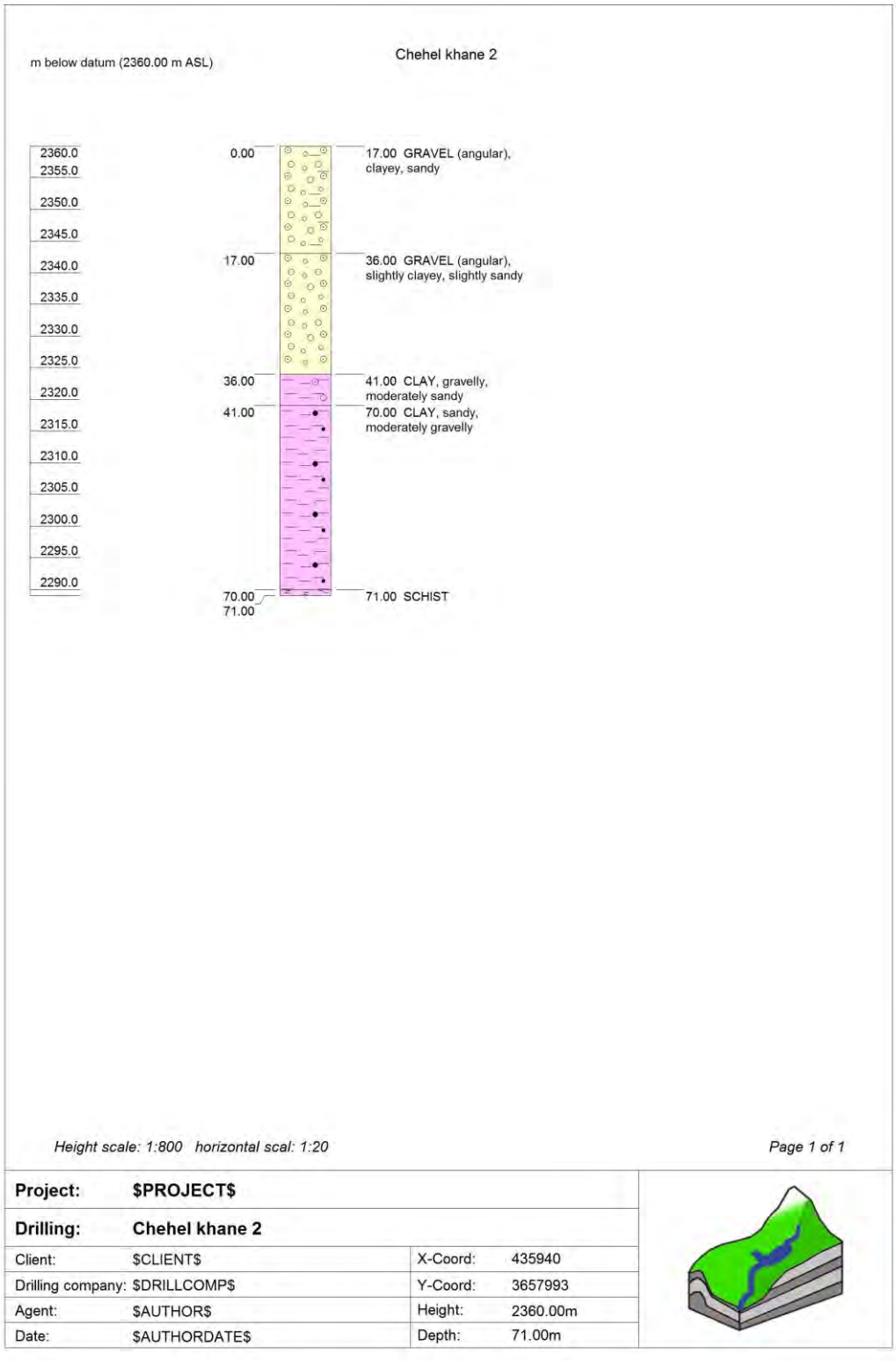
Appendix 9: Well log data of Daran 4



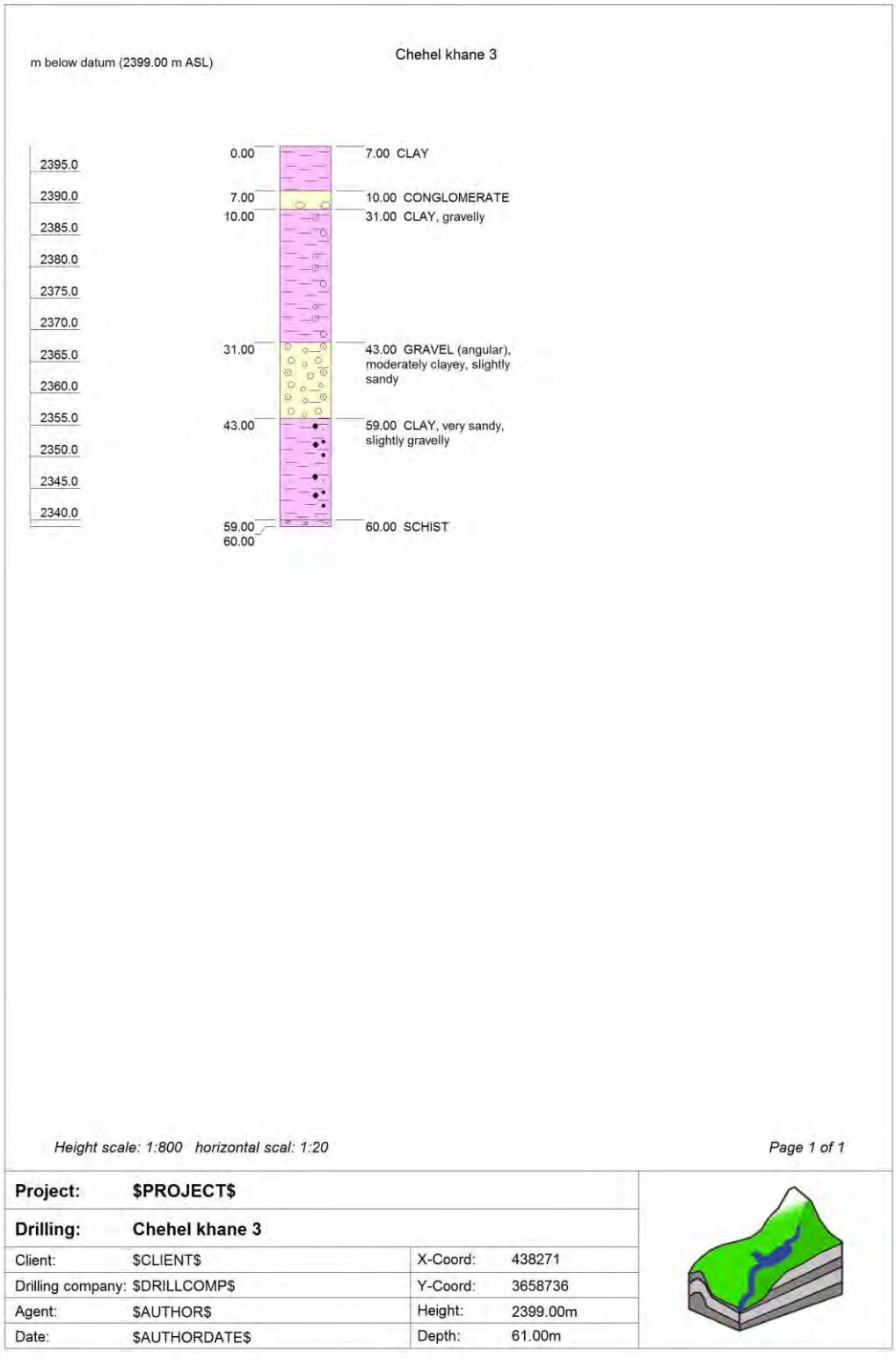
Appendix 10: Well log data of Daran 5



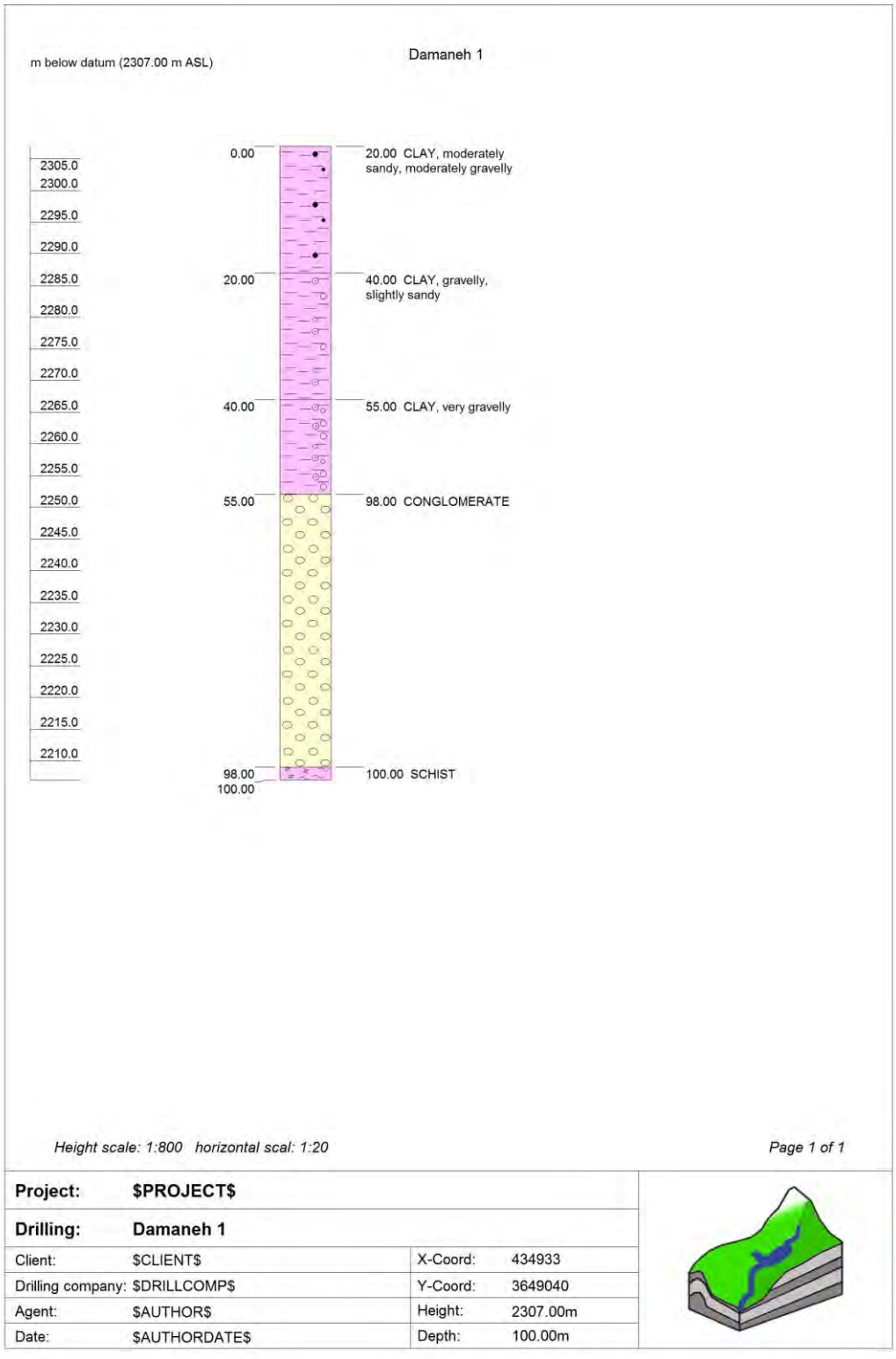
Appendix 11: Well log data of Chahel khane 1



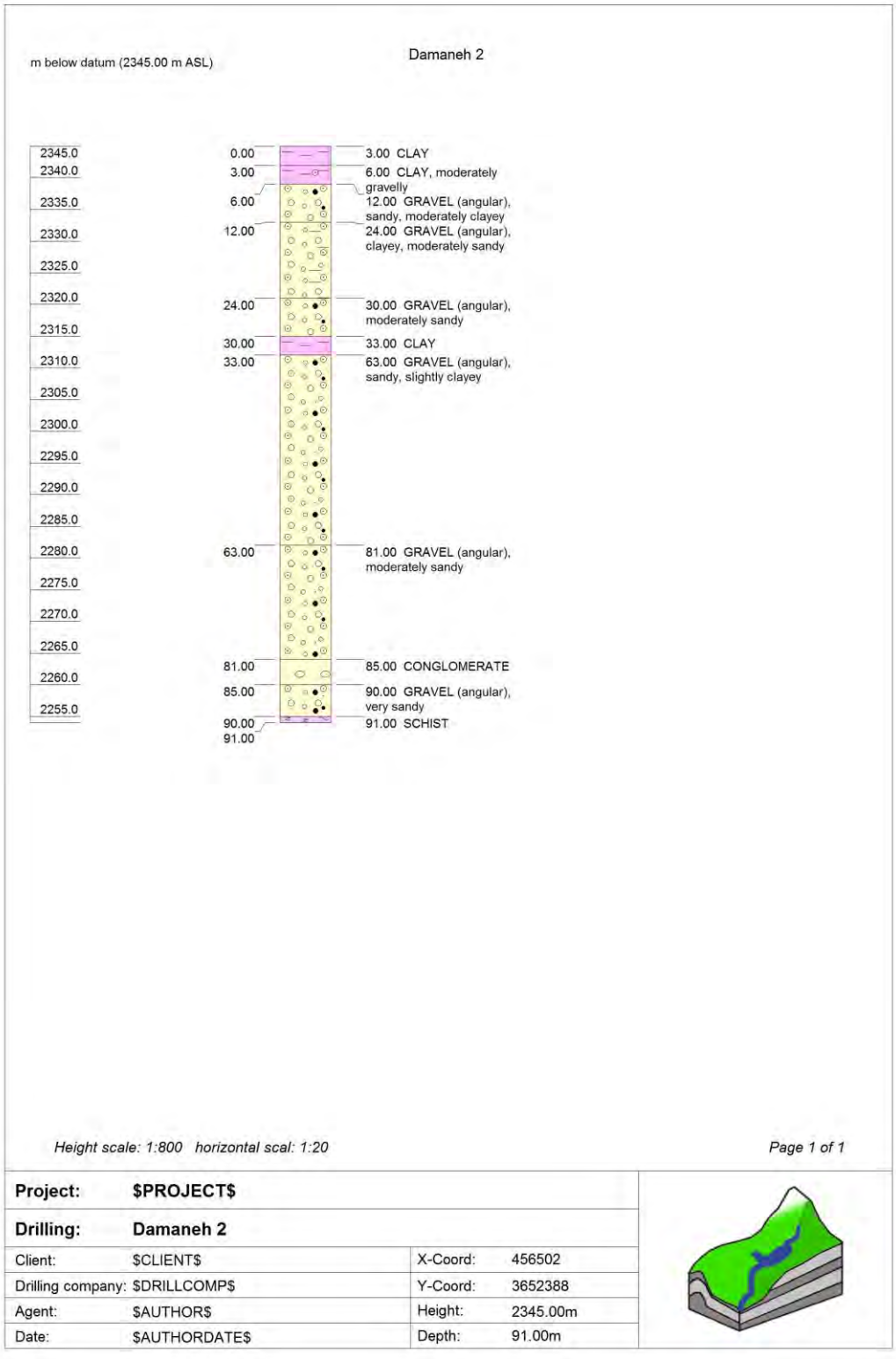
Appendix 12: Well log data of Chehel khane 2



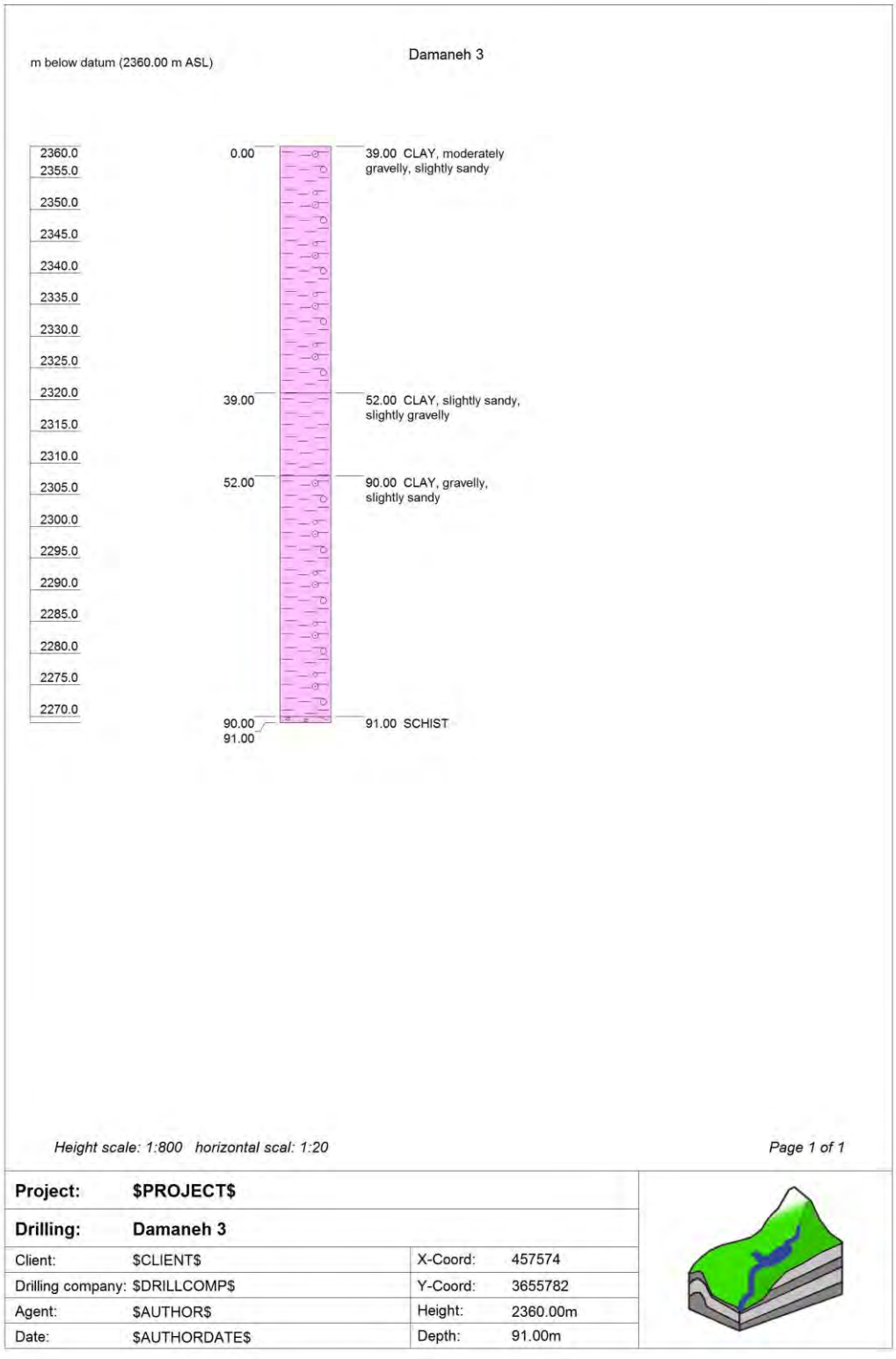
Appendix 13: Well log data of Chehel khane 3



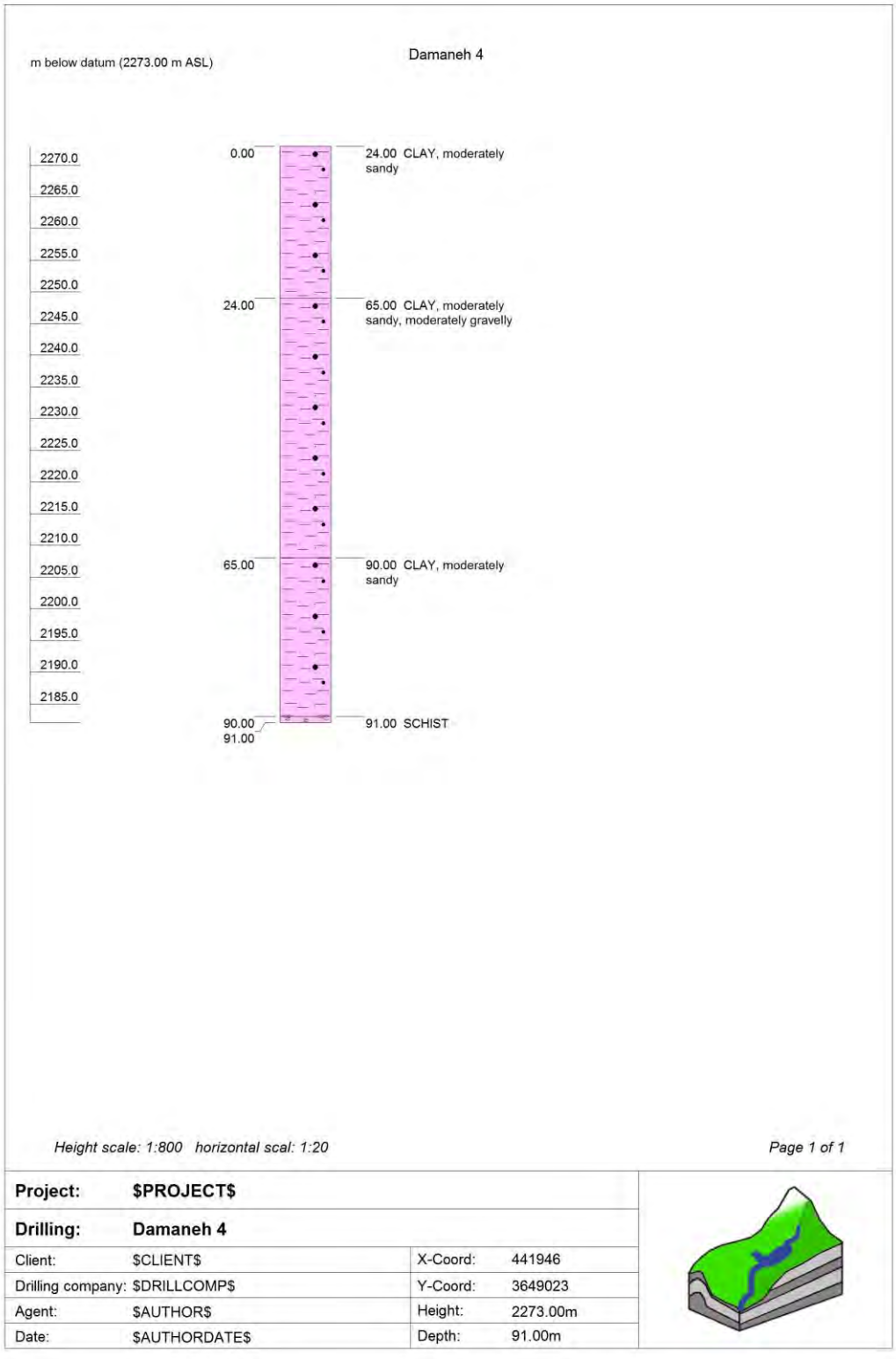
Appendix 14: Well log data of Damaneh 1



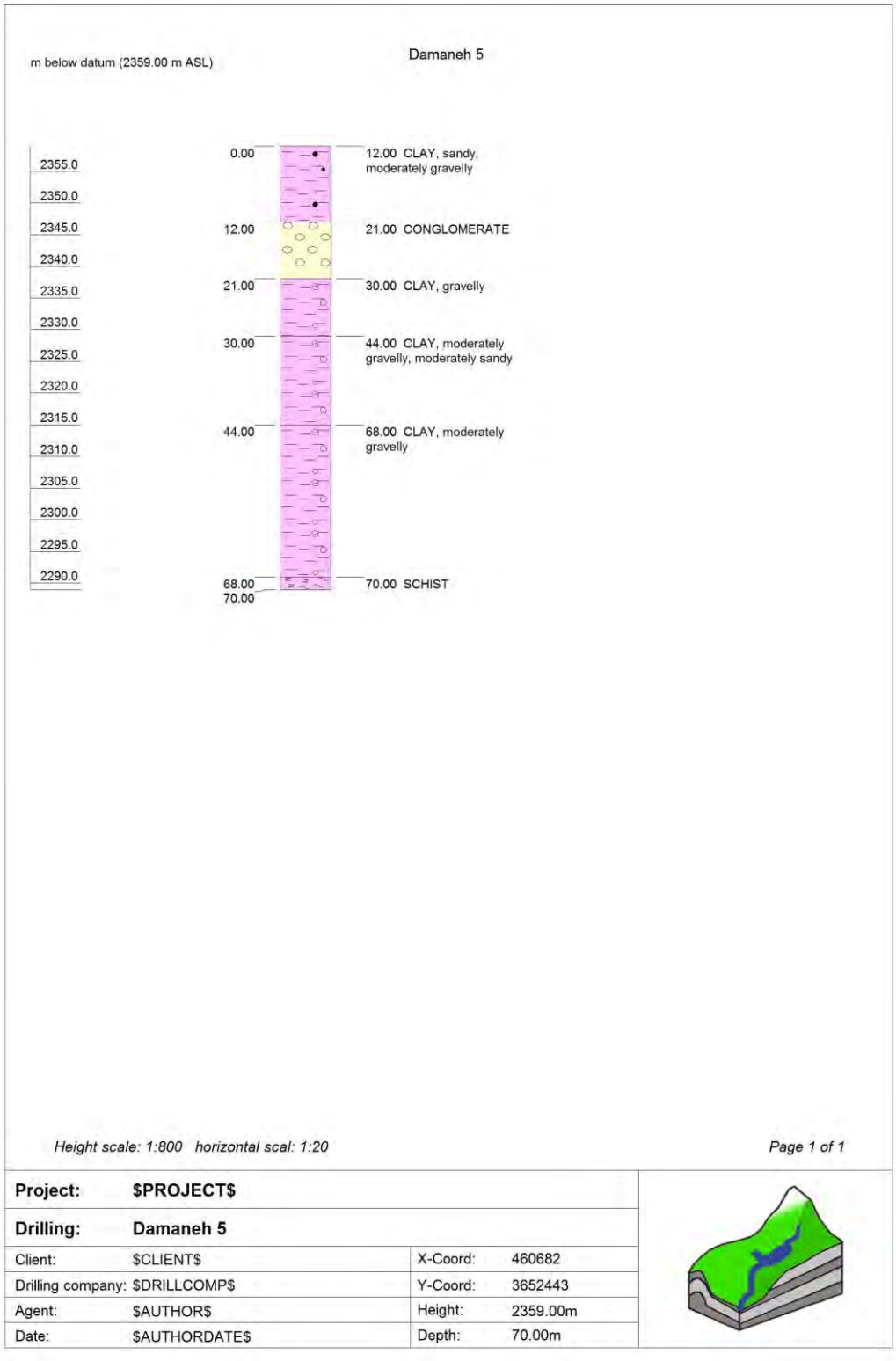
Appendix 15: Well log data of Dameneh 2



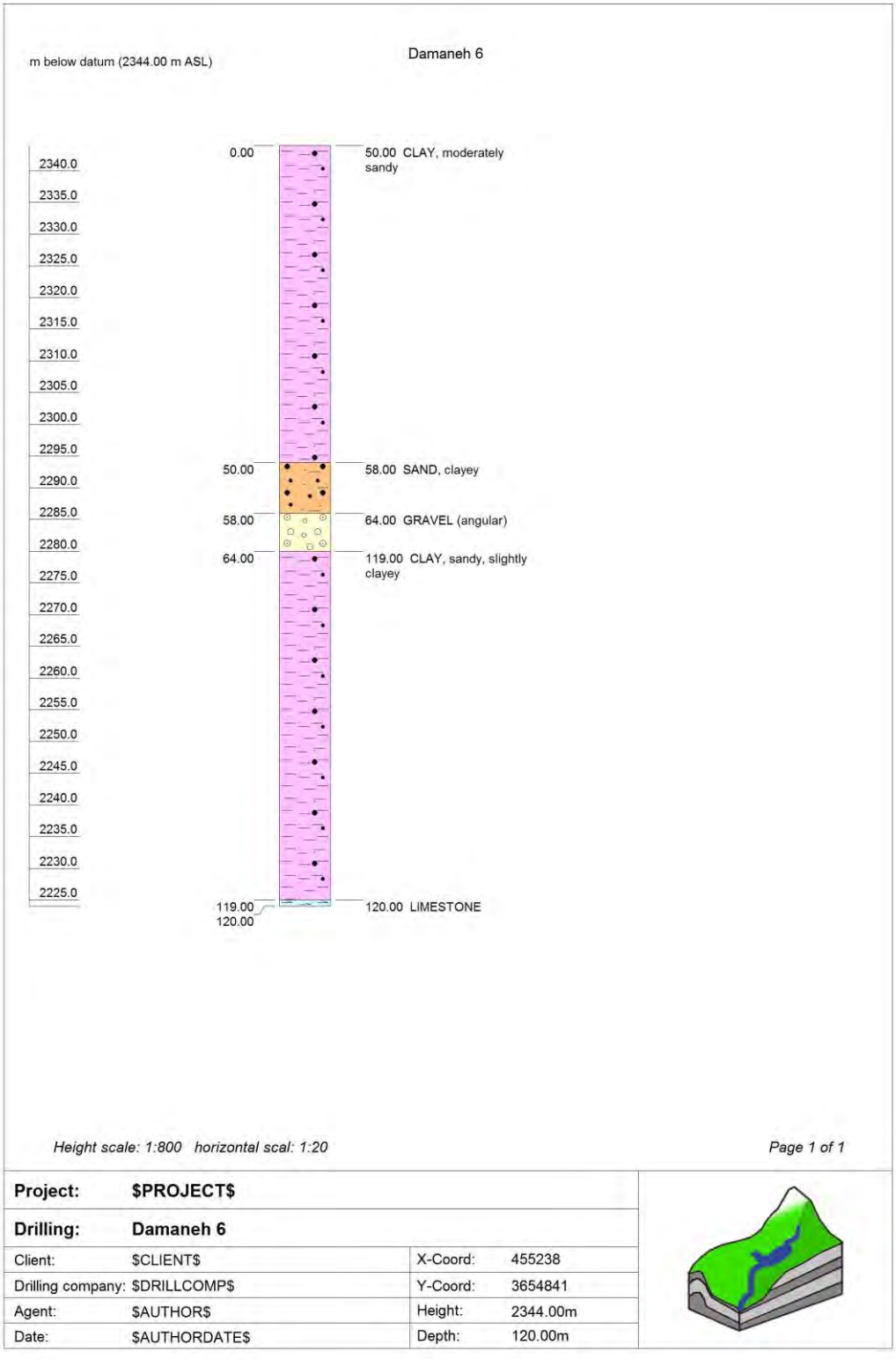
Appendix 16: Well log data of Damaneh 3



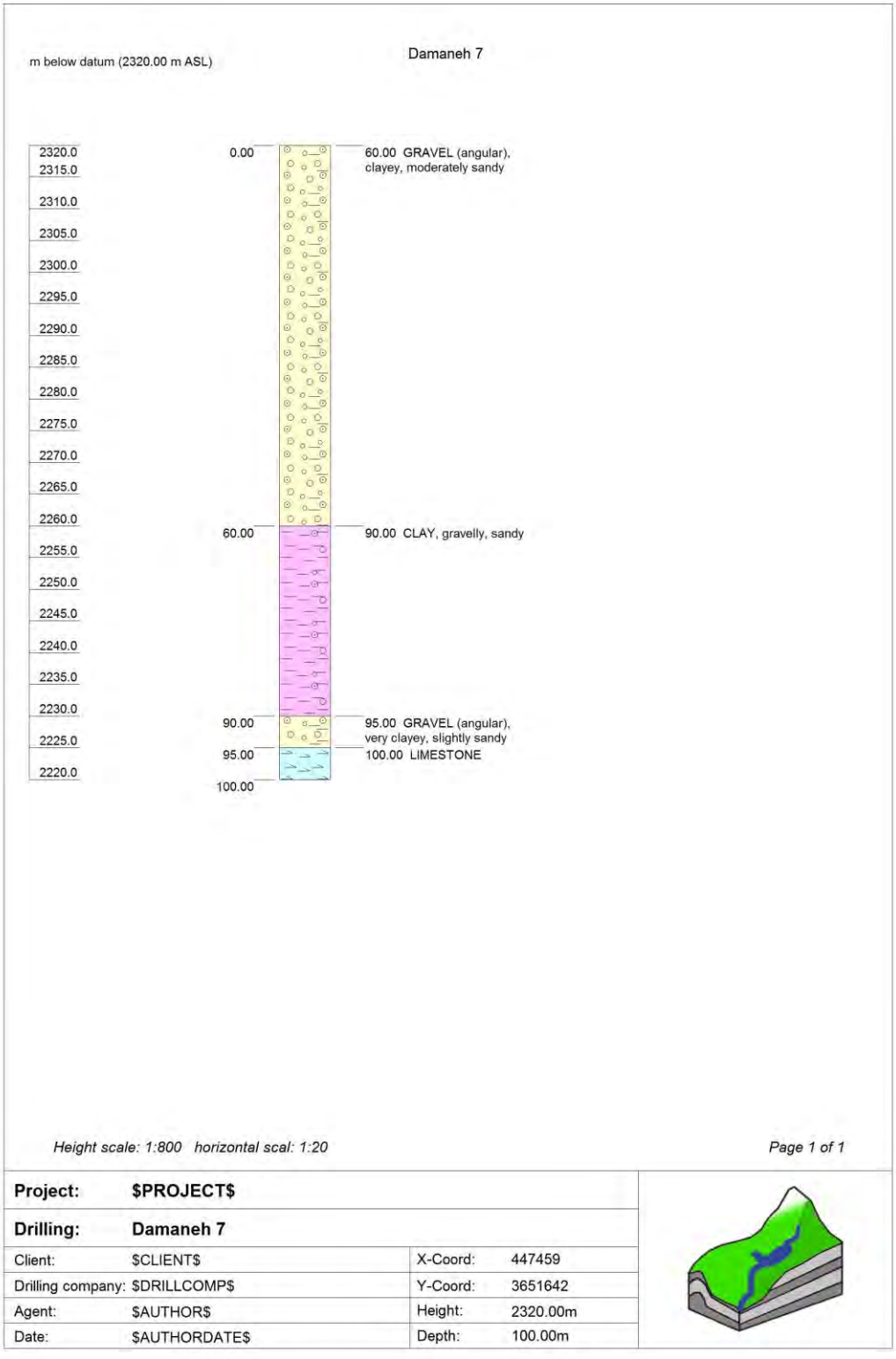
Appendix 17: Well log data of Damaneh 4



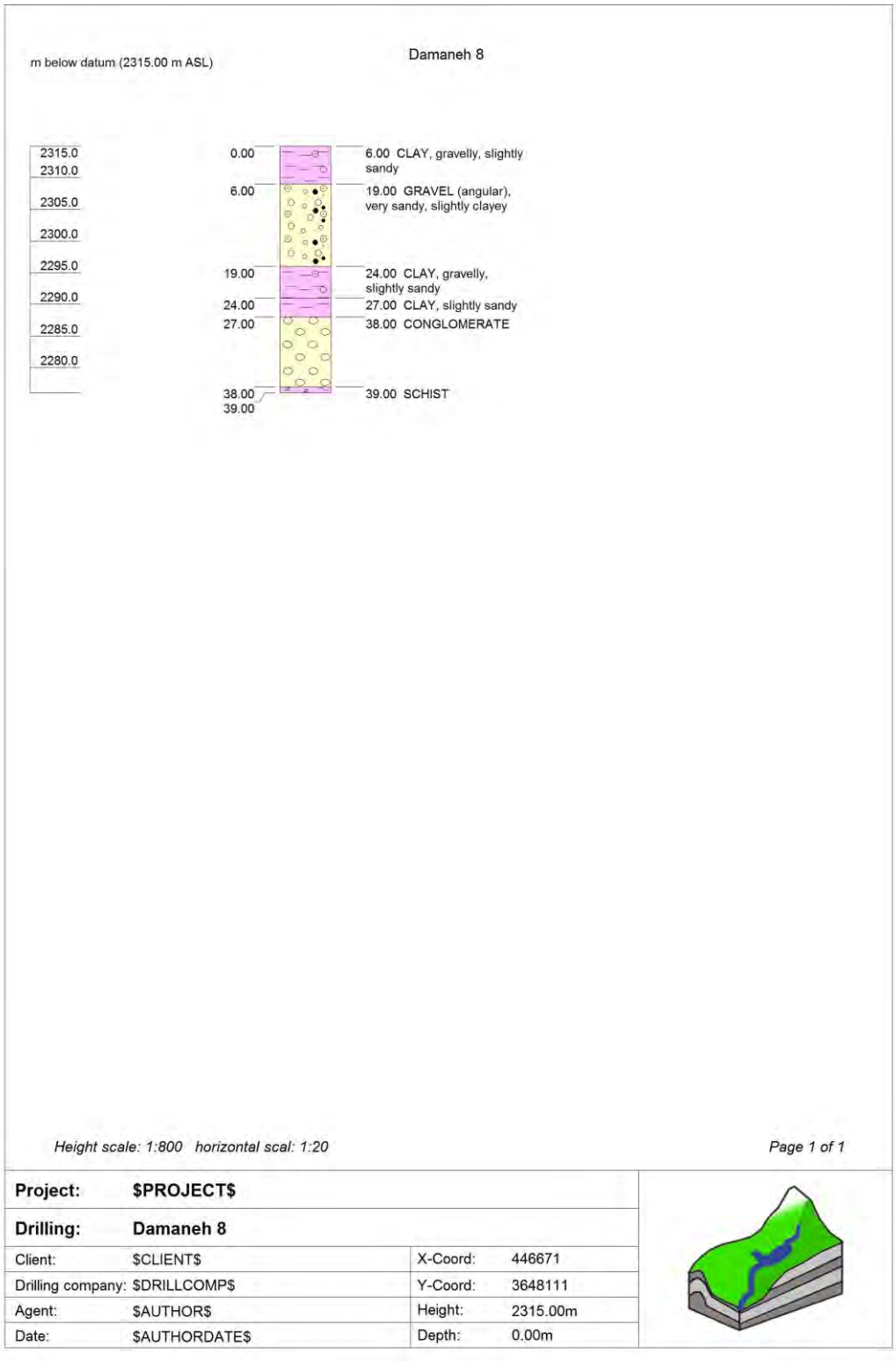
Appendix 18: Well log data of Damaneh 5



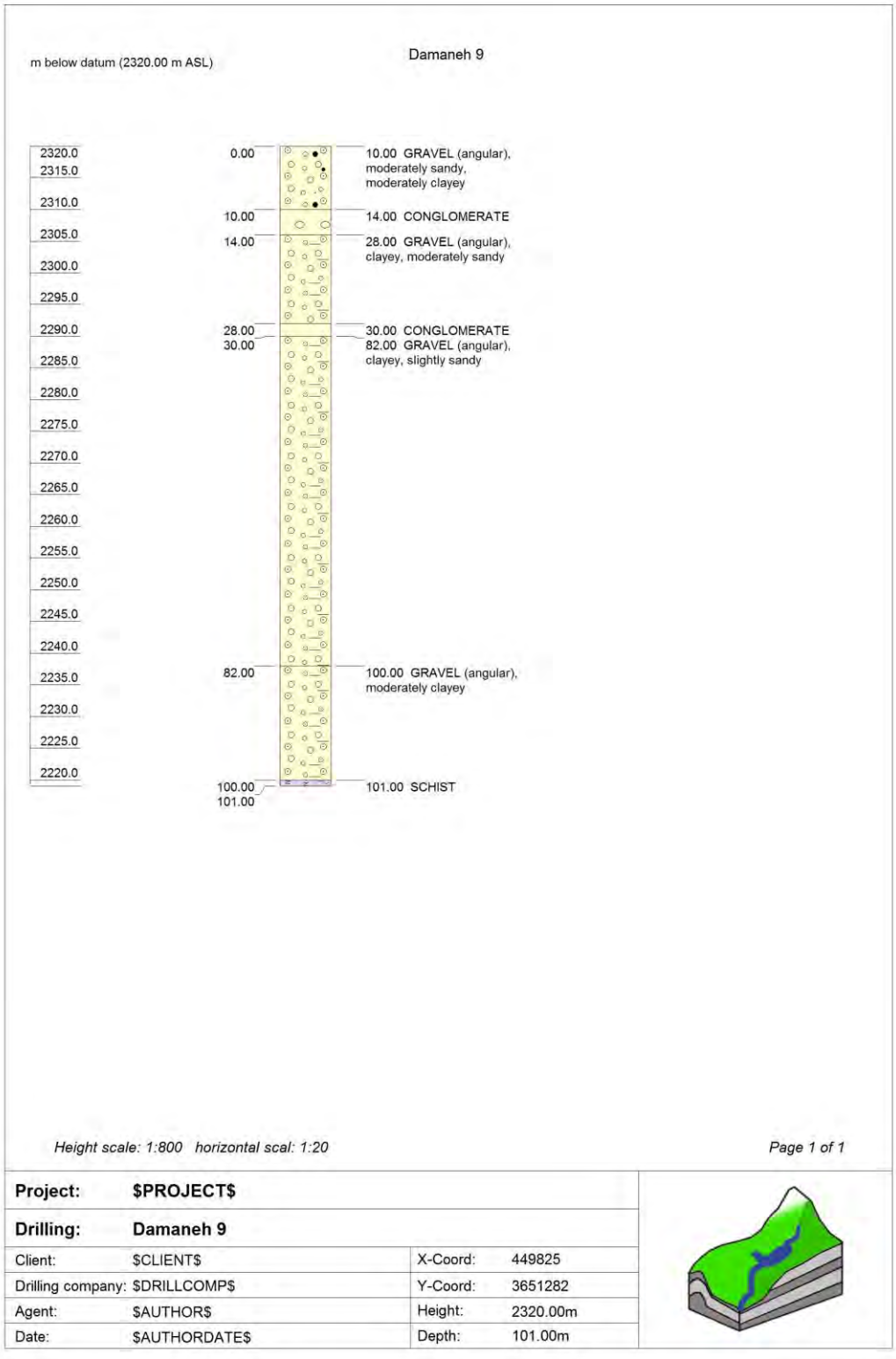
Appendix 19: Well log data of Damaneh 6



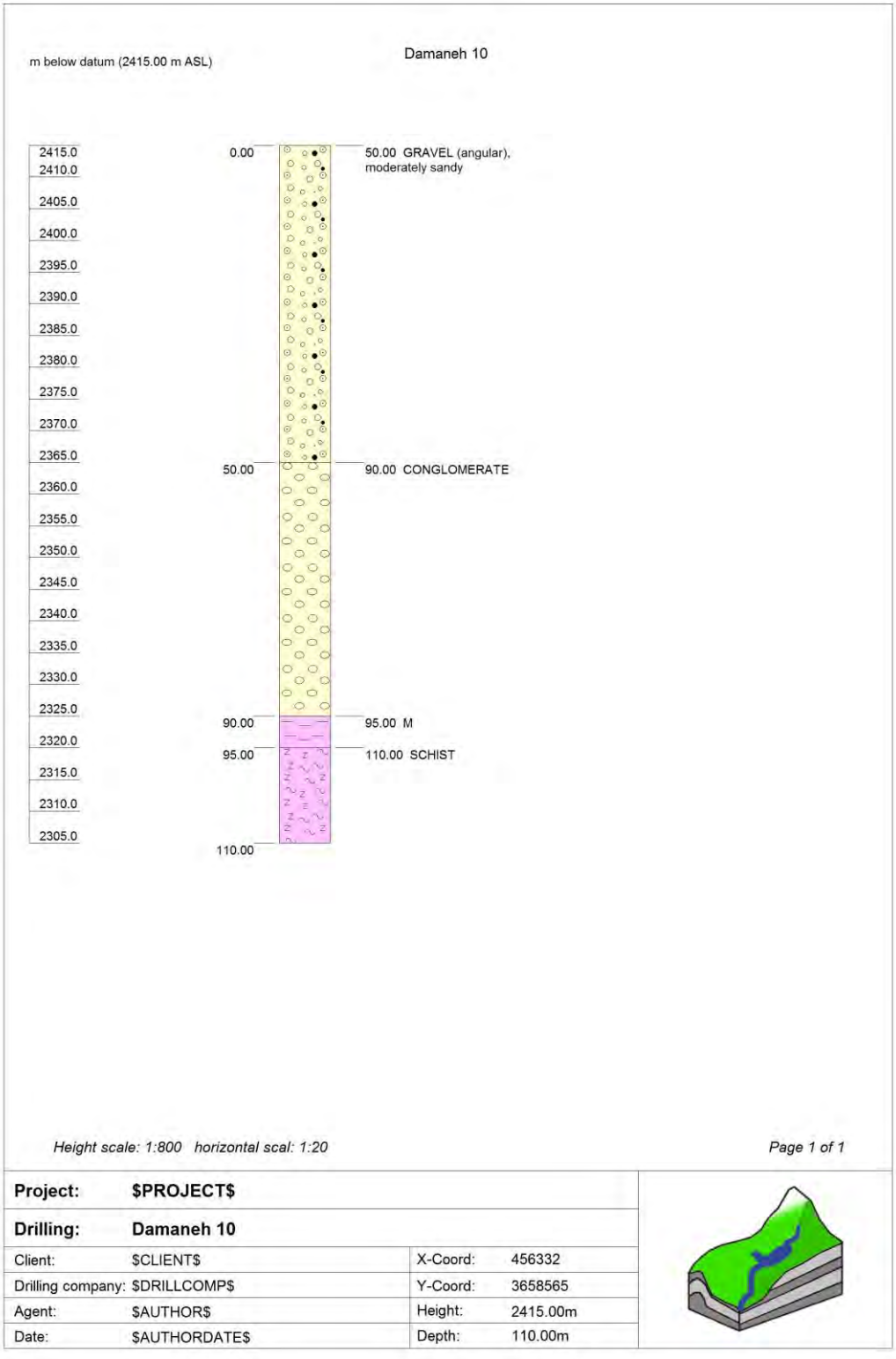
Appendix 20: Well log data of Damaneh 7



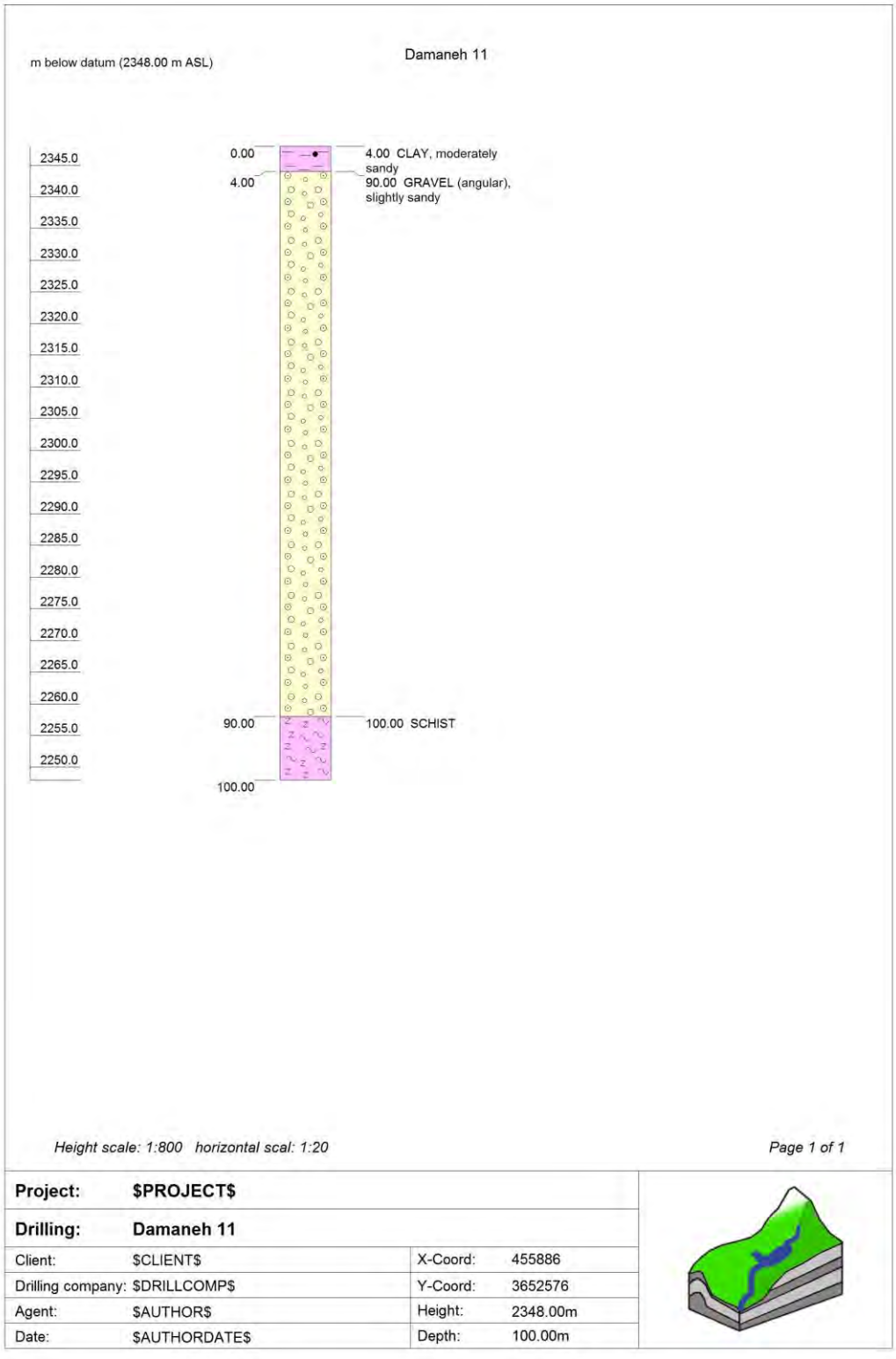
Appendix 21: Well log data of Damaneh 8



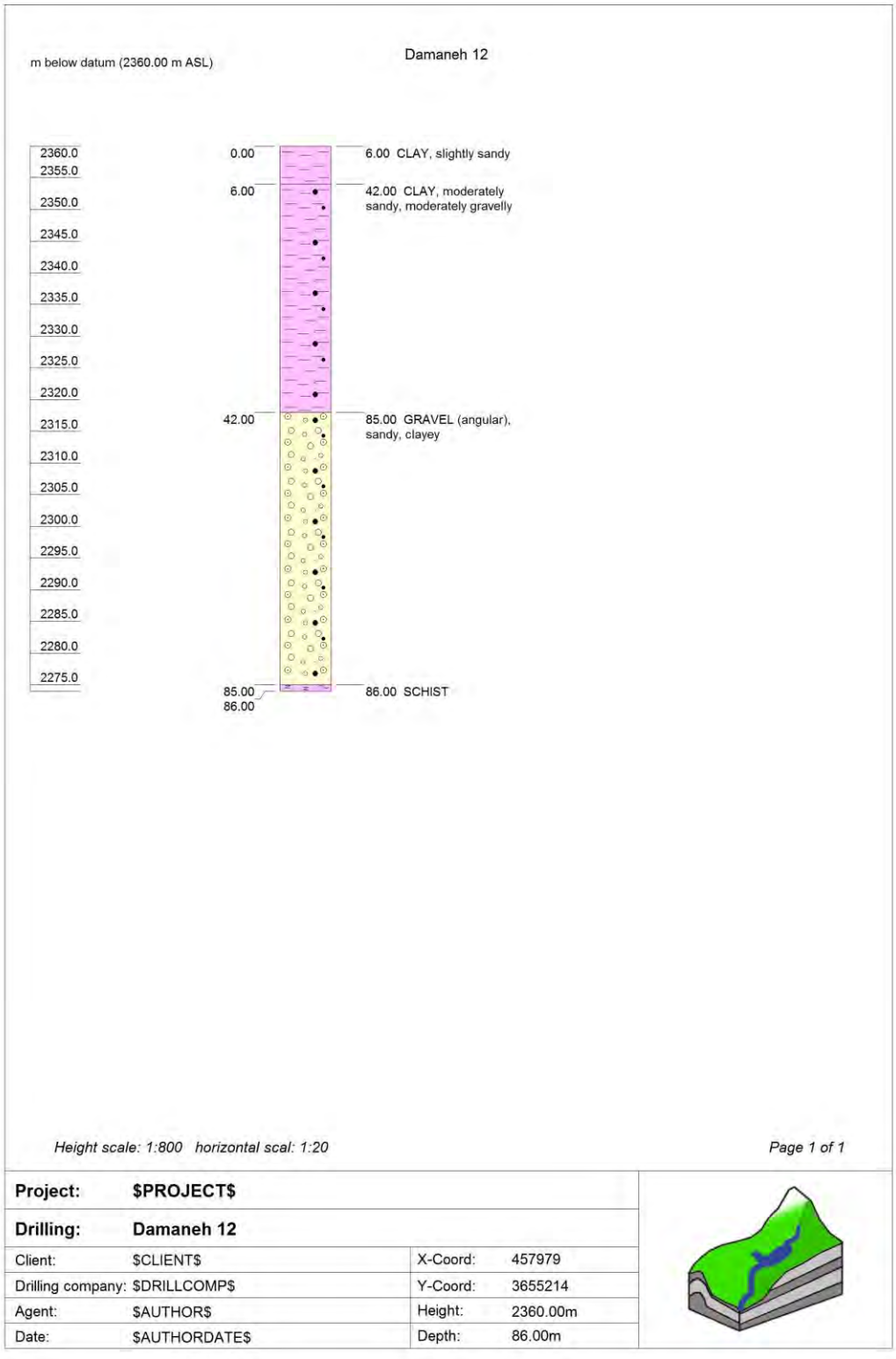
Appendix 22: Well log data of Damaneh 9



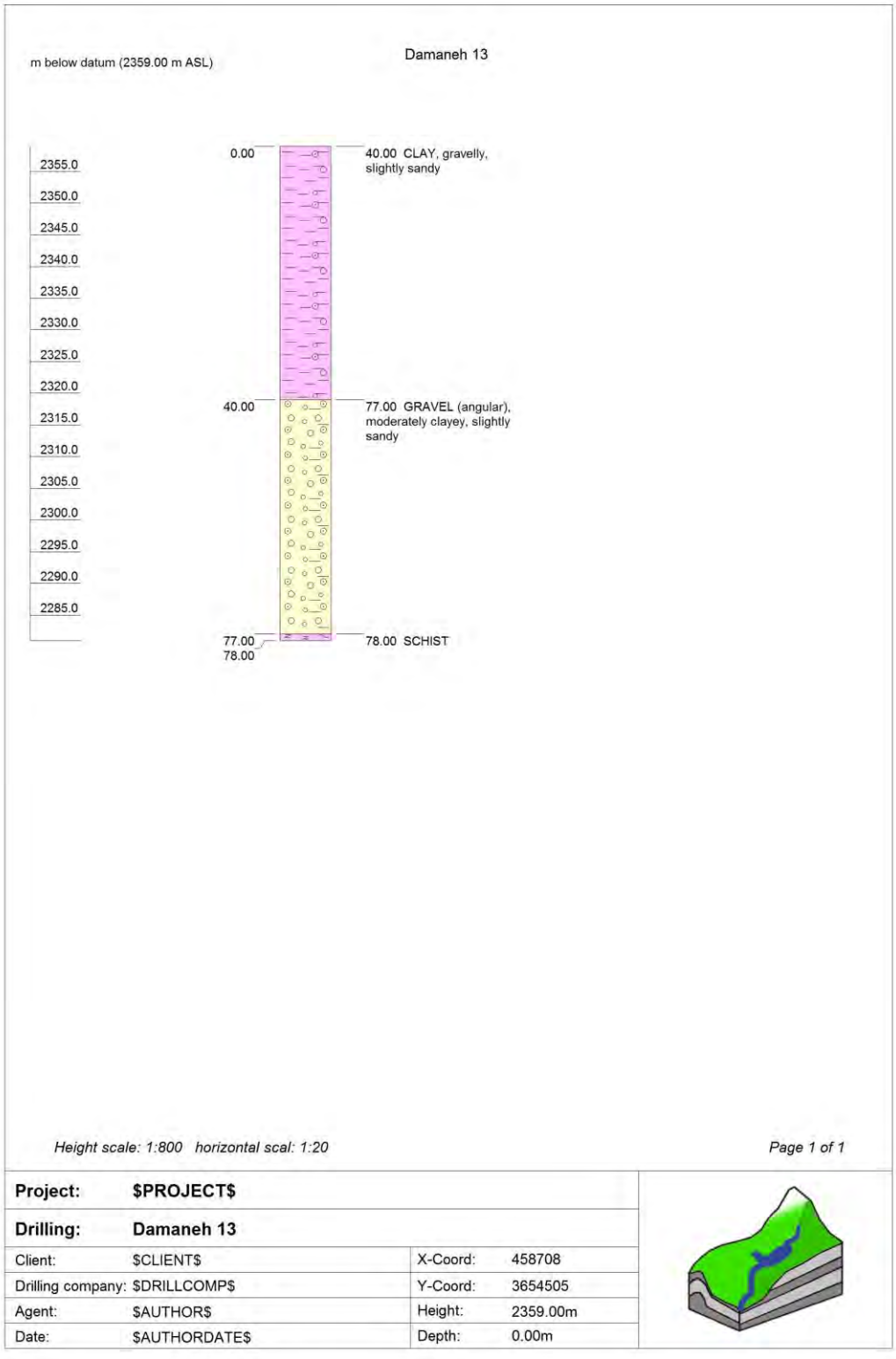
Appendix 23: Well log data of Damaneh 10



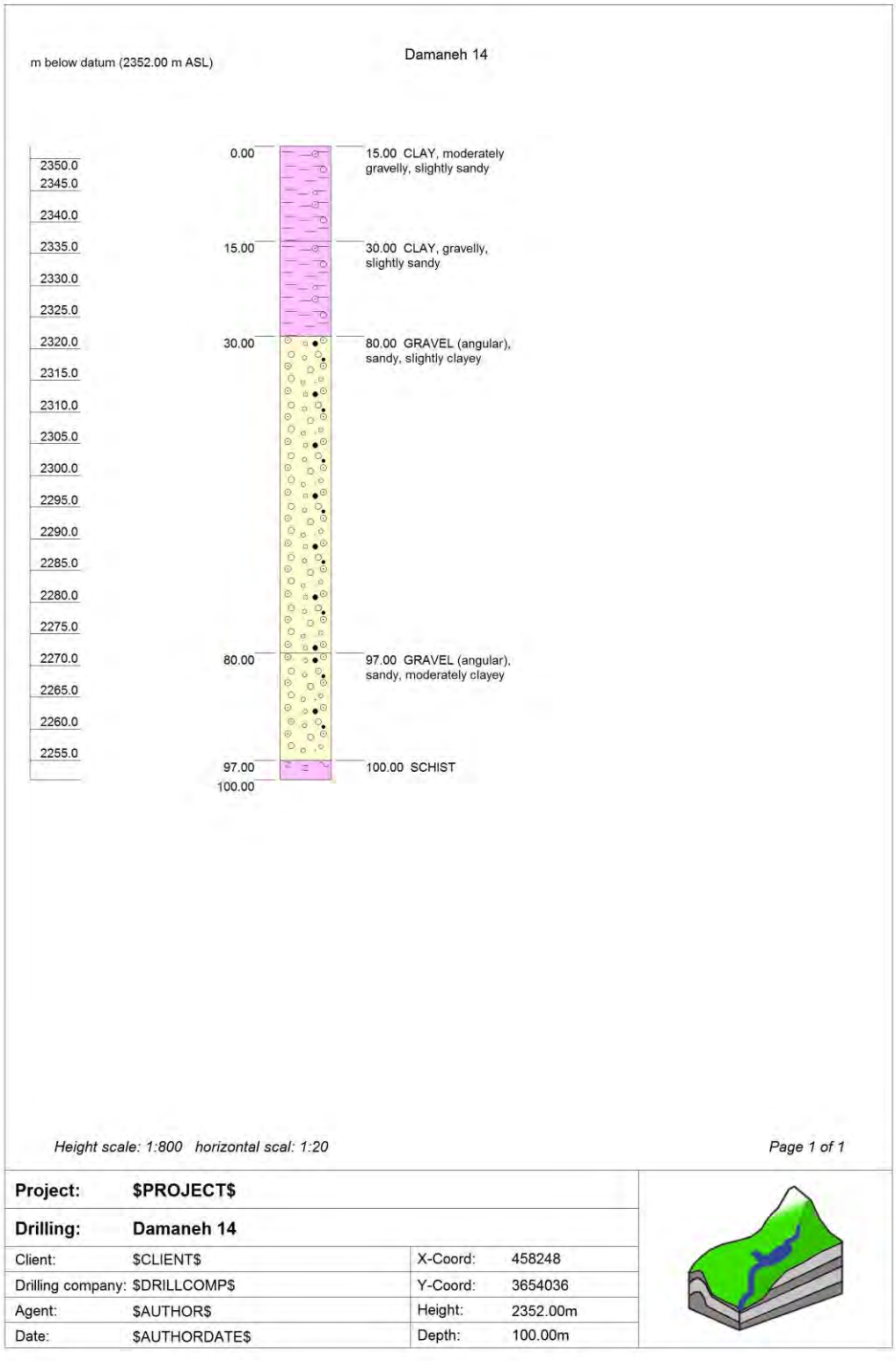
Appendix 24: Well log data of Damaneh 11



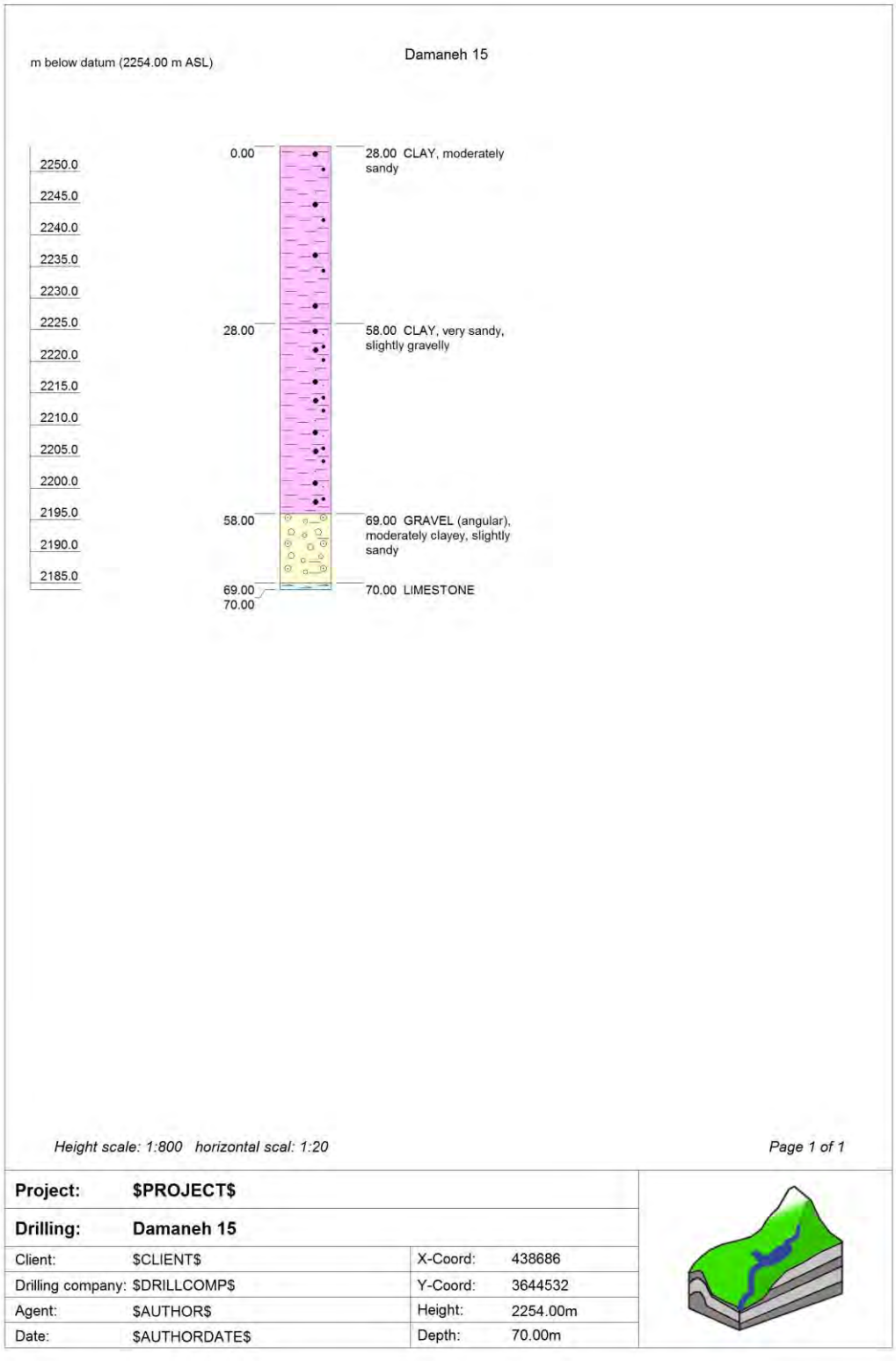
Appendix 25: Well log data of Damaneh 12



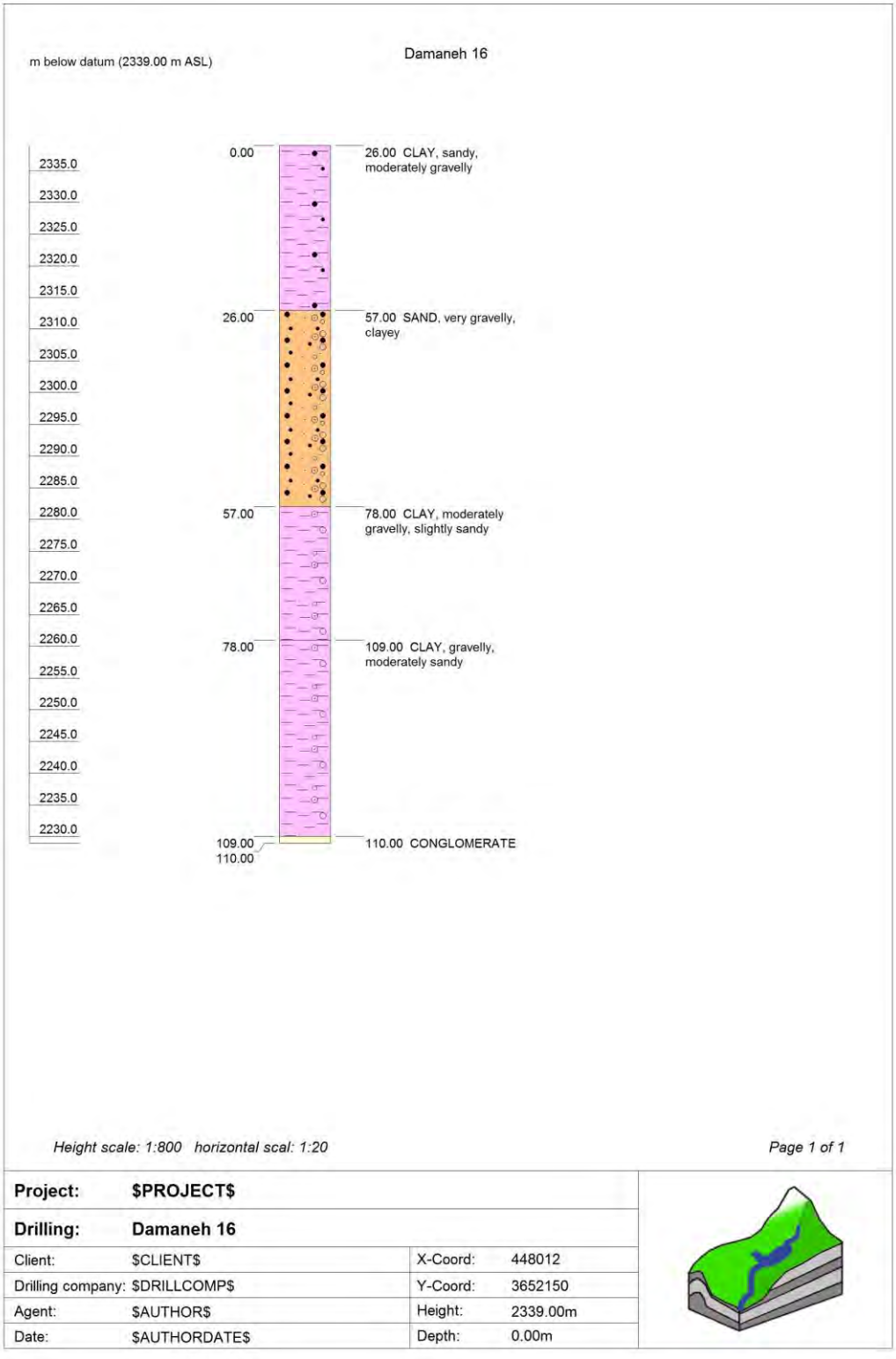
Appendix 26: Well log Data of Damaneh 13



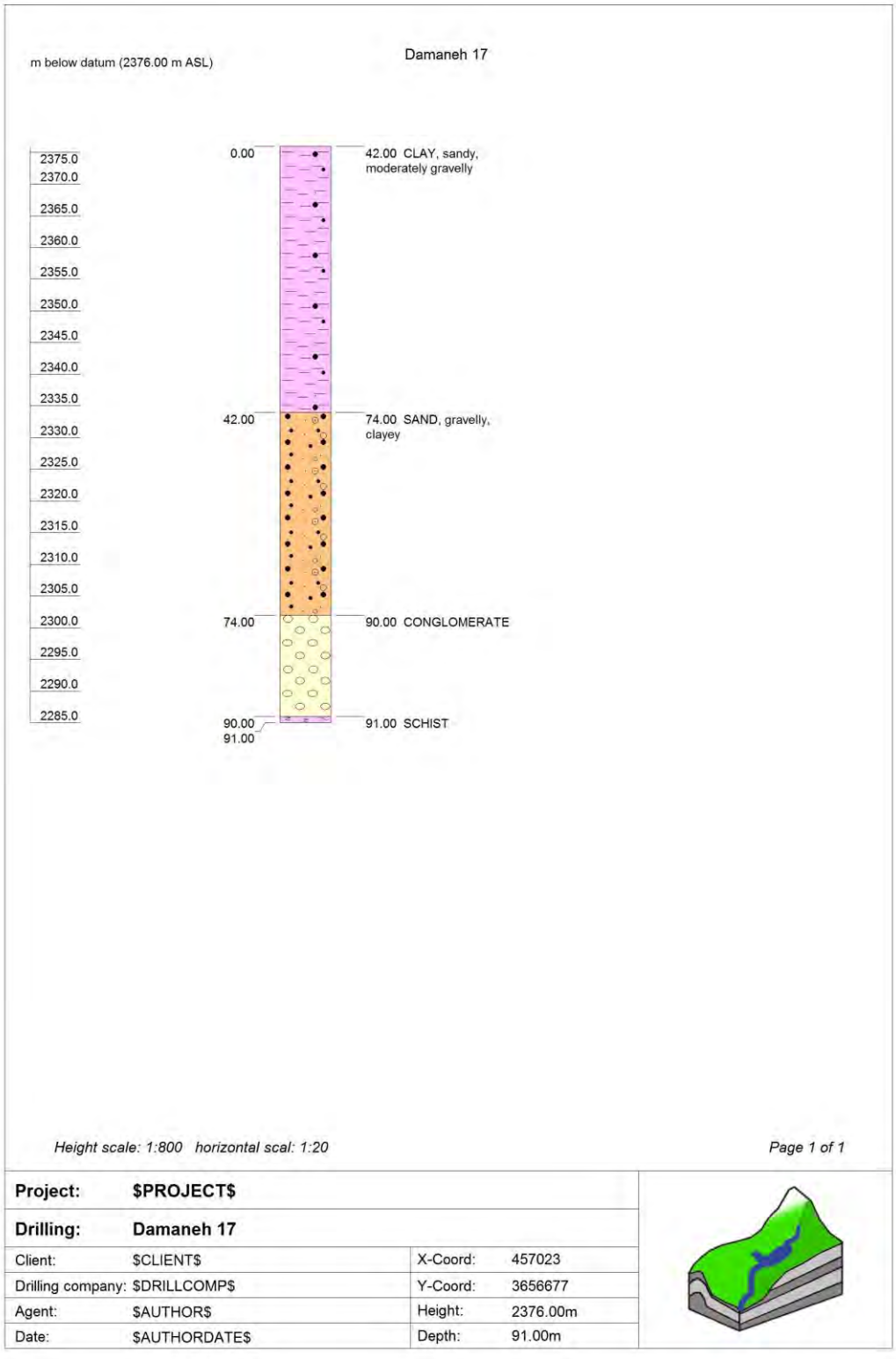
Appendix 27: Well log data of Damaneh 14



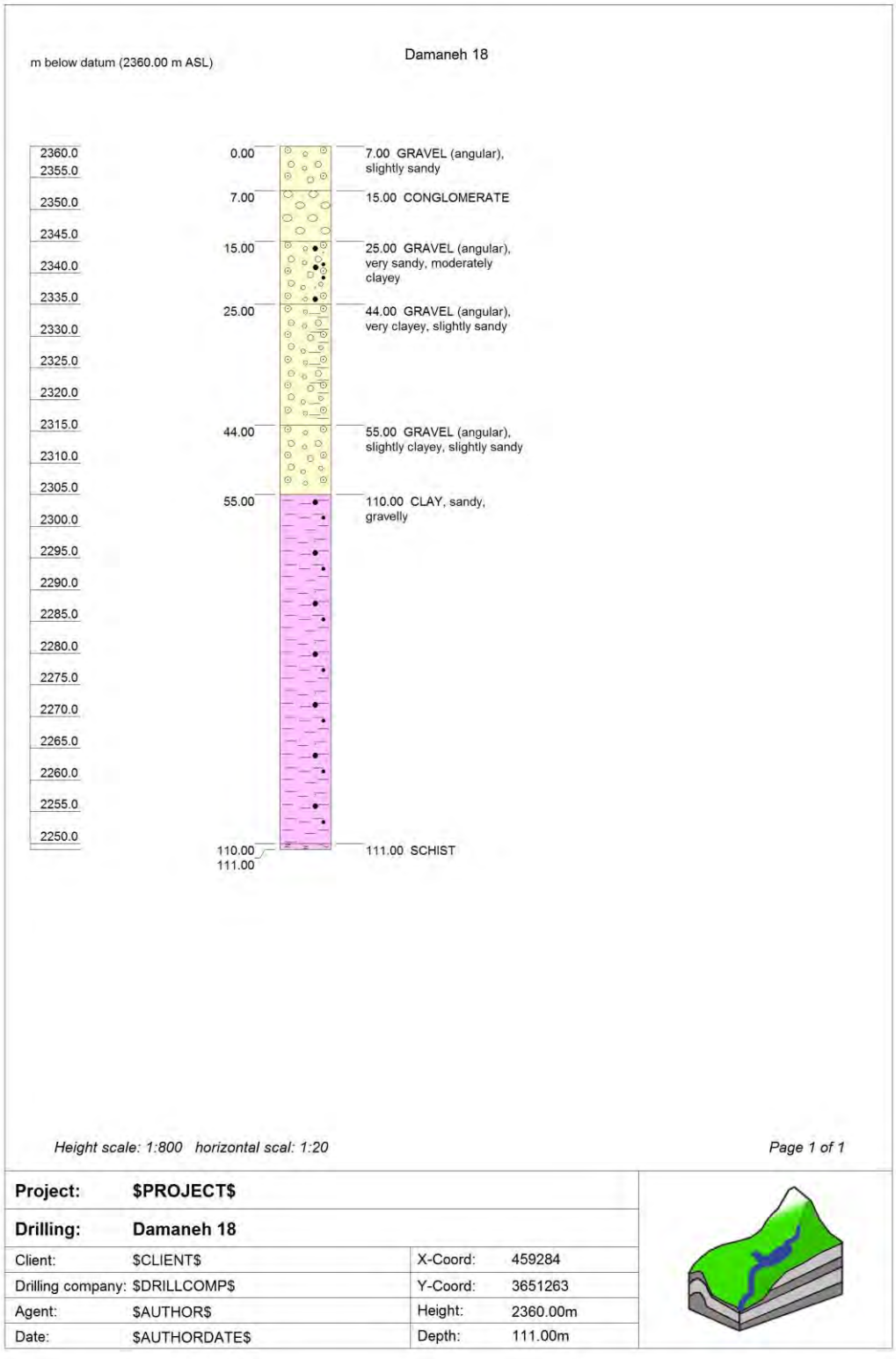
Appendix 28: Well log data of Damaneh 15



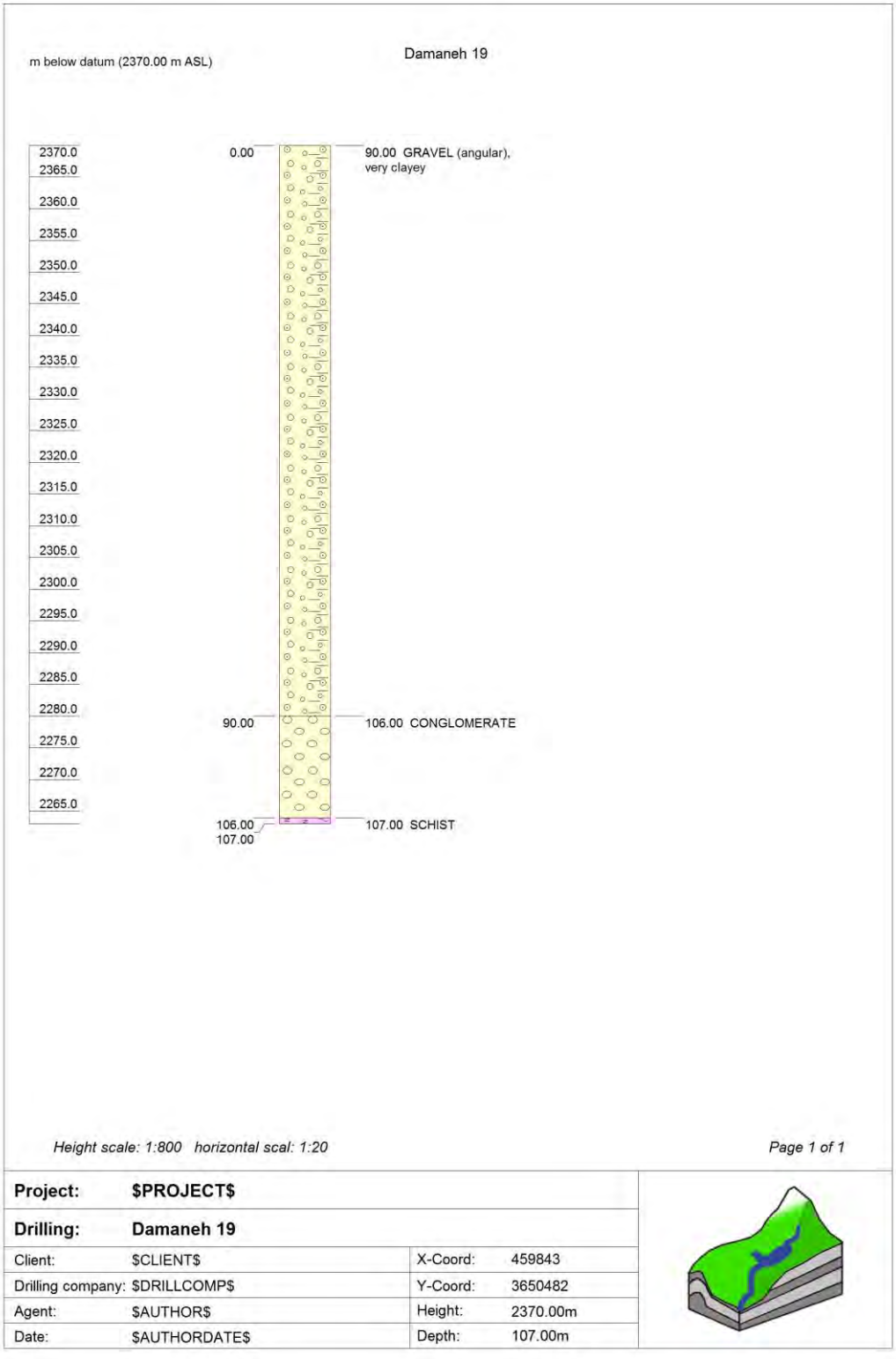
Appendix 29: Well log data of Damaneh 16



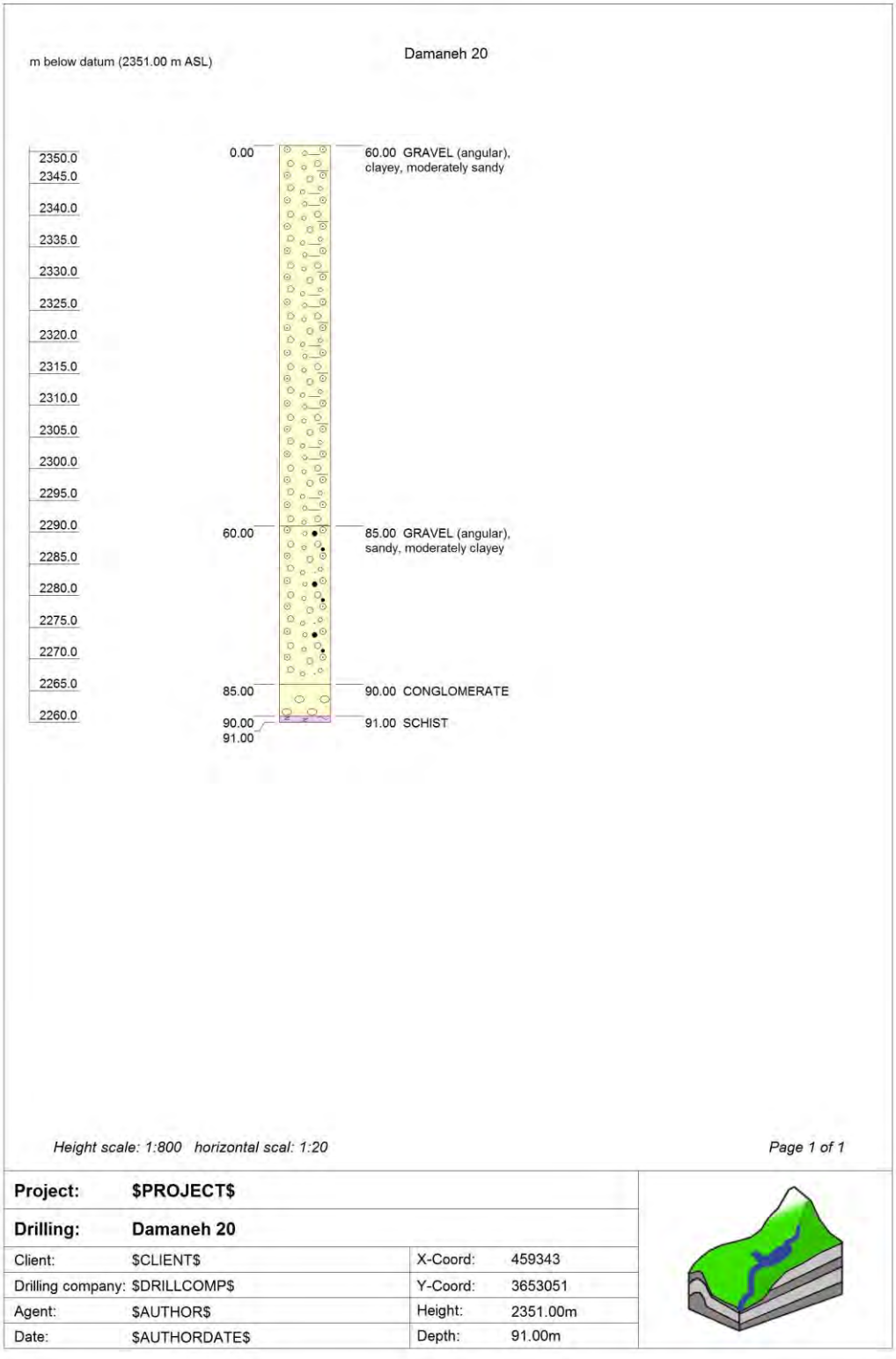
Appendix 30: Well log data of Damaneh 17



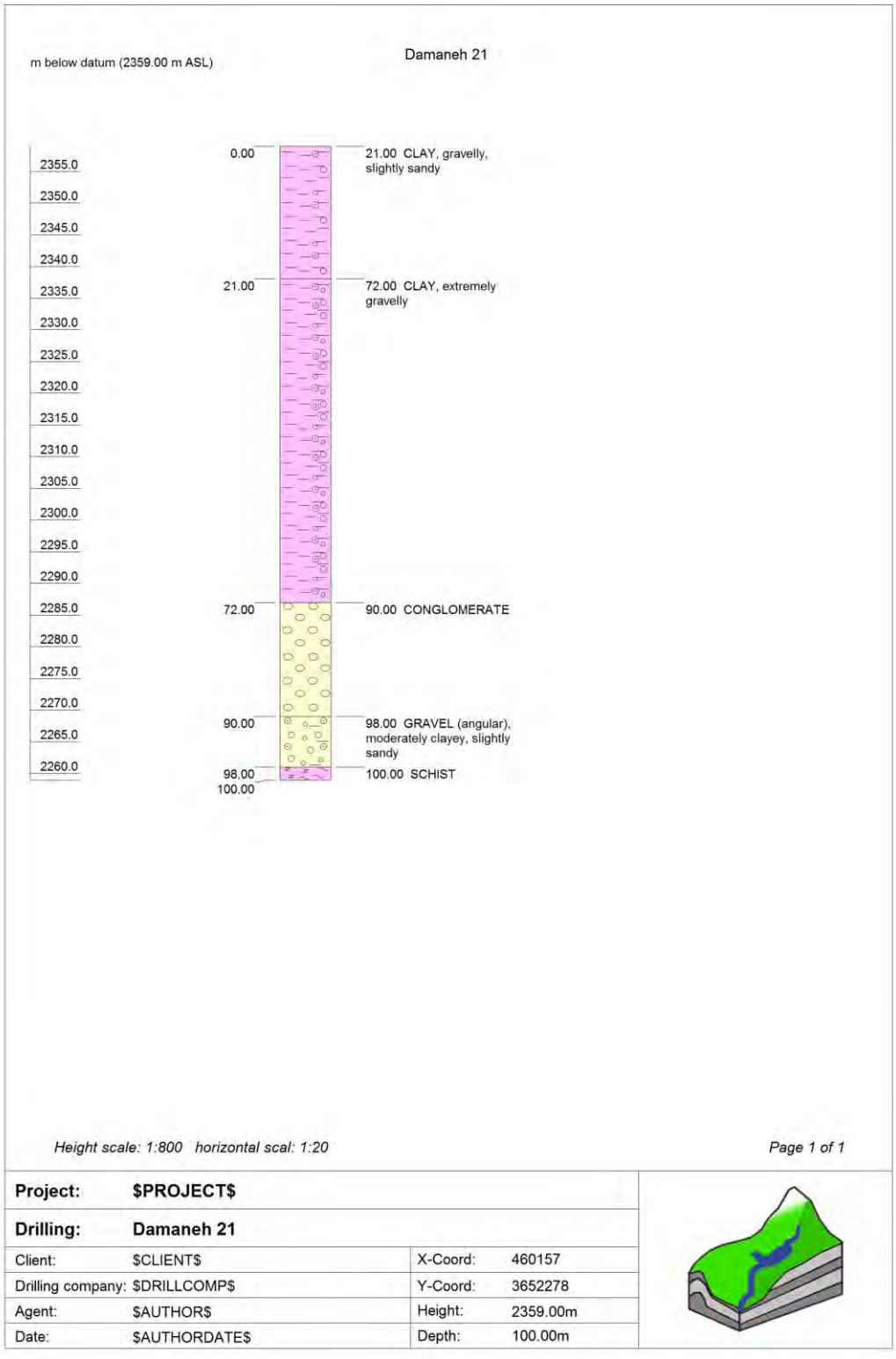
Appendix 31: Well log data of Damaneh 18



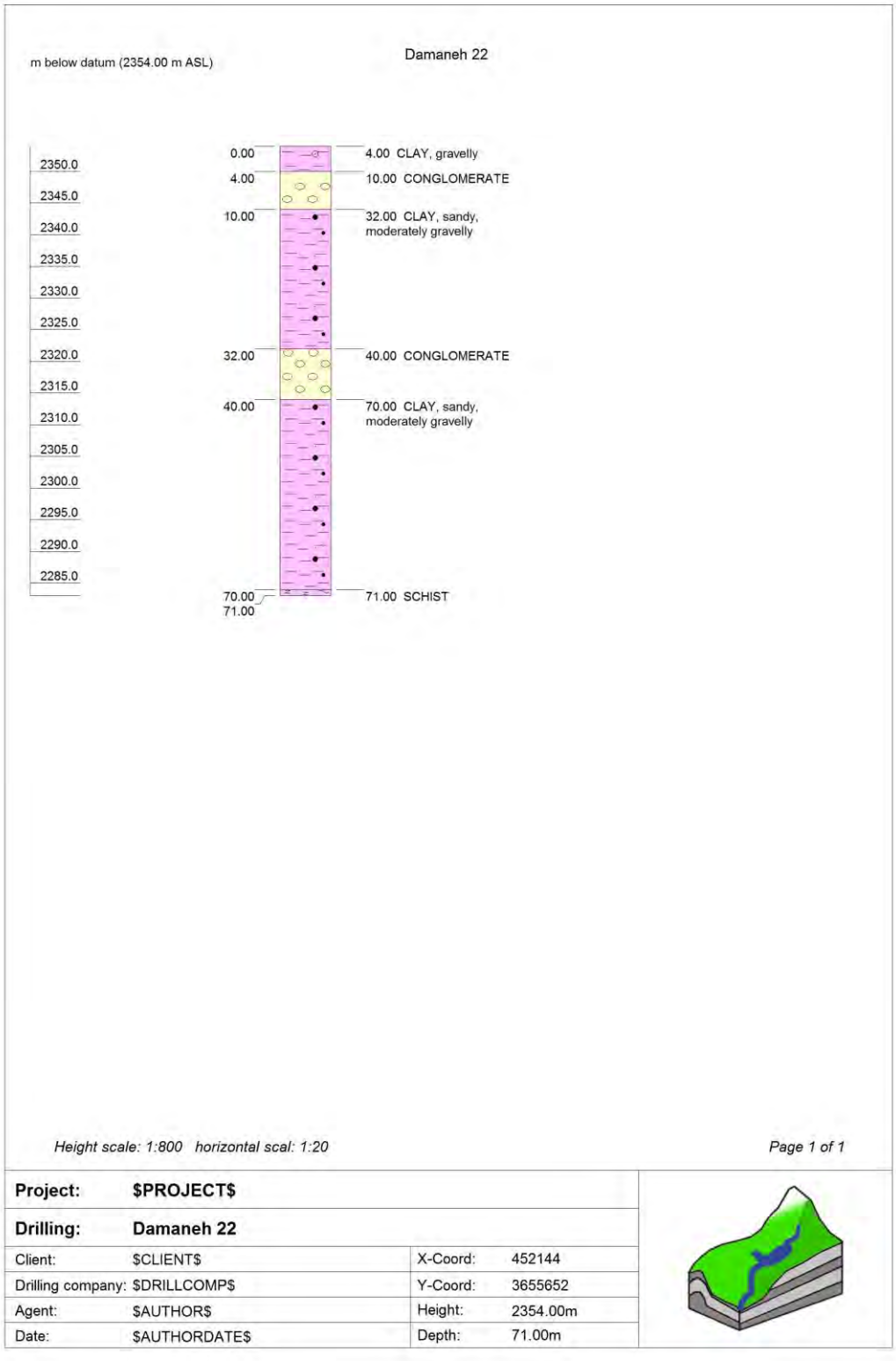
Appendix 32: Well log data of Damaneh 19



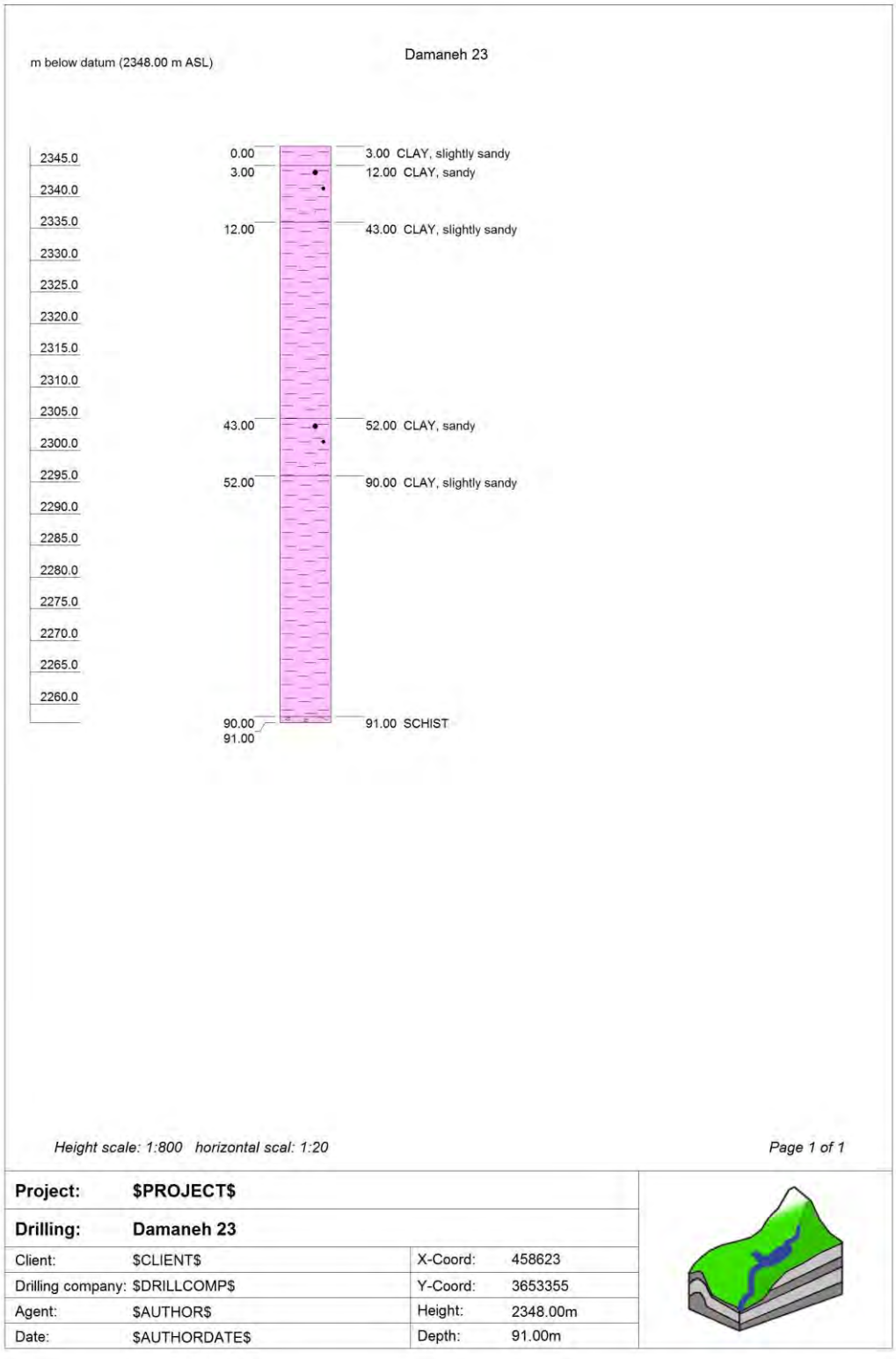
Appendix 33: Well log data of Damaneh 20



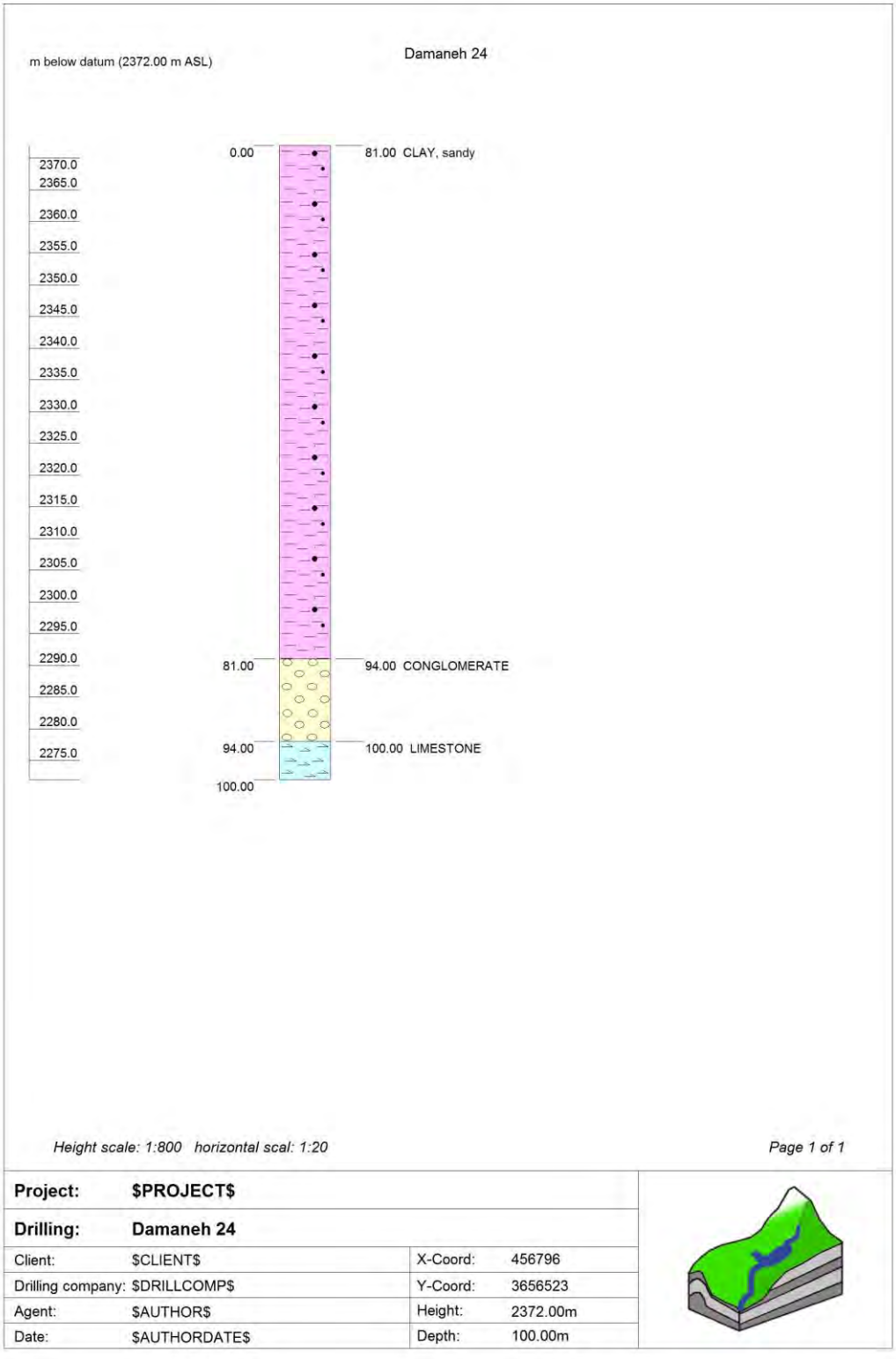
Appendix 34: Well log data of Damaneh 21



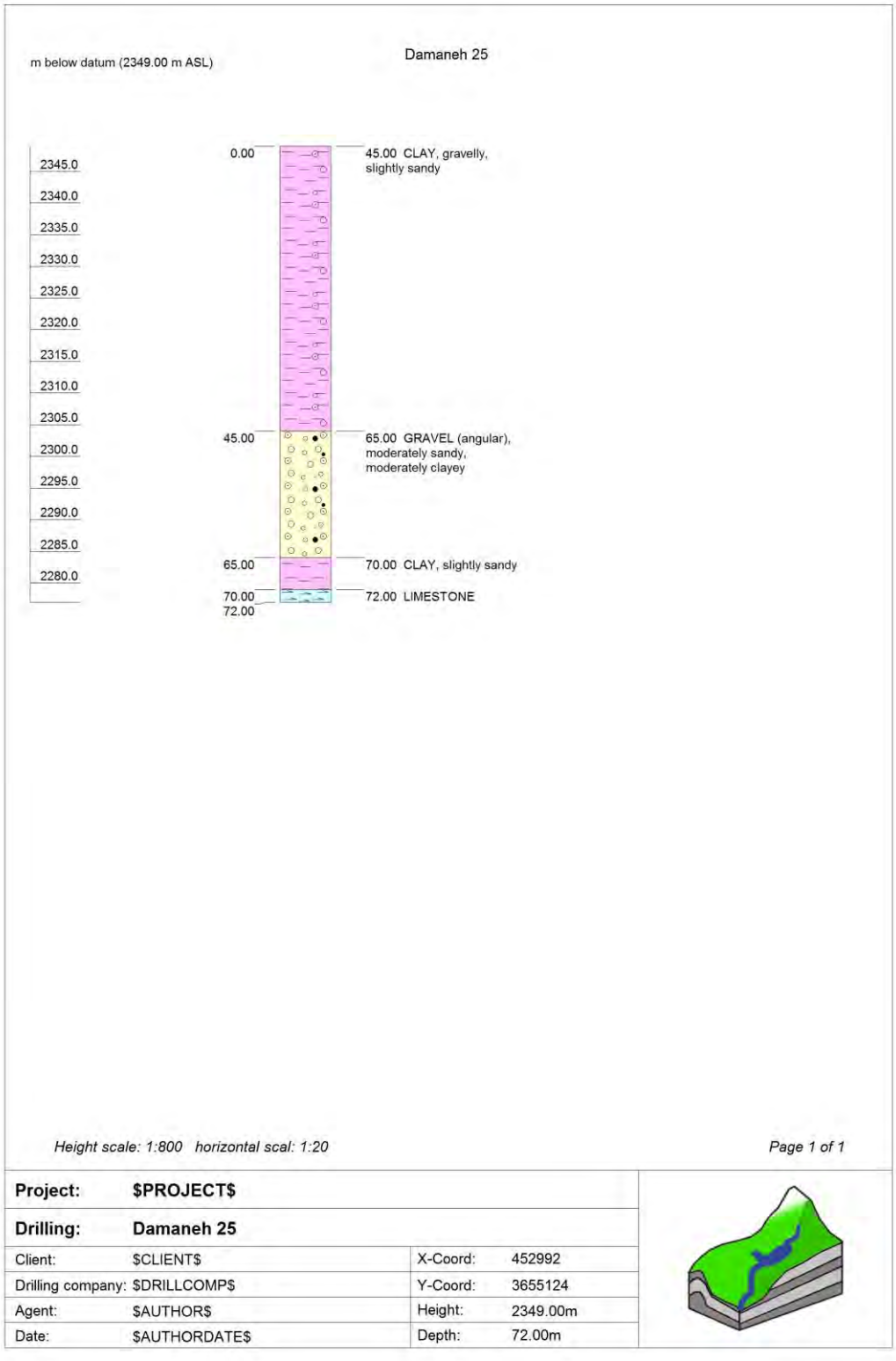
Appendix 35: Well log data of Damaneh 22



Appendix 36: Well log data of Damaneh 23



Appendix 37: Well log data of Damaneh 24



Appendix 38: Well log data of Damaneh 25